NURSERY PROPAGATION REFERENCE

Native Plant Seed Sowing Guide

Western Australian native species – Joondalup, WA

Version 1.9 · October 2026 · 106 species

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Easy · Easy–Mod. · Moderate · Mod.–Difficult · Difficult

Germination figures are nursery test averages – your own seed lots will vary. Record results to refine this guide.
Compiled from published sources (see References). Windows and ratings are starting points for local practice.

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How to use this guide

Each species has its own page set, listed alphabetically by scientific name. Every page follows the same layout: quick facts, a sowing calendar, a plant snapshot, seed collection and storage, the sowing and growing-on method, common problems, and a blank record table so the nursery can log germination and refine the advice over time.

Sowing windows are written for the Joondalup / Perth coastal climate. Most south-west WA seed germinates best at cool to mild temperatures (roughly 15–25°C), so autumn is the main sowing season. Germination figures come from routine tests at Alcoa's Marrinup Nursery and are a guide to what a good seed lot can do – not a guarantee.

Species index – at a glance

Species	Page	Difficulty	Pre-treatment	Best sowing	Depth	Germ. %*	Seed / 100 plants*
Acacia cochlearis Rigid Wattle	12	Easy	Boiling water 30 s	**Mar – May**	3–5 mm	60%	1.2 g
Acacia lasiocarpa Panjang	14	Easy	Boiling water 30 s	**Mar – May**	3–5 mm	64%	1.0 g
Acacia pulchella Prickly Moses	16	Easy	Boiling water 30 s	**Mar – May**	2–3 mm	61%	1.3 g
Acacia truncata West Coast Wattle	18	Easy–Moderate	Boiling water 30 s	**Mar – May**	3–5 mm	41%	1.1 g
Acanthocarpus preissii Prickle Lily	20	Moderate	Warm stratify 7+ weeks, then cool	**Dec – Feb**	3–5 mm	–	–
Allocasuarina fraseriana Sheoak / Western Sheoak	22	Easy	None	**Mar – May**	2–3 mm	37%	1.3 g
Allocasuarina humilis Dwarf Sheoak	24	Easy–Moderate	None	**Mar – May**	2–3 mm	25%	1.3 g
Allocasuarina lehmanniana Dune Sheoak	26	Easy–Moderate	None	**Mar – May**	2–3 mm	19%	0.5 g
Ammothryon grandiflorum Large-flowered Bog Rush	28	Difficult *	Heat + smoke (trial)	**Apr – May**	2–3 mm	–	–
Amphipogon turbinatus	30	Difficult *	Leach + smoke (trial)	**Feb – Apr**	Just covered	–	–
Anigozanthos humilis Catspaw	32	Moderate	Smoke water (optional)	**Mar – May**	1–2 mm	4%	1.9 g
Anigozanthos manglesii Red and Green Kangaroo Paw	34	Moderate	Dry heat 100°C 3 h, or aged seed	**Mar – May**	1–2 mm	70%	0.5 g
Atriplex cinerea Grey Saltbush	36	Easy–Moderate	Soak 4 days; smoke water	**Mar – May**	1–2 mm	21%	9.1 g
Atriplex isatidea Coast Saltbush	38	Easy–Moderate	Soak 4 days; smoke water	**Mar – May**	1–2 mm	22%	6.7 g
Austrostipa elegantissima Feather Speargrass	40	Moderate	Aged seed; smoke-water soak	**Mar – May**	Just covered	–	–
Austrostipa flavescens Coast Speargrass	42	Easy–Moderate	Smoke water; aged seed	**Mar – May**	0–2 mm	–	–
Banksia attenuata Slender Banksia / Candle Banksia	44	Easy–Moderate	None (extract from cones)	**Mar – May**	3–5 mm	60%	6.7 g
Banksia dallanneyi Couch Honeypot	46	Easy–Moderate	None (warm fruit to release)	**Mar – May**	3–5 mm	37%	3.0 g
Banksia grandis Bull Banksia	48	Easy	None	**Mar – May**	3–5 mm	78%	7.4 g

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Species	Page	Difficulty	Pre-treatment	Best sowing	Depth	Germ. %*	Seed / 100 plants*
Banksia menziesii Firewood Banksia	50	Easy–Moderate	None (extract from cones)	**Mar – May**	3–5 mm	56%	3.7 g
Banksia prionotes Acorn Banksia	52	Easy–Moderate	None (heat cones to open)	**Mar – May**	3–5 mm	57%	3.4 g
Banksia sessilis Parrot Bush	54	Easy–Moderate	None (smoke water optional)	**Mar – May**	3–5 mm	42%	3.0 g
Bossiaea eriocarpa Common Brown Pea	56	Easy–Moderate	Boiling water 30 s	**Mar – May**	2–3 mm	44%	0.4 g
Burchardia congesta Milkmaids	58	Moderate	None; smoke optional	**Mar – May**	Lightly covered	44%	0.5 g
Calothamnus quadrifidus One-sided Bottlebrush	60	Easy	None	**Mar – Apr, Sep – Oct**	Surface	26%	0.2 g
Carpobrotus virescens Coastal Pigface	62	Easy–Moderate	Ferment fruit; smoke water	**Mar – May**	Surface	21%	0.2 g
Clematis linearifolia Slender Clematis / Old Man's Beard	64	Moderate	Smoke-water soak 24 h	**Apr – Jun**	Just covered (1–2 mm)	–	–
Conostylis aculeata Prickly Conostylis	66	Moderate	Smoke water	**Mar – May**	Just covered (1–2 mm)	19%	0.1 g
Conostylis candicans Grey Cottonhead	68	Mod.–Difficult	Smoke water; aged seed	**Mar – May**	Just covered	3%	0.7 g
Conostylis setigera Bristly Cottonhead	70	Easy–Moderate	None; smoke water helps	**Mar – May**	Just covered	31%	0.2 g
Corymbia calophylla Marri	72	Easy	None	**Mar – Apr, Sep – Oct**	About 5 mm	60%	5.3 g
Daviesia decurrens Prickly Bitter-pea	74	Moderate	Dry heat or hot water	**Mar – May**	3–5 mm	25%	11 g
Daviesia divaricata Marno	76	Moderate	Boiling water 30 s	**Mar – May**	3–5 mm	28%	9.2 g
Daviesia nudiflora	78	Moderate	Boiling water 30 s	**Mar – May**	3–5 mm	22%	1.8 g
Daviesia triflora	80	Moderate	Boiling water 30 s	**Mar – May**	3–5 mm	26%	9.2 g
Dianella revoluta Blueberry Lily / Flax Lily	82	Difficult	GA 24 h + smoke water 24 h	**Mar – Apr**	3–5 mm	<1%†	–
Diplolaena angustifolia Yanchep Rose	84	Mod.–Difficult *	Smoke water (trial GA)	**Mar – May**	About 4 mm	–	–
Dodonaea aptera Coast Hop-bush	86	Moderate	Hot water	**Mar – May**	Lightly covered	47%	1.5 g
Eremaea pauciflora Orange-flowered Eremaea	88	Easy–Moderate *	None	**Mar – May**	Surface – light cover	–	–
Eryngium pinnatifidum Blue Devils	90	Moderate *	None	**Mar – May**	Lightly covered	3%	10 g
Eucalyptus decipiens Redheart / Redheart Moit	92	Easy	None	**Mar – Apr, Sep – Oct**	1–2 mm	46%	0.2 g
Eucalyptus gomphocephala Tuart	94	Easy	None	**Mar – Apr, Sep – Oct**	1–2 mm	42%	0.4 g

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Species	Page	Difficulty	Pre-treatment	Best sowing	Depth	Germ. %*	Seed / 100 plants*
Eucalyptus marginata Jarrah	96	Easy	None	**Mar – Apr, Sep – Oct**	2–3 mm	31%	3.0 g
Exocarpos sparteus Broom Ballart	98	Difficult *	Gibberellic acid 48 h	**Apr – Jun**	About 5 mm	–	–
Ficinia nodosa Knotted Club Rush	100	Easy	None; smoke water to water in	**Mar – May; Sep – Oct**	Surface	55%	<0.1 g
Frankenia pauciflora Seaheath	102	Easy *	None	**Apr – Jun**	Surface – light cover	–	–
Gastrolobium capitatum Bacon and Eggs	104	Moderate *	Hot water or dry heat	**Mar – May**	2–3 mm	–	–
Gastrolobium nervosum	106	Moderate *	Hot water or dry heat	**Mar – May**	2–3 mm	–	–
Gompholobium tomentosum Hairy Yellow Pea	108	Easy–Moderate	Dry heat or hot water	**Mar – May**	3–4 mm	34%	0.8 g
Grevillea vestita	110	Moderate	Nick seed coat; smoke water	**Mar – May**	3–5 mm	–	–
Guichenotia ledifolia	112	Mod.–Difficult	Boiling water over seed; smoke	**Mar – May**	2–3 mm	–	–
Haemodorum laxum Bloodroot	114	Moderate	None (smoke optional)	**Mar – May**	Lightly covered	73%	0.7 g
Hakea lissocarpha Honey Bush	116	Easy	None	**Mar – May**	3–5 mm	66%	2.0 g
Hakea prostrata Harsh Hakea	118	Easy	None	**Mar – May**	3–5 mm	79%	4.9 g
Hakea trifurcata Two-leaf Hakea	120	Easy	None	**Mar – May**	3–5 mm	69%	1.7 g
Hardenbergia comptoniana Native Wisteria	122	Moderate	Hot water (or boil 30 s); smoke	**Mar – May**	3–5 mm	31%	5.3 g
Hovea pungens Devil's Pins	124	Easy–Moderate	Boiling water 30 s	**Mar – May**	4–6 mm	41%	3.0 g
Hovea trisperma Common Hovea	126	Moderate	Boiling water 30 s	**Mar – May**	3–5 mm	31%	4.4 g
Hyalosperma cotula	128	Moderate	None, or smoke water	**Apr – May**	Surface – light cover	27%	<0.1 g
Jacksonia furcellata Grey Stinkwood	130	Moderate	Boiling water 30 s	**Mar – May**	2–3 mm	34%	2.1 g
Jacksonia sericea Waldjumi	132	Moderate	Boiling water 30 s	**Mar – May**	2–3 mm	26%	1.2 g
Jacksonia sternbergiana Stinkwood / Green Stinkwood	134	Moderate	Boiling water 30 s	**Mar – May**	3–5 mm	33%	3.0 g
Kennedia prostrata Scarlet Runner / Running Postman	136	Easy	Hot water + smoke water	**Mar – May**	About 5 mm	30%	8.3 g
Laxmannia squarrosa Paper Lily	138	Mod.–Difficult	Smoke water	**Mar – May**	Lightly covered	5%	0.6 g
Lepidosperma gladiatum Coast Sword-sedge	140	Mod.–Difficult *	None, or heat + smoke (trial)	**Apr – May**	3–5 mm	41%	0.3 g
Leucophyta brownii Cushion Bush	142	Easy–Moderate	None	**Mar – May**	Surface – light cover	–	–

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Species	Page	Difficulty	Pre-treatment	Best sowing	Depth	Germ. %*	Seed / 100 plants*
Leucopogon parviflorus Coast Beard-heath	144	Difficult *	Peroxide + smoke water	**Mar – May**	Just covered	–	–
Macrozamia fraseri Sandplain Zamia	146	Moderate	Remove fleshy coat; float test	**Feb – Mar (fresh seed)**	Surface	–	–
Melaleuca cardiophylla Tangling Melaleuca	148	Easy–Moderate	None (smoke water to water in)	**Mar – Apr, Sep – Oct**	Surface	19%	0.5 g
Melaleuca huegelii Chenille Honeymyrtle	150	Easy	None	**Mar – Apr, Sep – Oct**	Surface	58%	0.1 g
Melaleuca lanceolata Rottnest Teatree / Moonah	152	Easy	None (smoke water to water in)	**Mar – Apr, Sep – Oct**	Surface	55%	<0.1 g
Melaleuca systena Coastal Honeymyrtle	154	Easy–Moderate	None	**Mar – Apr, Sep – Oct**	Surface	14%	0.3 g
Mesomelaena pseudostygia Semaphore Sedge	156	Difficult *	Heat + smoke (trial)	**Apr – May**	2–3 mm	–	–
Myoporum insulare Blueberry Tree / Common Boobialla	158	Moderate	Ferment berries; smoke-water soak	**Mar – May**	2–3 mm	–	–
Nuytsia floribunda WA Christmas Tree / Moodjar	160	Mod.–Difficult	None – sow fresh	**Feb – Apr (fresh seed)**	Surface, upright	29%	2.4 g
Olearia axillaris Coastal Daisybush	162	Easy–Moderate	Air-dry 3 months; smoke water	**Aug – Sep**	Surface	21%	1.4 g
Opercularia vaginata Dog Weed	164	Difficult *	Smoke water (trial)	**Apr – May**	1–2 mm	<1%†	38 g
Orthrosanthus laxus Morning Iris	166	Mod.–Difficult	Smoke water	**Mar – May**	Lightly covered	7%	5.6 g
Patersonia occidentalis Purple Flag	168	Difficult	Smoke; aged seed	**May – Jun**	Lightly covered	2%	14 g
Petrophile axillaris	170	Difficult *	Remove hairs, nick coat; smoke	**Mar – May**	3–5 mm	–	–
Petrophile brevifolia	172	Difficult *	Remove hairs, nick coat; smoke	**Mar – May**	3–5 mm	–	–
Petrophile linearis Pixie Mops	174	Difficult	Remove hairs, nick coat; smoke	**Mar – May**	3–5 mm	<1%†	–
Petrophile macrostachya	176	Difficult *	Remove hairs, nick coat; smoke	**Mar – May**	3–5 mm	–	–
Philotheca spicata Pepper and Salt	178	Moderate	Smoke water	**Mar – May**	Lightly covered	35%	4.7 g
Pigea calycina Wild Violet	180	Difficult *	Smoke water	**Apr – May**	2–3 mm	–	–
Poa poiformis Coastal Poa	182	Easy–Moderate *	Chill 1–2 weeks	**Mar – May**	Just covered	–	–
Podotheca angustifolia Sticky Longheads	184	Easy–Moderate	None	**Apr – May**	Surface – light cover	–	0.7 g
Podotheca gnaphalioides Golden Long-heads	186	Moderate	Smoke water (optional)	**Apr – May**	Surface – light cover	7%	1.2 g
Ptilotus drummondii Narrowleaf Mulla Mulla	188	Mod.–Difficult	Remove woolly floral parts; smoke	**Mar – May**	1–2 mm	4%	8.4 g
Ptilotus manglesii Pom Poms	190	Mod.–Difficult	Remove woolly floral parts; smoke	**Mar – May**	1–2 mm	3%	18 g

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Species	Page	Difficulty	Pre-treatment	Best sowing	Depth	Germ. %*	Seed / 100 plants*
Ptilotus polystachyus Prince of Wales Feather	192	Moderate *	Remove woolly floral parts; smoke	**Mar – May**	1–2 mm	–	–
Rhagodia baccata Berry Saltbush	194	Moderate	Ferment berries; smoke-water soak	**Mar – May**	2–3 mm	11%	1.3 g
Ricinocarpos undulatus Wedding Bush	196	Difficult *	Smoke water (trial GA)	**Mar – May**	Covered ~5 mm (dark)	–	–
Santalum acuminatum Quandong	198	Mod.–Difficult	Aged seed; crack shell; GA	**May – Jul**	About the stone's thickness	53%	769 g
Spinifex hirsutus Hairy Spinifex	200	Moderate *	Cut off spines/awns; smoke water	**Mar – May**	10 mm (needs dark)	–	–
Spinifex longifolius Beach Spinifex	202	Moderate *	Cut off spines/awns; smoke water	**Mar – May**	10 mm (needs dark)	–	–
Spyridium globulosum Basket Bush	204	Moderate	Hot water (or dry heat)	**Apr – Jun**	2–3 mm (needs dark)	41%	0.5 g
Stirlingia latifolia Blueboy	206	Difficult	Smoke water (trial)	**Apr – Jun**	3–5 mm	2%	56 g
Templetonia retusa Cockies Tongues	208	Easy–Moderate	Hot water (or boil 30 s); smoke	**Mar – May**	3–5 mm	42%	5.0 g
Threlkeldia diffusa Coast Bonefruit	210	Easy–Moderate	24 h smoke-water soak	**Apr – Jun**	1–2 mm	–	–
Thysanotus manglesianus Mangles' Fringed Lily / Climbing Fringed Lily	212	Mod.–Difficult *	Smoke water	**Mar – May**	Lightly covered	–	2.0 g
Trachymene pilosa Native Parsnip / Dwarf Parsnip	214	Easy–Moderate	None; smoke optional	**Apr – May**	Lightly covered	41%	0.3 g
Tricoryne elatior Yellow Autumn Lily	216	Difficult	Smoke water (trial)	**Mar – May**	Lightly covered	1%	42 g
Trymalium ledifolium Star Buckthorn	218	Moderate	Hot water	**Apr – Jun**	2–3 mm (needs dark)	22%	0.6 g
Waitzia suaveolens Fragrant Waitzia	220	Easy–Moderate	Smoke water (optional)	**Apr – May**	Surface – light cover	21%	0.1 g
Xanthorrhoea preissii Balga / Grass Tree	222	Easy–Moderate	None	**Mar – May**	3–5 mm	47%	2.2 g

* Alcoa Marrinup Nursery routine test averages (after the pre-treatment shown). 'Seed / 100 plants' is the approximate weight of seed that produced 100 seedlings in those tests – sow extra to allow for losses. † Untreated seed – germinates poorly without the pre-treatment shown. '–' = no nursery test data.

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Sowing calendar

Species	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Acacia cochlearis	–	OK	Best	Best	Best	OK	OK	OK	Best	OK	–	–
Acacia lasiocarpa	–	OK	Best	Best	Best	OK	OK	OK	Best	OK	–	–
Acacia pulchella	–	OK	Best	Best	Best	OK	OK	OK	Best	OK	–	–
Acacia truncata	–	OK	Best	Best	Best	OK	OK	OK	Best	OK	–	–
Acanthocarpus preissii	Best	Best	OK	OK	–	–	–	–	–	–	OK	Best
Allocasuarina fraseriana	–	OK	Best	Best	Best	OK	OK	OK	OK	OK	–	–
Allocasuarina humilis	–	OK	Best	Best	Best	OK	OK	OK	OK	OK	–	–
Allocasuarina lehmanniana	–	OK	Best	Best	Best	OK	OK	OK	OK	OK	–	–
Ammothryon grandiflorum	–	–	OK	Best	Best	OK	OK	–	–	–	–	–
Amphipogon turbinatus	–	Best	Best	Best	OK	OK	–	–	–	–	–	–
Anigozanthos humilis	–	–	Best	Best	Best	OK	OK	OK	–	–	–	–
Anigozanthos manglesii	–	–	Best	Best	Best	OK	OK	OK	–	–	–	–
Atriplex cinerea	–	–	Best	Best	Best	OK	OK	OK	OK	OK	–	–
Atriplex isatidea	–	–	Best	Best	Best	OK	OK	OK	OK	OK	–	–
Austrostipa elegantissima	–	–	Best	Best	Best	OK	OK	OK	–	–	–	–
Austrostipa flavescens	–	OK	Best	Best	Best	OK	OK	OK	–	–	–	–
Banksia attenuata	–	–	Best	Best	Best	OK	OK	OK	OK	–	–	–
Banksia dallanneyi	–	–	Best	Best	Best	OK	OK	OK	OK	–	–	–
Banksia grandis	–	–	Best	Best	Best	OK	OK	OK	OK	–	–	–
Banksia menziesii	–	–	Best	Best	Best	OK	OK	OK	OK	–	–	–
Banksia prionotes	–	–	Best	Best	Best	OK	OK	OK	OK	–	–	–
Banksia sessilis	–	–	Best	Best	Best	OK	OK	OK	–	–	–	–
Bossiaea eriocarpa	–	OK	Best	Best	Best	OK	OK	OK	Best	OK	–	–
Burchardia congesta	–	–	Best	Best	Best	OK	OK	OK	–	–	–	–
Calothamnus quadrifidus	–	OK	Best	Best	OK	OK	OK	OK	Best	Best	OK	–
Carpobrotus virescens	–	OK	Best	Best	Best	OK	–	–	OK	OK	–	–
Clematis linearifolia	–	–	OK	Best	Best	Best	OK	–	–	–	–	–
Conostylis aculeata	–	–	Best	Best	Best	OK	OK	OK	–	–	–	–
Conostylis candicans	–	–	Best	Best	Best	OK	OK	OK	–	–	–	–
Conostylis setigera	–	–	Best	Best	Best	OK	OK	OK	–	–	–	–
Corymbia calophylla	–	OK	Best	Best	OK	OK	OK	OK	Best	Best	OK	–
Daviesia decurrens	–	OK	Best	Best	Best	OK	OK	OK	Best	OK	–	–
Daviesia divaricata	–	OK	Best	Best	Best	OK	OK	OK	Best	OK	–	–
Daviesia nudiflora	–	OK	Best	Best	Best	OK	OK	OK	Best	OK	–	–
Daviesia triflora	–	OK	Best	Best	Best	OK	OK	OK	Best	OK	–	–
Dianella revoluta	–	OK	Best	Best	OK	–	–	–	–	–	–	–
Diplolaena angustifolia	–	–	Best	Best	Best	OK	OK	OK	OK	OK	–	–
Dodonaea aptera	–	–	Best	Best	Best	OK	OK	OK	OK	OK	–	–
Eremaea pauciflora	–	OK	Best	Best	OK	OK	OK	OK	Best	Best	OK	–
Eryngium pinnatifidum	–	–	Best	Best	Best	OK	OK	OK	–	–	–	–
Eucalyptus decipiens	–	OK	Best	Best	OK	OK	OK	OK	Best	Best	OK	–
Eucalyptus gomphocephala	–	OK	Best	Best	OK	OK	OK	OK	Best	Best	OK	–
Eucalyptus marginata	–	OK	Best	Best	OK	OK	OK	OK	Best	Best	OK	–
Exocarpos sparteus	–	–	OK	Best	Best	Best	OK	–	–	–	–	–
Ficinia nodosa	–	OK	Best	Best	Best	OK	OK	OK	Best	Best	OK	–

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Species	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Frankenia pauciflora	–	–	OK	Best	Best	Best	OK	OK	OK	–	–	–
Gastrolobium capitatum	–	OK	Best	Best	Best	OK	OK	OK	Best	OK	–	–
Gastrolobium nervosum	–	OK	Best	Best	Best	OK	OK	OK	Best	OK	–	–
Gompholobium tomentosum	–	OK	Best	Best	Best	OK	OK	OK	Best	OK	–	–
Grevillea vestita	–	–	Best	Best	Best	OK	OK	OK	–	–	–	–
Guichenotia ledifolia	–	–	Best	Best	Best	OK	OK	OK	Best	OK	–	–
Haemodorum laxum	–	–	Best	Best	Best	OK	OK	OK	–	–	–	–
Hakea lissocarpha	–	–	Best	Best	Best	OK	OK	OK	OK	–	–	–
Hakea prostrata	–	–	Best	Best	Best	OK	OK	OK	OK	–	–	–
Hakea trifurcata	–	–	Best	Best	Best	OK	OK	OK	OK	–	–	–
Hardenbergia comptoniana	–	OK	Best	Best	Best	OK	OK	OK	Best	OK	–	–
Hovea pungens	–	OK	Best	Best	Best	OK	OK	OK	Best	OK	–	–
Hovea trisperma	–	OK	Best	Best	Best	OK	OK	OK	Best	OK	–	–
Hyalosperma cotula	–	–	OK	Best	Best	OK	–	–	–	–	–	–
Jacksonia furcellata	–	OK	Best	Best	Best	OK	OK	OK	Best	OK	–	–
Jacksonia sericea	–	OK	Best	Best	Best	OK	OK	OK	Best	OK	–	–
Jacksonia sternbergiana	–	OK	Best	Best	Best	OK	OK	OK	Best	OK	–	–
Kennedia prostrata	–	OK	Best	Best	Best	OK	OK	OK	Best	OK	–	–
Laxmannia squarrosa	–	–	Best	Best	Best	OK	OK	OK	–	–	–	–
Lepidosperma gladiatum	–	–	OK	Best	Best	OK	OK	–	–	–	–	–
Leucophyta brownii	–	–	Best	Best	Best	OK	OK	OK	OK	–	–	–
Leucopogon parviflorus	–	–	Best	Best	Best	OK	OK	OK	–	–	–	–
Macrozamia fraseri	OK	Best	Best	OK	OK	–	–	–	–	–	–	–
Melaleuca cardiophylla	–	OK	Best	Best	OK	OK	OK	OK	Best	Best	OK	–
Melaleuca huegelii	–	OK	Best	Best	OK	OK	OK	OK	Best	Best	OK	–
Melaleuca lanceolata	–	OK	Best	Best	OK	OK	OK	OK	Best	Best	OK	–
Melaleuca systena	–	OK	Best	Best	OK	OK	OK	OK	Best	Best	OK	–
Mesomelaena pseudostygia	–	–	OK	Best	Best	OK	OK	–	–	–	–	–
Myoporum insulare	–	–	Best	Best	Best	OK	OK	OK	OK	–	–	–
Nuytsia floribunda	OK	Best	Best	Best	OK	–	–	–	–	–	–	OK
Olearia axillaris	–	–	OK	OK	OK	OK	OK	Best	Best	OK	–	–
Opercularia vaginata	–	–	OK	Best	Best	OK	OK	–	–	–	–	–
Orthrosanthus laxus	–	–	Best	Best	Best	OK	OK	OK	–	–	–	–
Patersonia occidentalis	–	–	OK	OK	Best	Best	OK	OK	–	–	–	–
Petrophile axillaris	–	–	Best	Best	Best	OK	OK	OK	–	–	–	–
Petrophile brevifolia	–	–	Best	Best	Best	OK	OK	OK	–	–	–	–
Petrophile linearis	–	–	Best	Best	Best	OK	OK	OK	–	–	–	–
Petrophile macrostachya	–	–	Best	Best	Best	OK	OK	OK	–	–	–	–
Philotheca spicata	–	–	Best	Best	Best	OK	OK	OK	–	–	–	–
Pigea calycina	–	–	OK	Best	Best	OK	OK	–	–	–	–	–
Poa poiformis	–	–	Best	Best	Best	OK	OK	OK	OK	OK	–	–
Podotheca angustifolia	–	–	OK	Best	Best	OK	–	–	–	–	–	–
Podotheca gnaphalioides	–	–	OK	Best	Best	OK	–	–	–	–	–	–
Ptilotus drummondii	–	OK	Best	Best	Best	OK	OK	OK	OK	–	–	–
Ptilotus manglesii	–	OK	Best	Best	Best	OK	OK	OK	OK	–	–	–

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Species	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Ptilotus polystachyus	–	OK	Best	Best	Best	OK	OK	OK	OK	–	–	–
Rhagodia baccata	–	–	Best	Best	Best	OK	OK	OK	OK	–	–	–
Ricinocarpos undulatus	–	OK	Best	Best	Best	OK	OK	OK	OK	OK	–	–
Santalum acuminatum	–	–	–	OK	Best	Best	Best	OK	–	–	–	–
Spinifex hirsutus	OK	OK	Best	Best	Best	–	–	–	OK	OK	OK	OK
Spinifex longifolius	OK	OK	Best	Best	Best	–	–	–	OK	OK	OK	OK
Spyridium globulosum	–	–	OK	Best	Best	Best	OK	OK	–	–	–	–
Stirlingia latifolia	–	–	OK	Best	Best	Best	OK	–	–	–	–	–
Templetonia retusa	–	OK	Best	Best	Best	OK	OK	OK	Best	OK	–	–
Threlkeldia diffusa	–	–	OK	Best	Best	Best	OK	OK	–	–	–	–
Thysanotus manglesianus	–	–	Best	Best	Best	OK	OK	OK	–	–	–	–
Trachymene pilosa	–	–	OK	Best	Best	OK	–	–	–	–	–	–
Tricoryne elatior	–	–	Best	Best	Best	OK	OK	OK	–	–	–	–
Trymalium ledifolium	–	–	OK	Best	Best	Best	OK	OK	–	–	–	–
Waitzia suaveolens	–	–	OK	Best	Best	OK	–	–	–	–	–	–
Xanthorrhoea preissii	–	–	Best	Best	Best	OK	OK	OK	–	–	–	–
Perth season	Summer		Autumn			Winter			Spring			Sum.

Best window Workable – slower or less reliable Avoid / low success

Work back from your planting date: most of these species need roughly 3–6 months from sowing to a plantable tube.

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Pre-treatments explained

Many WA species have seed dormancy that must be broken before they will germinate. The species pages say which treatment to use.

Treatment	How to do it	Used for
None	Sow untreated seed. Remove from cones or follicles first where needed.	*Allocasuarina*, *Banksia*
Boiling water (30 s)	Drop seed into boiling water for 30 seconds, then drain and cool (Alcoa Marrinup method).	Hard-seeded peas, e.g. *Daviesia*, *Hovea*
Hot water soak	Pour near-boiling water over the seed (at least 10 parts water to 1 part seed) and leave to cool and soak overnight. Sow the seed that has swollen (1.5–2× its size); re-treat the rest.	Hard-seeded peas
Dry heat	Spread seed on a tray in a pre-heated oven. Peas: 80–120°C for 10 min to 1 hour. Kangaroo paw: 100°C for 3 hours.	*Gompholobium*, *Anigozanthos*
Smoke water	Soak seed for up to 24 hours in diluted smoke water (e.g. 1:9), or water trays with it after sowing. Also has slight antifungal properties.	Fire-responsive species, e.g. *Anigozanthos* (optional for others)
Aged seed	Store seed dry at room temperature for one or more years before sowing – dormancy fades with time (after-ripening).	*Anigozanthos*
Ferment	Soak ripe fleshy fruit or berries in water for 4–5 days, wash the pulp off through a sieve, then dry the seed.	*Carpobrotus*, *Myoporum*, *Rhagodia*
Long soak	Soak for 4 days, changing the water 3 times a day, to wash germination inhibitors out of the fruit.	*Atriplex*
Clean or nick	Remove whatever blocks germination: the woolly floral parts (*Ptilotus*), the fleshy coat (*Macrozamia*), or nick the seed coat with a scalpel (*Grevillea*, *Petrophile*).	*Ptilotus*, *Macrozamia*, *Grevillea*, *Petrophile*

Safety: boiling water and hot ovens scald – use tongs, strainers, heat-proof containers and gloves.

Difficulty ratings

Rating	Meaning
Easy	Germinates reliably with the standard method; few losses.
Easy–Moderate	Germinates well, but one factor needs care (e.g. damping off, extraction, low fill).
Moderate	Needs the right pre-treatment and timing; germination may be low or variable.
Moderate–Difficult	Low or erratic germination; specialist treatment usually needed.
Difficult	Very low germination from seed; cuttings or other methods usually preferred.

Confidence

Level	Meaning
High	Species-specific germination data and handling guidance are published.
Medium	Some species data (e.g. few tests) plus genus-level guidance. Verify with trials.
Low	Little information available; treat as a trial protocol.

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Standard sowing method

Species pages refer back to these steps and only note where a species differs.

Step	What to do
1. Seed mix	Sterile, free-draining, low-nutrient mix: about **80–85% washed river sand + 15–20% coir or peat**, or a commercial native seed-raising mix. Use fresh mix and clean trays. Pre-moisten before sowing.
2. Check the seed	Label every lot (species, provenance, collection date). Cut-test a few seeds – good seed is firm and white inside. For peas, a float test removes damaged seed.
3. Pre-treat	Use the treatment on the species page (see Pre-treatments explained). Sow soon after hot-water treatment.
4. Sow	Cover to about **twice the seed's thickness**. Fine seed: sprinkle on the surface and press in lightly, or cover with a thin layer of fine sand or vermiculite. Large seed (e.g. *Banksia*, peas): sow direct into tubes to avoid pricking-out damage.
5. Water in (post-treatment)	Water with a fine spray – smoke water is a useful first watering (slight antifungal effect, more even germination). Keep moist, never wet. A capillary bed lets you water from below.
6. Germinate	Shaded, sheltered, protected from rain and drying wind. Check daily. Watch for damping off, snails and slugs.
7. Prick out	When seedlings can be handled – usually when the **second pair of leaves** appears; earlier for taprooted species. Lift with a narrow spatula, hold by the leaves, not the stem, and firm mix around the roots with a gentle spray.
8. Shade house	About **2 weeks** in the shade house after pricking out. Don't let tubes dry out (a capillary bed helps).
9. Harden off	Move gradually into full sun over **2–4 weeks**. Pot on to larger containers after 1–3 months if needed; don't let roots become pot-bound.

Seed storage

Topic	Guidance
General rule	Dry the seed (air-dry or desiccator, never an oven), seal it in labelled airtight containers and keep it cool and dark, away from vermin. A sealed fridge is good; freezing suits many species but not all.
Exceptions	*Anigozanthos manglesii* lost germination at −18°C – store at room temperature or cooler, not frozen. *Allocasuarina* and *Banksia* seed is short-lived once out of the cone – store in the cone, or dry and cool. *Macrozamia fraseri* seed is shed moist and is damaged by cold, dry storage – sow it fresh.
Long-lived	Hard-coated pea seed keeps many years dry (related *Acacia* seed has stayed viable 50+ years).
Collecting	Collecting from bushland generally needs a DBCA licence and landholder permission. Collect only ripe seed from many plants to keep genetic diversity.

Hygiene and damping off

Damping off (*Pythium*, *Rhizoctonia*) rots seedlings at soil level, especially in wet, crowded trays. Prevent it with sterile mix, clean trays, thin sowing, good airflow and bottom watering. Remove affected seedlings quickly and use a registered fungicide only if the problem persists. Keep pots off the ground to reduce *Phytophthora* risk.

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Acacia cochlearis

Rigid Wattle

Family **Fabaceae (Mimosoideae)**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Easy	Boiling water 30 s	Mar – May	3–5 mm	60%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	OK	Best	Best	Best	OK	OK	OK	Best	OK	–	–

Best window | Workable – slower or less reliable | Avoid / low success

Plant snapshot

Habit	Bushy, erect to sprawling shrub, (0.3–) 0.5–3 m, with stiff, sharp-pointed phyllodes.
Flowering	Jul – Oct (yellow balls).
Habitat	White, yellow or red sand; coastal and near-coastal sandplains and dunes, often on limestone.
Local note	Recorded in Joondalup and Wanneroo. Tolerates coastal exposure. Prickly – wear gloves.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Possible	Half-hardened wattle cuttings, 5–15 cm (genus). Untested.	–
Seed	Main method	Boiling water 30 s; about 60% (Alcoa).	This guide
Division	Not used	Shrub – not divided.	–

Seed collection & storage

Item	Details
Seed collection	Seed ripens in late spring–early summer, when pods dry, change colour, turn brittle and split – they split quickly, so bag branches or check often.
Extraction & cleaning	Dry pods in paper bags; rub and sieve out the seed.
Seed quality	Hard-seeded. Float test: discard seed that floats.
Storage life	**Very long** – Acacia seed can stay viable for 50+ years stored dry.
Typical germination	About **60%** after boiling-water treatment (Alcoa nursery average, 11 tests). About **82 seedlings per gram** – roughly 1.2 g of seed per 100 seedlings.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May** and **Sep**. Feb, Jun – Aug and Oct workable. Avoid Nov – Jan.
Seed mix	Standard sterile native seed mix.
Pre-treatment	**Boiling water for 30 seconds** (Alcoa method), or a hot-water overnight soak. Don't overdo it – very hot water can kill softer seed. Sow the swollen seed.
Sowing depth & method	Cover about **3–5 mm**. Direct sow 2–3 seeds per tube and thin.
After sowing	Water in; keep moist, not wet.
Germination time	From a few days to a few weeks.
Pricking out	Early, at the cotyledon to first true leaf stage.

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Item	Details
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 3–4 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Low germination	Seed coat not broken. Re-treat unswollen seed.
Swollen seed rots	Boiled too long or mix too wet.
Seedlings collapse	Damping off. Sterile mix, good airflow.

Notes

Other notes	Wattle cuttings of half-hardened wood can also work, especially for small-phyllode species.
Confidence	**High.** Nursery data from 11 tests and WA Acacia seed notes available.
Sources	See References: 3, 4, 15, 43, 96

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Acacia lasiocarpa

Panjang

Family **Fabaceae (Mimosoideae)**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Easy	Boiling water 30 s	Mar – May	3–5 mm	64%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	OK	Best	Best	Best	OK	OK	OK	Best	OK	–	–

Best window · Workable – slower or less reliable · Avoid / low success

Plant snapshot

Habit	Shrub, 0.15–1.5 (–2.5) m, often spiny. Coastal dune plants are low and compact with few spines.
Flowering	May – Oct (yellow balls).
Habitat	Sandy soils and laterite; winter-wet areas, swamps, coastal dunes and flats.
Local note	Recorded in Joondalup and Wanneroo (var. *lasiocarpa*). Often short-lived.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Possible	Growers report softwood cuttings work; no strike data.	–
Seed	Main method	Boiling water 30 s; about 64% germination.	This guide
Division	Not used	Woody shrub.	–

Seed collection & storage

Item	Details
Seed collection	Pods are ripe when they dry, change colour, turn brittle and start to split – and they split quickly. Bag branches with fine mesh, or check plants often as pods brown.
Extraction & cleaning	Dry pods in paper bags; rub and sieve or winnow out the seed.
Seed quality	Hard-seeded. Float test: discard seed that floats.
Storage life	**Very long.** Acacia seed can stay viable for 50+ years at room temperature; drying and freezing extend it further.
Typical germination	About **64%** after boiling-water treatment (Alcoa nursery average, 21 tests). About **105 seedlings per gram** – roughly 1 g of seed per 100 seedlings. In one Alcoa trial, untreated seed gave 31% and 8% against 70% and 94% after a 30 s boil.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May** and **Sep**. Feb, Jun – Aug and Oct are workable. Avoid Nov – Jan.
Seed mix	Standard sterile native seed mix.
Pre-treatment	**Boiling water:** 30 seconds (Alcoa method), or pour boiling water over the seed and leave to stand overnight, then sow the seed that has swollen. A 2-minute boil lowered germination in the Alcoa trial – don't overdo it.
Sowing depth & method	Cover about **3–5 mm**. Direct sow 2–3 seeds per tube and thin, or sow in trays.
After sowing	Water in; keep moist, not wet.

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Item	Details
Germination time	Can start about **one week** after treatment.
Pricking out	Early, at the cotyledon to first true leaf stage.
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 3–4 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Low germination	Seed coat not broken. Re-treat unswollen seed.
Swollen seed rots	Boiled too long or mix too wet.
Seedlings collapse	Damping off. Sterile mix, good airflow.

Notes

Other notes	Growers report softwood cuttings also work.
Confidence	**High.** Large nursery data set (21 tests) including a treatment trial; WA Acacia seed notes available.
Sources	See References: 3, 4, 15, 42, 43

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Acacia pulchella

Prickly Moses

Family **Fabaceae (Mimosoideae)**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Easy	Boiling water 30 s	Mar – May	2–3 mm	61%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	OK	Best	Best	Best	OK	OK	OK	Best	OK	–	–

Best window · Workable – slower or less reliable · Avoid / low success

Plant snapshot

Habit	Spiny shrub, 0.3–3 m, with one or two spines at the base of each leaf.
Flowering	May – Dec (yellow balls).
Habitat	Sandy soils and clay loam over laterite; low-lying areas, swamps, near watercourses.
Local note	The variety recorded in Joondalup and Wanneroo is var. *glaberrima*. Wear gloves – very prickly.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Possible	Growers report cuttings of new growth work; no strike data.	–
Seed	Main method	Boiling water 30 s; about 61% germination.	This guide
Division	Not used	Woody shrub.	–

Seed collection & storage

Item	Details
Seed collection	Pods split quickly once dry and brittle. Bag branches with fine mesh or collect as pods brown.
Extraction & cleaning	Dry pods in paper bags; rub and sieve out the seed.
Seed quality	Hard-seeded. Float test: discard seed that floats.
Storage life	**Very long.** Acacia seed can stay viable for 50+ years stored dry.
Typical germination	var. *glaberrima*: about **61%** after boiling (Alcoa, 69 tests), about **78 seedlings per gram** – roughly 1.3 g of seed per 100 seedlings. Untreated seed gave only 4–5% in an Alcoa trial, against 88–94% after a 30 s boil.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May** and **Sep**. Feb, Jun – Aug and Oct are workable. Avoid Nov – Jan.
Seed mix	Standard sterile native seed mix.
Pre-treatment	**Boiling water for 30 seconds** (Alcoa method), or a boiling-water overnight soak. Sow the swollen seed; re-treat the rest.
Sowing depth & method	Cover about **2–3 mm**. Direct sow 2–3 seeds per tube and thin.
After sowing	Water in; keep moist, not wet.
Germination time	Can start about **one week** after treatment.
Pricking out	Early, at the cotyledon to first true leaf stage. Gloves once spines develop.

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Item	Details
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 3–4 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Low germination	Seed coat not broken. Re-treat unswollen seed.
Swollen seed rots	Boiled too long or mix too wet.
Seedlings collapse	Damping off. Sterile mix, good airflow.

Notes

Other notes	Label seed by variety – four varieties exist and only var. *glaberrima* is recorded locally. Growers report cuttings of new growth after pruning also work.
Confidence	**High.** Large nursery data set (69 tests for the local variety) including treatment trials; WA Acacia seed notes available.
Sources	See References: 3, 4, 15, 42, 43

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Acacia truncata

West Coast Wattle

Family **Fabaceae (Mimosoideae)**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Easy–Moderate	Boiling water 30 s	Mar – May	3–5 mm	41%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	OK	Best	Best	Best	OK	OK	OK	Best	OK	–	–

Best window · Workable – slower or less reliable · Avoid / low success

Plant snapshot

Habit	Usually a dense, domed shrub, 0.5–2.3 m.
Flowering	Jun – Sep (yellow to cream balls).
Habitat	Sandy soils; coastal limestone and dunes.
Local note	Recorded in Joondalup and Wanneroo. Tolerates salt spray – good for seaside plantings.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Possible	Not documented for this species; many acacias strike from 75–100 mm cuttings.	–
Seed	Main method	Boiling water 30 s or scarify.	This guide
Division	Not used	Woody shrub.	–

Seed collection & storage

Item	Details
Seed collection	Pods split quickly once dry and brittle. Bag branches or collect as pods brown.
Extraction & cleaning	Dry pods in paper bags; rub and sieve out the seed.
Seed quality	Hard-seeded. Float test: discard seed that floats.
Storage life	**Very long.** Acacia seed can stay viable for 50+ years stored dry.
Typical germination	About **41%** after boiling-water treatment (Alcoa nursery average, 12 tests). About **94 seedlings per gram** – roughly 1.1 g of seed per 100 seedlings.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May** and **Sep**. Feb, Jun – Aug and Oct are workable. Avoid Nov – Jan.
Seed mix	Standard sterile native seed mix.
Pre-treatment	**Boiling water for 30 seconds**, or scarify the seed coat (ANBG). Sow the swollen seed.
Sowing depth & method	Cover about **3–5 mm**. Direct sow 2–3 seeds per tube and thin.
After sowing	Water in; keep moist, not wet.
Germination time	Can start about **one week** after treatment.
Pricking out	Early, at the cotyledon to first true leaf stage.

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Item	Details
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 3–4 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Low germination	Seed coat not broken. Re-treat unswollen seed.
Swollen seed rots	Boiled too long or mix too wet.
Seedlings collapse	Damping off. Sterile mix, good airflow.

Notes

Other notes	A tough coastal wattle for exposed sites.
Confidence	**High.** Nursery germination data (12 tests); WA Acacia seed notes and ANBG notes available.
Sources	See References: 3, 4, 15, 43, 44

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Acanthocarpus preissii

Prickle Lily

Family **Asparagaceae (Lomandroideae)**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Moderate	Warm stratify 7+ weeks, then cool	Dec – Feb	3–5 mm	80%+ after warm stratification

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	Best	Best	OK	OK	–	–	–	–	–	–	OK	Best

Best window · Workable – slower or less reliable · Avoid / low success

Plant snapshot

Habit	Rhizomatous, tufted perennial herb, 0.2–0.7 (–2) m, with stiff, prickly leaves.
Flowering	Apr – May (white).
Habitat	White, grey, red or brown sand, sandstone and limestone; coastal dunes.
Local note	Recorded in Joondalup and Wanneroo. Breeding plant for the silver-spotted ochre butterfly.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Not used	Not documented.	–
Seed	Main method	Warm-stratify 7+ weeks, then cool: over 80% in research.	This guide
Division	Possible	Rhizomatous clumps; division not documented.	–

Seed collection & storage

Item	Details
Seed collection	Collect fruits once they ripen and dry after the April – May flowering.
Extraction & cleaning	Clean away any fruit material and dry the seed.
Seed quality	Fresh seed viability is high (over 90% in research), but it is dormant.
Storage life	Stores dry; the seed after-ripens in dry storage, which loosens dormancy.
Typical germination	Fresh seed germinated **under 8%** even with smoke or gibberellic acid, but **over 80%** after at least 7 weeks warm and moist (18/33°C) followed by cool conditions (7/18°C) (Turner et al. 2006). Alcoa's single untreated test gave 0%.

Sowing & growing on

Item	Details
Best sowing time	**Dec – Feb**, so the trays get the summer warmth they need, then germinate as it cools in autumn. Mar – Apr and Nov workable.
Seed mix	Standard sterile native seed mix, free-draining.
Pre-treatment	**Warm stratification:** sow into moist mix and keep the trays warm (a sunny, sheltered bench or a warm glasshouse) for at least 7 weeks, then move them somewhere cool. Smoke-water soak (24 h, 1:9) is the Geraldton nursery method.
Sowing depth & method	Cover about **3–5 mm**. Sow in deep trays or 2–3 seeds per deep tube.
After sowing	Keep the mix moist the whole time – dry trays won't stratify.

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Item	Details
Germination time	Geraldton recorded **28–56 days**; with summer sowing expect emergence as autumn cools.
Pricking out	Early, at the first leaf, into deep tubes.
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 6–9 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
No germination by autumn	Trays too dry or too cool in summer. Keep moist and warm for 7+ weeks, then cool.
Seedlings rot	Too wet once cool. Water less; good drainage.
Slow growth	Normal. Deep tubes and patience.

Notes

Other notes	Rhizomatous – division of clumps may also work but isn't documented.
Confidence	**High.** Species-specific dormancy research (Turner et al. 2006) plus a WA community-nursery method.
Sources	See References: 4, 5, 71

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Allocasuarina fraseriana

Sheoak / Western Sheoak

Family **Casuarinaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Easy	None	Mar – May	2–3 mm	37%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	OK	Best	Best	Best	OK	OK	OK	OK	OK	–	–

Best window · Workable – slower or less reliable · Avoid / low success

Plant snapshot

Habit	Erect tree, 5–15 m. Male and female flowers usually on separate trees (dioecious).
Flowering	May – Oct. Woody cones up to 40 mm long follow on female trees.
Habitat	Lateritic soils and white, grey or yellow sand; jarrah forest and sand dunes.
Local note	Recorded in the Wanneroo LGA. Only female trees bear cones – note which trees you collect from.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Not used	Rarely used for sheoaks.	–
Seed	Main method	No pre-treatment; 2–5 weeks.	This guide
Division	Not used	Tree.	–

Seed collection & storage

Item	Details
Seed collection	Cones mature within about a year of flowering. Collect cones that have turned pale to dark brown **with the valves still closed**. Use secateurs or a pole pruner.
Extraction & cleaning	Place cones in a paper bag or tray in a warm, dry spot – they open readily and drop the winged seeds (samaras). Sieve off cone debris.
Seed quality	Many seeds look sound but are empty. **Cut-test** a sample before sowing and discard lots with empty or shrivelled seed.
Storage life	**Short once released from the cone** unless kept dry and cool. Store dried seed sealed in the fridge, or keep seed in closed cones until the sowing season.
Typical germination	About **37%** (untreated, Alcoa nursery average). About **76 seedlings per gram** – roughly 1.3 g of seed per 100 seedlings.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May** (cooling autumn weather, 15–25°C). Feb and Jun – Oct are workable. Avoid Nov – Jan (heat).
Seed mix	Standard sterile native seed mix (see Standard method).
Pre-treatment	**None needed.** Seed becomes sticky (mucilaginous) when wet – mix dry seed with a little dry, sterile yellow sand so it spreads evenly and can be seen.

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Item	Details
Sowing depth & method	Scatter thinly and cover lightly with about **2–3 mm** of fine sand or vermiculite. **Direct sowing into tubes or cells** (a few seeds each, thinned later) reduces losses – sheoak seedlings are prone to damping off after pricking out.
After sowing	Water in with a fine spray (smoke water is optional – aerosol smoke gave similar results to untreated seed in Alcoa trials). Keep moist, not wet. Watch for damping off.
Germination time	Seedlings emerge after about **2–5 weeks**.
Pricking out	If sown in trays, prick out early – once seedlings are about 2–3 cm tall and can be handled – disturbing roots as little as possible.
Shade house & hardening	About **2 weeks** in the shade house after pricking out or thinning, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 4–6 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Seedlings collapse at soil level	Damping off. Sow thinly, use sterile mix, water from below where possible, direct sow into tubes.
Very low germination	Seed lot mostly empty. Cut-test before sowing; collect from closed, mature cones.
Seed clumps together	Mucilage. Mix dry seed with dry sand before sowing; don't pre-soak.

Notes

Other notes	Cuttings are rarely used. Seed from several trees keeps genetic diversity in revegetation stock.
Confidence	**High.** Published nursery test data and WA seed-handling notes are available.
Sources	See References: 1, 2, 3, 4, 5

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Allocasuarina humilis

Dwarf Sheoak

Family **Casuarinaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Easy–Moderate	None	Mar – May	2–3 mm	25%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	OK	Best	Best	Best	OK	OK	OK	OK	OK	–	–

Best window · Workable – slower or less reliable · Avoid / low success

Plant snapshot

Habit	Erect or spreading shrub, 0.2–2 m. Dioecious or monoecious.
Flowering	May – Nov.
Habitat	Sand (often over laterite), sandy clay and gravel on sandplains and dunes.
Local note	Widespread in the South West. Collect from local-provenance plants and record the source.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Not used	Rarely used for sheoaks.	–
Seed	Main method	No pre-treatment; sow generously.	This guide
Division	Not used	Woody shrub.	–

Seed collection & storage

Item	Details
Seed collection	Collect mature brown cones **with valves still closed**, usually within a year of flowering. On dioecious plants only female plants bear cones.
Extraction & cleaning	Dry cones in a paper bag in a warm, dry place until they open; sieve out the small winged seeds.
Seed quality	Germination is often low – **cut-test** a sample. Many seeds can be empty.
Storage life	**Short once released from the cone** unless kept dry and cool. Store sealed in the fridge, or keep in closed cones until needed.
Typical germination	About **25%** (untreated, Alcoa nursery average). About **77 seedlings per gram** – roughly 1.3 g of seed per 100 seedlings.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May** (15–25°C). Feb and Jun – Oct are workable. Avoid Nov – Jan.
Seed mix	Standard sterile native seed mix.
Pre-treatment	**None needed.** Mix dry seed with dry, sterile sand for even spreading (seed turns sticky when wet).
Sowing depth & method	Cover lightly, about **2–3 mm**. Direct sow a few seeds per tube/cell and thin, or sow thinly in trays. Sow more generously than for *A. fraseriana* to allow for lower germination.
After sowing	Water in with a fine spray; keep moist, not wet. Good airflow to limit damping off.
Germination time	About **2–5 weeks**.

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Item	Details
Pricking out	Prick out early, at about 2–3 cm tall, with minimal root disturbance.
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 4–6 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Seedlings collapse at soil level	Damping off. Sow thinly, sterile mix, avoid overwatering.
Very low germination	Empty seed. Cut-test; collect from closed, fully mature cones.
Seed clumps together	Mix dry seed with dry sand; don't pre-soak.

Notes

Other notes	Slower-growing shrub – allow extra time to reach planting size.
Confidence	**High.** Published nursery test data and WA seed-handling notes are available.
Sources	See References: 1, 2, 3, 4, 5

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Allocasuarina lehmanniana

Dune Sheoak

Family **Casuarinaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Easy–Moderate	None	Mar – May	2–3 mm	19%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	OK	Best	Best	Best	OK	OK	OK	OK	OK	–	–

Best window · Workable – slower or less reliable · Avoid / low success

Plant snapshot

Habit	Shrub, 0.5–4 m. Male and female flowers on separate plants (dioecious).
Flowering	Sep – Dec (female flowers red).
Habitat	Coastal sites and winter-wet areas.
Local note	Recorded in Joondalup and Wanneroo. Only female plants bear cones – note which plants you collect from.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Not used	Rarely used for sheoaks.	–
Seed	Main method	No pre-treatment; like other sheoaks.	This guide
Division	Not used	Woody shrub.	–

Seed collection & storage

Item	Details
Seed collection	Collect mature brown cones **with the valves still closed**, usually within a year of flowering. Use secateurs.
Extraction & cleaning	Dry cones in a paper bag in a warm, dry spot until they open; sieve out the small winged seeds.
Seed quality	Many seeds can be empty – **cut-test** a sample before sowing.
Storage life	**Short once released from the cone** unless kept dry and cool. Store sealed in the fridge, or keep seed in closed cones until needed.
Typical germination	About **19%** (untreated, Alcoa nursery average, 3 tests). About **193 seedlings per gram** – roughly 0.5 g of seed per 100 seedlings.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May** (15–25°C). Feb and Jun – Oct are workable. Avoid Nov – Jan.
Seed mix	Standard sterile native seed mix.
Pre-treatment	**None needed.** Mix dry seed with dry, sterile sand for even spreading (seed turns sticky when wet).
Sowing depth & method	Cover lightly, about **2–3 mm**. Direct sow a few seeds per tube or cell and thin, or sow thinly in trays.
After sowing	Water in with a fine spray; keep moist, not wet. Good airflow to limit damping off.
Germination time	About **2–5 weeks**.

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Item	Details
Pricking out	Prick out early, at about 2–3 cm tall, with minimal root disturbance.
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 4–6 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Seedlings collapse at soil level	Damping off. Sow thinly, sterile mix, avoid overwatering.
Very low germination	Empty seed. Cut-test; collect from closed, fully mature cones.
Seed clumps together	Mix dry seed with dry sand; don't pre-soak.

Notes

Other notes	A good choice for damp, low-lying coastal sites. Collect from several female plants to keep genetic diversity.
Confidence	**Medium.** Nursery germination data (3 tests); handling follows the other *Allocasuarina* in this guide.
Sources	See References: 2, 3, 4

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Ammothryon grandiflorum

Large-flowered Bog Rush

Family
Cyperaceae (sedges)

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Difficult *	**Heat + smoke (trial)**	**Apr – May**	**2–3 mm**	**0% untreated (1 test)**

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	OK	Best	Best	OK	OK	–	–	–	–	–

Best window · Workable – slower or less reliable · Avoid / low success

Plant snapshot

Habit	Rhizomatous, tufted perennial sedge, 0.4–1.5 m.
Flowering	Jan or Apr – Jul (brown).
Habitat	White, grey, brown or yellow sand and sandy clay; coastal dunes and sandplains.
Local note	Grows on the local coastal dunes (Cottesloe Coastcare). Renamed from *Schoenus grandiflorus* in 2021.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Not used	Sedge.	–
Seed	Difficult	0% in the only test; heat + smoke is a trial.	This guide
Division	Trial	Rhizomatous; division not documented. Tube stock is usually tissue-cultured.	–

Seed collection & storage

Item	Details
Seed collection	Collect ripe nutlets when the heads dry and they come away easily.
Extraction & cleaning	Rub heads over a sieve.
Seed quality	Expect many empty nutlets – cut-test a sample (common in sedges).
Storage life	Not well documented. Store dry and cool.
Typical germination	0% untreated in the only Alcoa nursery test.

Sowing & growing on

Item	Details
Best sowing time	**Apr – May** (trial). Mar, Jun – Jul workable.
Seed mix	Free-draining, sterile mix.
Pre-treatment	**Trial only:** no treatment has been published for this species. Research on a related sedge (*Lepidosperma scabrum*) found a heat shock (100°C for 10 minutes) plus smoke water worked best – a reasonable trial here.
Sowing depth & method	Cover about **2–3 mm**. Sow generously.
After sowing	Keep moist, not wet. Keep trays going for months.
Germination time	Not recorded.
Pricking out	At the first leaf, into deep tubes.

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Item	Details
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Slow – record your own.

Common problems

Problem	Likely cause and fix
No germination	Expected. Try heat plus smoke water, older seed, and larger sowings.
Empty seed	Cut-test before sowing; collect only fully ripe heads.
Seedlings rot	Too wet. Water less.

Notes

Other notes	Tube stock is usually bought in from tissue-culture labs. Record any trial results so this page can be updated.
Confidence	**Low.** No reliable seed method published: 0% in the only nursery test. Commercial stock is produced by tissue culture. Trial protocol.
Sources	See References: 3, 4, 74, 78, 80

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Amphipogon turbinatus

No widely used common name (a grass)

Family **Poaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Difficult *	Leach + smoke (trial)	Feb – Apr	Just covered	0% in one test

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	Best	Best	Best	OK	OK	–	–	–	–	–	–

Best window · Workable – slower or less reliable · Avoid / low success

Plant snapshot

Habit	Tufted perennial grass, 0.2–0.8 m, with short rhizomes.
Flowering	Jul – Dec (purple-green, in short dense heads about 1.5 cm long).
Habitat	White, grey, lateritic or granitic sand and brown loam; winter-wet flats.
Local note	Recorded in the Wanneroo LGA (not listed for Joondalup). A useful understorey grass in Banksia woodland restoration.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Not used	Grasses are not grown from cuttings.	–
Seed	Difficult	0% in one nursery test; leaching + smoke helped a relative.	This guide
Division	Trial	Short rhizomes – small clumps may be worth a trial.	–

Seed collection & storage

Item	Details
Seed collection	Collect heads when the florets come away easily (estimate: Nov – Jan, after flowering). The unit is the hairy floret with three awns (6–8 mm) and a sharp, hairy callus.
Extraction & cleaning	Rub heads gently and sieve; keep the florets whole.
Seed quality	Expect many empty florets – a related species tested 25% viable (Seeds of SA). A filled floret holds a grain about 2.5 mm long.
Storage life	Not documented. Store dry and cool.
Typical germination	Only one nursery test: **0%** with smoke (Alcoa). The related *A. amphipogonoides* also scored 0% in routine tests, but in trials punnets held at warm days and cool nights (34/12°C, then 18/10°C) and leached gave 48%, and leached plus aerosol smoke 58%, against 33% untreated; smoke water added little.

Sowing & growing on

Item	Details
Best sowing time	**Feb – Apr** (estimate) – late summer to autumn, to copy the warm-then-cool trial. May – Jun workable. Avoid Jul – Jan.
Seed mix	Standard sterile native seed mix, free-draining.
Pre-treatment	Trial on part of a batch, sown in late summer so the trays get warm days then cool nights: **leach** the trays – water them heavily four times, 4 days apart – and smoke them with aerosol smoke if you have a smoke tent (the related species' trial).

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Item	Details
Sowing depth & method	Just cover the florets.
After sowing	Keep moist, not wet; keep trays at least 3 months.
Germination time	Not documented.
Pricking out	At 2–3 leaves.
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 4–6 months from germination (record your own).

Common problems

Problem	Likely cause and fix
Nothing comes up	Many florets are empty. Cut-test a sample; collect later in the season when heads are fully ripe.
Florets wash out of trays	Cover lightly and water gently.
Seedlings collapse	Damping off. Sterile mix, airflow.

Notes

Other notes	Division is untested. It has short rhizomes, so small clumps may be worth a trial.
Confidence	**Low.** One nursery test (0%). Method based on trials with the related *A. amphipogonoides* – treat as a trial.
Sources	See References: 3, 4, 126, 127, 138

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Anigozanthos humilis

Catspaw

Family **Haemodoraceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Moderate	Smoke water (optional)	Mar – May	1–2 mm	4%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	Best	Best	Best	OK	OK	OK	–	–	–	–

Best window | Workable – slower or less reliable | Avoid / low success

Plant snapshot

Habit	Low rhizomatous herb, 0.1–1 m (usually 0.1–0.3 m for the local subspecies). Often short-lived.
Flowering	Jul – Oct (yellow, orange and red furry paws).
Habitat	Sand, sandy loam, clay, laterite and limestone; from well-drained ground to winter-wet swamps.
Local note	Recorded in Joondalup and Wanneroo (subsp. *humilis*). Subsp. *chrysanthus* is Priority Four and not local – label seed by subspecies. Prone to rust.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Not used	Kangaroo paws are not grown from cuttings.	–
Seed	Main method	No pre-treatment needed; smoke lifts germination.	This guide
Division	Possible	After flowering, vigorous plants only.	→ Division Guide

Seed collection & storage

Item	Details
Seed collection	Capsules ripen a few months after flowering (late spring–summer). Collect as they dry and brown, before they split.
Extraction & cleaning	Dry in paper bags; rub and sieve out the small seed.
Seed quality	Cut-test a few seeds.
Storage life	Stores dry. For the related *A. manglesii*, Geraldton sows seed kept at room temperature for several years, as aged seed germinates better.
Typical germination	Low in nursery tests: about **4%** untreated (Alcoa, 2 tests), about 53 seedlings per gram. Glyceronitrile, a smoke compound, significantly increased germination in research – so smoke water is worth using.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May**. Jun – Aug workable. Avoid Sep – Feb.
Seed mix	Standard sterile native seed mix, very free-draining.
Pre-treatment	None needed (ANPSA), but **smoke water** (24 h soak or first watering) is likely to lift germination. Dry heat and aged seed help the related *A. manglesii* – worth trialling.
Sowing depth & method	Sow thinly; cover about **1–2 mm**.

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Item	Details
After sowing	Keep moist, not wet. Protect from snails.
Germination time	Not recorded for this species. Check trays from 3 weeks and record yours.
Pricking out	At the second-leaf stage, gently.
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 4–6 months. Can flower in its first year.

Common problems

Problem	Likely cause and fix
Almost no germination	Dormant seed. Use smoke water; trial aged or heat-treated seed.
Orange pustules on leaves	Rust – this species is particularly prone. Remove affected leaves; good airflow; keep foliage dry.
Plants die after a year or two	Normal – often short-lived. Keep sowing each year.

Notes

Other notes	Vigorous clumps can be divided after flowering (see the Division Guide).
Confidence	**Medium.** ANPSA says it grows from untreated seed; nursery tests (2) were low; a smoke compound raised germination in research.
Sources	See References: 3, 4, 5, 72, 81, 84

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Anigozanthos manglesii

Red and Green Kangaroo Paw

Family **Haemodoraceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Moderate	Dry heat 100°C 3 h, or aged seed	Mar – May	1–2 mm	70%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	Best	Best	Best	OK	OK	OK	–	–	–	–

Best window | Workable – slower or less reliable | Avoid / low success

Plant snapshot

Habit	Rhizomatous perennial herb, 0.2–1.1 m (flower stems). Often short-lived – best treated as a 2–3 year plant.
Flowering	Aug – Nov (green and red). WA's floral emblem.
Habitat	White, yellow or grey sand and sandy loam.
Local note	Recorded in the Wanneroo LGA. Also called Mangles Kangaroo Paw.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Not used	Clumping herb – no stems to cut.	–
Seed	Main method	Dry heat or aged seed; autumn sowing.	This guide
Division	Good	Split vigorous clumps into rhizome pieces with shoots and roots.	→ Division Guide

Seed collection & storage

Item	Details
Seed collection	Seed ripens roughly three months after flowering (about Nov – Feb). Collect when capsules dry and brown, before they split and shed.
Extraction & cleaning	Dry capsules in paper bags; rub and sieve to separate the small seed.
Seed quality	Fresh seed is strongly dormant – untreated germination can be very low (about 1% in one Alcoa trial).
Storage life	**Stores well for several years** dry at room temperature, and germination **improves with age** (after-ripening). Research found **freezing (–18°C) reduced germination** – store dry at 23°C or cooler, not in the freezer.
Typical germination	About **70%** after heat treatment (Alcoa nursery average). About **211 seedlings per gram** – roughly 0.5 g of seed per 100 seedlings.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May** – germinates in cool conditions (tested at 15–20°C). Jun – Aug workable (slower). Avoid Sep – Feb.
Seed mix	Standard sterile native seed mix; very free-draining.

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Item	Details
Pre-treatment	Choose one: • **Dry heat:** spread seed on a tray in an oven at **100°C for 3 hours** (Geraldton community nursery; Alcoa). • **Aged seed:** use seed stored at room temperature for a few years. • **Smoke water** (e.g. 24 h soak at 1:9) – response improves the longer seed has been stored.
Sowing depth & method	Sow thinly on the surface of trays and cover very lightly, about **1–2 mm** of fine sand or vermiculite.
After sowing	Water in with a fine spray (smoke water optional). Keep moist but not wet. Protect from snails and slugs.
Germination time	About **3–6 weeks** (around 20 days in the Geraldton nursery).
Pricking out	When the **second leaf** appears and seedlings can be handled (about 2–3 cm). Lift gently with a narrow spatula.
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks. Keep foliage dry and airflow good.
Time to planting size	Est. 4–6 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Almost no germination	Seed still dormant. Heat-treat, use older seed, or add smoke water.
Black spots spreading on leaves	Ink disease (fungal). Remove affected plants, improve airflow, avoid wetting foliage.
Seedlings chewed	Snails and slugs. Bait or hand-remove.

Notes

Other notes	Division of clumps also works. Seed-grown plants vary slightly; keep local provenance for revegetation.
Confidence	**High.** Well researched: dormancy, heat, smoke and storage responses are published.
Sources	See References: 1, 3, 5, 6, 7, 8, 9

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Atriplex cinerea

Grey Saltbush

Family **Amaranthaceae (chenopods)**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Easy–Moderate	Soak 4 days; smoke water	Mar – May	1–2 mm	21%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	Best	Best	Best	OK	OK	OK	OK	OK	–	–

Best window | Workable – slower or less reliable | Avoid / low success

Plant snapshot

Habit	Grey, erect to spreading shrub, 0.2–1.5 m (to 2 m). Plants are usually male or female, sometimes both.
Flowering	Sep – Oct or Dec (cream-yellow).
Habitat	Saline sand; coastal areas, limestone cliffs, salt lakes and dunes.
Local note	Recorded in the Wanneroo LGA (not listed for Joondalup). Tolerates drought, salt spray and saline soils.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Main method	Strikes easily; no mist. Keeps the parent's sex.	→ Cuttings Guide
Seed	Good	Wash out salts (4-day soak), smoke water; 7–21 days.	This guide
Division	Possible	Low stems layer themselves – lift rooted pieces.	–

Seed collection & storage

Item	Details
Seed collection	Collect fruits that are starting to dry and turn brown, from the bush or the ground underneath – Oct – Mar in SA. Seed is shed 3–14 days after it matures. **Only female plants** set seed (males sometimes carry a few female flowers).
Extraction & cleaning	No cleaning needed if you collect only the fruits – sow the bracteoles whole. Or rub and sieve.
Seed quality	Viability varies. **Brown seeds germinate readily; black seeds only after 3–6 years in storage** (Tasmania). Salts in the fruit stop germination, so they must be washed out before sowing.
Storage life	Store the seed in its fruit, in an airtight container with a desiccant (SA). Soaked seed can be dried again and stored (Geraldton).
Typical germination	Untreated: about **21%** for the upright form (about **11 seedlings per gram**) and 34% for the prostrate form (about 52 per gram) – Alcoa, one test each. Plan with the lower figure unless you know which form you have.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May** (estimate). Jun – Oct workable. Avoid Nov – Feb. Cool conditions favour germination (Geraldton).
Seed mix	Standard sterile native seed mix.
Pre-treatment	**Soak for 4 days, changing the water at least 3 times a day**, then water in with smoke water (Geraldton). A quicker option: rub the fruits under running water for several minutes, or soak 8 hours with 2–3 water changes (Tasmania).

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Item	Details
Sowing depth & method	Cover about **1–2 mm**. Sow thinly in trays, or 2–3 fruits per tube and thin.
After sowing	Water in with smoke water; keep moist, not wet.
Germination time	About **7–21 days** (Geraldton).
Pricking out	When the second pair of leaves appears.
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 3–4 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Poor, patchy germination	Salts not washed out. Soak the full 4 days with regular water changes.
Seedlings collapse	Damping off. Thinner sowing, sterile mix, good airflow.
No seed on stock plants	Male plants. Collect from females, or grow female plants from cuttings.

Notes

Other notes	Cuttings strike easily and keep the parent's sex – a way to raise female plants for seed (see the Cuttings Guide). The nursery records list prostrate and upright forms – keep them separate.
Confidence	**Medium.** WA community-nursery method; only one nursery test for each form.
Sources	See References: 3, 4, 5, 122, 123, 137

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Atriplex isatidea

Coast Saltbush

Family **Amaranthaceae (chenopods)**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Easy–Moderate	Soak 4 days; smoke water	Mar – May	1–2 mm	22%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	Best	Best	Best	OK	OK	OK	OK	OK	–	–

Best window | Workable – slower or less reliable | Avoid / low success

Plant snapshot

Habit	Grey-leaved, spreading shrub, 0.5–2.5 m. Male and female flowers on the same plant.
Flowering	Mainly Mar – Jun.
Habitat	White sand on coastal dunes.
Local note	Recorded in the Wanneroo LGA. Very salt- and wind-tolerant.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Good	Saltbush cuttings are generally reliable (provisional for this species).	→ Cuttings Guide
Seed	Main method	Soak 4 days in changed water, then smoke water.	This guide
Division	Not used	Woody shrub.	–

Seed collection & storage

Item	Details
Seed collection	Each seed is enclosed in a pair of papery bracts. Collect the fruiting bracts when they turn dry and straw-coloured and come away easily.
Extraction & cleaning	Rub off stems and leaves and sieve. Sow the seed still inside its bracts.
Seed quality	Soaking or washing removes germination inhibitors (salts) from the bracts – the reason for the long soak below.
Storage life	Not well documented. Store dry and cool. Soaked seed can be dried again and stored until needed (Geraldton community nursery).
Typical germination	About **22%** (untreated, Alcoa nursery average, 3 tests). About **15 seedlings per gram** – roughly 6.5 g of seed per 100 seedlings.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May**. Jun – Oct is workable (autumn, early winter and spring suit saltbush). Avoid Nov – Feb.
Seed mix	Standard sterile native seed mix.
Pre-treatment	**Soak for 4 days, changing the water 3 times a day**, then water in with smoke water (Geraldton method). Presoaked seed germinates faster.
Sowing depth & method	Cover about **1–2 mm**. Sow thinly in trays, or 2–3 bracts per tube and thin.
After sowing	Water in with smoke water; keep moist, not wet.

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Item	Details
Germination time	About **11–20 days** (Geraldton nursery). Germinates best in cool conditions.
Pricking out	When the second pair of leaves appears.
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 3–4 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Poor, patchy germination	Inhibitors not washed out. Soak the full 4 days with regular water changes.
Seedlings collapse	Damping off. Thinner sowing, sterile mix, good airflow.
Leaf spots on young plants	Foliage kept wet. Water the mix, not the leaves; improve airflow.

Notes

Other notes	Tip cuttings are a workable alternative (see the Cuttings Guide).
Confidence	**Medium.** Nursery germination data (3 tests) and a WA community-nursery method are available.
Sources	See References: 3, 4, 5, 33, 38

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Austrostipa elegantissima

Feather Speargrass

Family **Poaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Moderate	Aged seed; smoke-water soak	Mar – May	Just covered	Better with 1–2 yr old seed

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	Best	Best	Best	OK	OK	OK	–	–	–	–

Best window | Workable – slower or less reliable | Avoid / low success

Plant snapshot

Habit	Rhizomatous, tufted, often sprawling perennial grass, 0.35–2 (–3) m.
Flowering	Aug – Dec or Jan (fluffy, feathery green heads).
Habitat	Sand, loam, clay and granite, occasionally saline.
Local note	Recorded around Perth (Cottesloe, Cambridge, Cockburn); FloraBase has no Joondalup or Wanneroo specimens.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Not used	Grasses are not grown from cuttings.	–
Seed	Main method	1–2 year old seed; smoke-water soak.	This guide
Division	Good	Late winter – early spring (genus).	→ Division Guide

Seed collection & storage

Item	Details
Seed collection	Collect Oct – Feb. Gently strip seed that is turning brown – mature seed comes off easily.
Extraction & cleaning	Rub gently and sieve out the chaff.
Seed quality	Viability varies – one stored lot was 35% viable. Cut-test a sample.
Storage life	**Keep it 1–2 years before sowing** – older seed germinates better (Geraldton). Many *Austrostipa* need 2–24 months of after-ripening. Store dry and airtight with a desiccant.
Typical germination	Not recorded with treatment. Fresh untreated seed gave 0% in the only Alcoa test.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May**. Jun – Aug workable. Avoid Sep – Feb.
Seed mix	Standard sterile native seed mix.
Pre-treatment	**Soak in smoke water (1:9) for 24 hours** (Geraldton method). If only fresh seed is available, add gibberellic acid (GA) as well as smoke water.
Sowing depth & method	Just cover the seed, in trays or a few seeds per tube.
After sowing	Keep moist, not wet.
Germination time	About **5 days** for aged, smoke-treated seed at Geraldton; 3–10 weeks is typical for the genus.

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Item	Details
Pricking out	When seedlings have 2–3 leaves.
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 3–4 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
No germination	Seed too fresh. Store 1–2 years, or add GA to the smoke soak.
Seedlings collapse	Damping off. Sow thinly; airflow.
Floppy, tangled plants	Normal habit – give tubes space.

Notes

Other notes	Established clumps can also be divided (see the Division Guide).
Confidence	**Medium.** WA community-nursery method available; genus after-ripening well documented. The only nursery test (fresh, untreated) gave 0%.
Sources	See References: 3, 4, 5, 88, 89

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Austrostipa flavescens

Coast Speargrass

Family **Poaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Easy–Moderate	Smoke water; aged seed	Mar – May	0–2 mm	Not recorded

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	OK	Best	Best	Best	OK	OK	OK	–	–	–	–

Best window | Workable – slower or less reliable | Avoid / low success

Plant snapshot

Habit	Tufted perennial grass, 0.2–1.2 m.
Flowering	Sep – Oct.
Habitat	Grey, red or calcareous sand, sandy loam and limestone; sand dunes.
Local note	Coastal species. Collect from local-provenance stands and record the source.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Not used	Tufted grass.	–
Seed	Main method	Aged seed and smoke water; sow shallow.	This guide
Division	Good	Divide established tussocks into small clumps.	→ Division Guide

Seed collection & storage

Item	Details
Seed collection	Seed ripens in late spring to early summer after flowering. Collect when seed heads turn straw-coloured and seed pulls away easily; it sheds quickly once ripe.
Extraction & cleaning	Strip seed by hand and dry in paper bags. Long awns can be trimmed to make sowing easier.
Seed quality	Fresh seed is often dormant (after-ripening needed).
Storage life	Store dry at room temperature. **Older seed (1–2 years) gives higher germination** in *Austrostipa* (Geraldton community nursery).
Typical germination	No Alcoa data for this species. A related speargrass (*A. caespitosa*) gave 73% after aerosol smoke.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May** (autumn). Feb and Jun – Aug workable. Avoid Sep – Jan.
Seed mix	Standard sterile native seed mix.
Pre-treatment	• Use seed stored for a year or more, **or** • Water trays with smoke water after sowing (Geraldton method). • With fresh seed only, gibberellic acid plus smoke water can lift germination.
Sowing depth & method	**Sow very shallowly** – on or just below the surface (0–2 mm), pressed in lightly. Seed needs light to germinate.
After sowing	Water in with smoke water; keep moist.

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Item	Details
Germination time	About **21 days** (Geraldton nursery).
Pricking out	Prick out or divide small clumps when seedlings are 3–5 cm with several leaves.
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 3–4 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Poor germination from fresh seed	Dormant. Store seed a year, or use smoke water (and GA if available).
Seed buried too deep	Needs light. Sow on the surface and press in.
Seedlings rot	Overwatering. Keep moist, not wet.

Notes

Other notes	Can also be multiplied by dividing established clumps.
Confidence	**Medium.** WA community-nursery method and genus-level dormancy research; no Alcoa data for this species.
Sources	See References: 3, 5, 23

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Banksia attenuata

Slender Banksia / Candle Banksia

Family **Proteaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Easy–Moderate	None (extract from cones)	Mar – May	3–5 mm	60%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	Best	Best	Best	OK	OK	OK	OK	–	–	–

Best window | Workable – slower or less reliable | Avoid / low success

Plant snapshot

Habit	Lignotuberous tree or shrub, usually 2–4 m (up to 10 m). Resprouts after fire.
Flowering	Oct – Feb (yellow candles).
Habitat	White, yellow, brown or pale red sand, sometimes over laterite; sand dunes and sandplains.
Local note	Recorded in Joondalup and Wanneroo. Highly susceptible to *Phytophthora* dieback – strict hygiene.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Difficult	Banksias are rarely grown from cuttings.	–
Seed	Main method	Extract from cones; no pre-treatment.	This guide
Division	Not used	Tree / shrub.	–

Seed collection & storage

Item	Details
Seed collection	Follicles can take a year or more to mature. Collect **old, dry, dark grey-brown cones** with closed follicles. Cones hold seed for some time, so collection isn't tied to one month.
Extraction & cleaning	Open follicles with heat: oven at **120–130°C for 1 hour** (or 200°C for 10 min), or briefly heat then soak in water for several hours and dry in a warm oven. Separate the winged black seed from the woody separator. Good seed is white and firm inside. Never microwave cones.
Seed quality	Seed from older cones tends to have lower viability than from younger cones – favour recent cones.
Storage life	**Short-lived once extracted** unless stored dry and cool. Best kept in the cone until the sowing season; extracted seed can remain viable for some years sealed, dry and cool (10–25°C, 40–60% RH).
Typical germination	About **60%** (untreated, Alcoa nursery average). About **15 seedlings per gram** – roughly 7 g of seed per 100 seedlings.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May** – sow once the weather cools after March. Jun – Sep workable. Avoid Oct – Feb (heat raises damping-off losses).
Seed mix	Free-draining, **pathogen-free, low-phosphorus** mix (pasteurise home-made mix at 60°C for 30 min). Slightly acid (about pH 6).
Pre-treatment	**None needed** once seed is out of the follicle.

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Item	Details
Sowing depth & method	**Direct sow 1–2 seeds per deep tube** (preferred), or sow thinly in deep trays. **Just cover** with fine mix or vermiculite (about 3–5 mm).
After sowing	Water in with a fine spray. This species is **highly vulnerable to damping off** – water from below (capillary bed) if possible, or use a registered fungicide at germination (e.g. furalaxyl or captan, as per label).
Germination time	About **16–49 days**.
Pricking out	If tray-sown, prick out **at the cotyledon stage, no later than a few weeks after germination**, before the taproot lengthens.
Shade house & hardening	About **2 weeks** under shade cloth after pricking out, then harden off gradually to full sun over 2–4 weeks.
Time to planting size	Est. 4–6 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Seedlings collapse soon after emerging	Damping off. Sterile mix, bottom watering, fungicide at germination, less crowding.
Yellowing, dying seedlings in pots	Phosphorus toxicity or *Phytophthora*. Use very low-P fertiliser; keep pots off the ground.
Few seeds from cones	Immature or old cones. Collect ripe, recent cones; check follicles have developed.

Notes

Other notes	Feed only with half-strength, very low-phosphorus fertiliser once established (Proteaceae are P-sensitive).
Confidence	**High.** Published nursery test data and Banksia seed-handling guidance are available.
Sources	See References: 1, 3, 10, 11, 12, 13

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Banksia dallanneyi

Couch Honeypot

Family **Proteaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Easy–Moderate	None (warm fruit to release)	Mar – May	3–5 mm	37%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	Best	Best	Best	OK	OK	OK	OK	–	–	–

Best window Workable – slower or less reliable Avoid / low success

Plant snapshot

Habit	Low, lignotuberous shrub to 0.5 m, often prostrate and spreading by underground stems.
Flowering	May – Oct (cream, yellow-brown or pink honeypot heads at ground level).
Habitat	Sand or sandy loam, laterite, granite, limestone; flats and rises.
Local note	Recorded in the Wanneroo LGA (subsp. *dallanneyi* var. *dallanneyi* on the Swan Coastal Plain). Label seed by subspecies – there are five.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Possible	ANPSA reports cuttings are successful; no details.	–
Seed	Main method	Warm heads to release seed; no pre-treatment.	This guide
Division	Not used	Spreads by underground stems; not documented as a method.	–

Seed collection & storage

Item	Details
Seed collection	Follicles turn dark brown or black when ripe. Collect whole heads, checking for insect and bird damage.
Extraction & cleaning	Warm the heads in a paper bag – a low oven (**no more than 60°C**) or a car dashboard – until the follicles open. Separate the winged seed from the woody separators.
Seed quality	Cut-test a sample – good seed is white and firm inside.
Storage life	**Long** – seed stays viable for many years stored dry and cool.
Typical germination	About **37%** (Alcoa nursery average, 23 tests, fruit warmed to release seed). About **33 seedlings per gram** – roughly 3 g of seed per 100 seedlings. Many dryandras can reach close to 100% with good seed.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May**. Jun – Sep workable. Avoid Oct – Feb (heat and damping off).
Seed mix	Free-draining, pathogen-free, **low-phosphorus** mix.
Pre-treatment	**None needed** once the seed is out of the follicle.
Sowing depth & method	Direct sow 1–2 seeds per deep tube, just covered (about **3–5 mm**).
After sowing	Water in; keep moist, not wet. Water from below if you can – damping off is the main risk.

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Item	Details
Germination time	Several weeks to several months – keep trays going.
Pricking out	If tray-sown, prick out at the cotyledon stage, before the taproot lengthens.
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 4–6 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Seedlings collapse	Damping off. Sterile mix, bottom watering, good airflow.
Yellowing seedlings	Phosphorus toxicity or *Phytophthora*. Very low-P feed; pots off the ground.
Few seeds from heads	Insect-damaged or immature heads. Collect dark, ripe follicles.

Notes

Other notes	One of the more reliable dryandras in cultivation. ANPSA notes cuttings are also successful.
Confidence	**High.** Large nursery data set (23 tests) and WA *Dryandra* seed notes available.
Sources	See References: 3, 4, 20, 55

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Banksia grandis

Bull Banksia

Family **Proteaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Easy	None	Mar – May	3–5 mm	78%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	Best	Best	Best	OK	OK	OK	OK	–	–	–

Best window · Workable – slower or less reliable · Avoid / low success

Plant snapshot

Habit	Tree, 1.5–10 m (a shrub on the south coast), with huge, deeply lobed leaves.
Flowering	Sep – Dec or Jan (yellow-green spikes).
Habitat	White or grey sand and laterite.
Local note	Recorded in the Wanneroo LGA. **Very phosphorus-sensitive** and susceptible to *Phytophthora* dieback.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Difficult	Slow to strike, well under 100% success.	–
Seed	Main method	No pre-treatment; about 78% germination.	This guide
Division	Not used	Tree.	–

Seed collection & storage

Item	Details
Seed collection	Unlike many banksias, the follicles **open as they mature** without fire. Tie old stockings or muslin over maturing cones, or collect cones just before the follicles split.
Extraction & cleaning	Dry cones in a warm spot; if follicles stay shut, oven at **120–130°C for 1 hour**. Separate the winged seed from the separators.
Seed quality	Cut-test a sample – good seed is white and firm inside.
Storage life	**Short-lived once extracted** unless stored dry and cool.
Typical germination	About **78%** (untreated, Alcoa nursery average, 54 tests). About **14 seedlings per gram** – roughly 7 g of seed per 100 seedlings.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May** (about 20–25°C). Jun – Sep workable. Avoid Oct – Feb.
Seed mix	Free-draining, pathogen-free, **low-phosphorus** mix.
Pre-treatment	**None needed** – reliable without treatment (ANPSA).
Sowing depth & method	Direct sow 1–2 seeds per deep tube, just covered (about **3–5 mm**).
After sowing	Water in. Water from below if possible; watch closely for damping off.
Germination time	About **3–6 weeks** (22–42 days reported).
Pricking out	If tray-sown, prick out at the cotyledon stage, before the taproot lengthens.

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Item	Details
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 4–6 months from sowing. Slow to flower – many years from seed.

Common problems

Problem	Likely cause and fix
Seedlings collapse	Damping off. Sterile mix, bottom watering, less crowding.
Yellowing or dying in pots	Phosphorus toxicity or dieback. No P in feed; strict hygiene; pots off the ground.
Empty follicles	Cones collected too early. Wait until follicles are fully developed.

Notes

Other notes	Cuttings are slow and unreliable – seed is the method.
Confidence	**High.** Large nursery data set (54 tests) and Banksia seed-handling guidance available.
Sources	See References: 3, 4, 10, 11, 53, 56, 57

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Banksia menziesii

Firewood Banksia

Family **Proteaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Easy–Moderate	None (extract from cones)	Mar – May	3–5 mm	56%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	Best	Best	Best	OK	OK	OK	OK	–	–	–

Best window | Workable – slower or less reliable | Avoid / low success

Plant snapshot

Habit	Tree or shrub, 1.3–7 m, from a lignotuber. Resprouts after fire.
Flowering	Feb – Oct, peaking May – Jul (pink, red or yellow).
Habitat	White, grey or yellow sand.
Local note	Recorded in Joondalup and Wanneroo. Medium to high susceptibility to *Phytophthora* dieback.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Difficult	Banksias are rarely grown from cuttings.	–
Seed	Main method	Extract from cones; no pre-treatment.	This guide
Division	Not used	Tree / shrub.	–

Seed collection & storage

Item	Details
Seed collection	Up to 25 follicles per cone, each with 1–2 seeds. Follicles open mainly after fire (or release seed after about two years), so collect **mature, closed cones**.
Extraction & cleaning	Heat cones to open follicles: oven at **120–130°C for 1 hour** (or 200°C for 10 min), or briefly heat, soak for several hours, then dry in a warm oven. Separate seed from the woody separators.
Seed quality	Good seed is white and firm with a dark coat. Recent cones generally give better viability than old cones.
Storage life	**Short-lived once extracted** unless stored dry and cool. Keep in the cone until needed where possible.
Typical germination	About **56%** (untreated, Alcoa nursery average). About **27 seedlings per gram** – roughly 3.7 g of seed per 100 seedlings.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May**, once the weather cools. Jun – Sep workable. Avoid Oct – Feb.
Seed mix	Free-draining, pathogen-free, **low-phosphorus** mix (about pH 6).
Pre-treatment	**None needed** once seed is out of the follicle.
Sowing depth & method	**Direct sow 1–2 seeds per deep tube** (preferred), or thinly in deep trays. Just cover (about 3–5 mm).

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Item	Details
After sowing	Water in with a fine spray; bottom watering where possible. Watch for damping off; fungicide at germination if needed (as per label).
Germination time	About **26–40 days**.
Pricking out	If tray-sown, prick out at the cotyledon stage within a few weeks of germination.
Shade house & hardening	About **2 weeks** under shade cloth after pricking out, then harden off gradually to full sun over 2–4 weeks.
Time to planting size	Est. 4–6 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Seedlings collapse soon after emerging	Damping off. Sterile mix, bottom watering, less crowding.
Yellowing or dying in pots	Phosphorus toxicity or *Phytophthora*. Very low-P feed; strict hygiene; pots off the ground.
Follicles won't open	Not enough heat. Repeat heating, or heat then soak and re-dry.

Notes

Other notes	Takes about five to seven years to flower from seed.
Confidence	**High.** Published nursery test data and Banksia seed-handling guidance are available.
Sources	See References: 1, 3, 10, 11, 12, 13

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Banksia prionotes

Acorn Banksia

Family **Proteaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Easy–Moderate	None (heat cones to open)	Mar – May	3–5 mm	57%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	Best	Best	Best	OK	OK	OK	OK	–	–	–

Best window | Workable – slower or less reliable | Avoid / low success

Plant snapshot

Habit	Tree or shrub to 10 m. No lignotuber – killed by fire and regenerates only from seed.
Flowering	Feb – Aug (cream and orange 'acorn' spikes).
Habitat	Sand, sandy loam, clay, laterite and granite; sandplains, dunes and slopes.
Local note	Recorded in the Wanneroo LGA. Highly susceptible to *Phytophthora* dieback. Good cut flower.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Difficult	Cuttings are difficult to strike.	–
Seed	Main method	Heat old cones to open; no pre-treatment.	This guide
Division	Not used	Tree.	–

Seed collection & storage

Item	Details
Seed collection	Seed is mostly held in the cone until fire, releasing only slowly otherwise. Collect cones **at least 12 months old**, with closed follicles.
Extraction & cleaning	Open follicles with heat: oven at **120–130°C for 1 hour** (or 200°C for 10 minutes), or pass over a burner, soak for several hours and dry in a warm oven; repeat if needed. Never microwave cones.
Seed quality	Cut-test a sample – good seed is white and firm inside.
Storage life	**Short-lived once extracted** – keep seed in the cone until the sowing season.
Typical germination	About **57%** (untreated, Alcoa, 3 tests). About **29 seedlings per gram** – roughly 3.4 g of seed per 100 seedlings.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May** (about 20–25°C). Jun – Sep workable. Avoid Oct – Feb.
Seed mix	Free-draining, pathogen-free, **low-phosphorus** mix.
Pre-treatment	**None needed** once the seed is out of the follicle.
Sowing depth & method	Direct sow 1–2 seeds per deep tube, just covered (about **3–5 mm**).
After sowing	Water in. Water from below if possible; watch for damping off.
Germination time	About **3–5 weeks** (21–35 days reported).

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Item	Details
Pricking out	If tray-sown, prick out at the cotyledon stage, before the taproot lengthens.
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 4–6 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Seedlings collapse	Damping off. Sterile mix, bottom watering, less crowding.
Yellowing or dying in pots	Phosphorus toxicity or dieback. Very low-P feed; strict hygiene.
Follicles won't open	Not enough heat. Repeat the heat-and-soak cycle.

Notes

Other notes	Cuttings are difficult; grafting onto *B. integrifolia* or *B. serrata* has been used for humid climates.
Confidence	**Medium.** Nursery germination data from 3 tests plus Banksia seed-handling guidance.
Sources	See References: 3, 4, 10, 11, 54, 56

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Banksia sessilis

Parrot Bush

Family **Proteaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Easy–Moderate	None (smoke water optional)	Mar – May	3–5 mm	42%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	Best	Best	Best	OK	OK	OK	–	–	–	–

Best window · Workable – slower or less reliable · Avoid / low success

Plant snapshot

Habit	Upright, prickly shrub or small tree to 6 m. Killed by fire and regrows from seed.
Flowering	Mainly Jul – Nov (cream-yellow domes).
Habitat	Sandy soils over laterite or limestone.
Local note	The Swan Coastal Plain form is var. *cygnorum*. Formerly *Dryandra sessilis*.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Not used	Cuttings have proved virtually impossible.	–
Seed	Main method	No pre-treatment; sow as it cools after March.	This guide
Division	Not used	Shrub.	–

Seed collection & storage

Item	Details
Seed collection	Not serotinous: most seed is **released in autumn** even without fire. Collect ripe heads in summer, before release.
Extraction & cleaning	Put heads in a paper bag in a warm place or a low oven (**no hotter than 60°C**) until the follicles open; cut stubborn follicles with secateurs.
Seed quality	Good seed is firm and white inside.
Storage life	Seed in the soil is short-lived, but cleaned seed keeps well **stored dry and cool**.
Typical germination	About **42%** (untreated, Alcoa nursery average). About **33 seedlings per gram** – roughly 3 g of seed per 100 seedlings.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May** – sow once the weather cools after March. Jun – Aug workable. Avoid Sep – Feb.
Seed mix	Free-draining, sterile, **low-phosphorus** mix.
Pre-treatment	**None needed.** Geraldton soaks seed in smoke water (1:9) for 24 hours.
Sowing depth & method	Direct sow 1–2 seeds per tube, just covered (about 3–5 mm).
After sowing	Water in; avoid excess water – damping off can be a problem.
Germination time	About **2–6 weeks** (14 days at Geraldton).

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Item	Details
Pricking out	If tray-sown, prick out at the cotyledon stage, within a few weeks of germination.
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 4–6 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Seedlings collapse	Damping off. Water less, sterile mix, good airflow.
Few seeds in heads	Collected too late – seed already released in autumn. Collect in summer.
Yellowing seedlings	Phosphorus toxicity. Use very low-P fertiliser.

Notes

Other notes	Cuttings have proved virtually impossible – seed is the only practical method. Flowers within about two years.
Confidence	**High.** Nursery test data (as *Dryandra sessilis*) and WA seed-handling notes available.
Sources	See References: 3, 5, 20, 25

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Bossiaea eriocarpa

Common Brown Pea

Family **Fabaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Easy–Moderate	Boiling water 30 s	Mar – May	2–3 mm	44%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	OK	Best	Best	Best	OK	OK	OK	Best	OK	–	–

Best window · Workable – slower or less reliable · Avoid / low success

Plant snapshot

Habit	Erect or straggly, spreading shrub, 0.2–1 m.
Flowering	Jul – Nov (yellow and brown-red pea flowers).
Habitat	Usually sandy soils.
Local note	Recorded in Joondalup and Wanneroo.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Not used	Not documented.	–
Seed	Main method	Boiling water 30 s.	This guide
Division	Not used	–	–

Seed collection & storage

Item	Details
Seed collection	Pods ripen in summer and shed seed when ripe. **Bag developing pods** with stockings or muslin.
Extraction & cleaning	Rub and sieve the small seed from pod fragments.
Seed quality	Hard-seeded – fresh seed is dormant until the coat is treated. Float test to remove damaged seed.
Storage life	**Long.** Keeps many years when dry (typical of hard-coated pea seed).
Typical germination	About **44%** after boiling-water treatment (Alcoa nursery average). About **229 seedlings per gram** – under 0.5 g per 100 seedlings.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May** and **Sep**. Feb, Jun – Aug and Oct workable. Avoid Nov – Jan.
Seed mix	Standard sterile native seed mix.
Pre-treatment	**Boiling water for 30 seconds** (Alcoa method), or pour near-boiling water over and soak overnight. Sow swollen seed; re-treat the rest.
Sowing depth & method	Small seed – cover about **2–3 mm**. Direct sow a few seeds per tube and thin.
After sowing	Water in; keep moist, not wet.
Germination time	Can start about **one week** after treatment.
Pricking out	At the cotyledon to first true leaf stage, before the taproot lengthens.

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Item	Details
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 3–5 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Low germination	Seed coat not broken. Re-treat unswollen seed.
Swollen seed rots	Over-boiled or too wet. Keep to 30 s; water less.
Seedlings collapse	Damping off. Sterile mix, airflow, avoid crowding.

Notes

Other notes	Very small seed for a pea – a little goes a long way.
Confidence	**High.** Nursery germination data (40 tests) and WA hard-seed guidance available.
Sources	See References: 3, 5, 14, 15, 16

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Burchardia congesta

Milkmaids

Family **Colchicaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Moderate	None; smoke optional	Mar – May	Lightly covered	44%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	Best	Best	Best	OK	OK	OK	–	–	–	–

Best window Workable – slower or less reliable Avoid / low success

Plant snapshot

Habit	Erect, tuberous perennial herb to 0.8 m; dies back to its tubers over summer.
Flowering	Aug – Oct (white, with yellow anthers).
Habitat	A wide range of sandy, loamy and gravelly soils; dunes, plains, slopes and ridges.
Local note	Recorded in Joondalup and Wanneroo. WA plants were called *B. umbellata* until 2005 – older lists and nursery records use that name.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Not used	Not grown from cuttings.	–
Seed	Main method	No pre-treatment; about 44% (Alcoa, as B. umbellata). Sow cool.	This guide
Division	Not used	Tubers are replaced each year – grow from seed.	–

Seed collection & storage

Item	Details
Seed collection	Collect in **December – January**. In a study at Woodvale, seed was mature and capsules opened in late November but still held their seed at the end of December; by late January about a fifth had shed, and by the end of February all had.
Extraction & cleaning	Dry in paper bags until the capsules open, then thresh lightly and sieve (related species).
Seed quality	Good seed is dark brown and hard, about 1.5–2.5 mm long (genus).
Storage life	Store dry and cool in an airtight container. Eastern plants need about 3 months of after-ripening before seed germinates.
Typical germination	About **44%** untreated (Alcoa, 6 tests, recorded as *B. umbellata*), about **199 seedlings per gram**. A single test under the name *B. congesta* gave 0%. Kings Park found smoke promotes germination in *Burchardia*.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May** – cool conditions; germination may fail above about 18°C (related species). Jun – Aug workable. Avoid Sep – Feb.
Seed mix	Standard sterile native seed mix, free-draining.
Pre-treatment	None needed. Smoke water (a 12-hour soak in 10% smoke water, Dixon) is worth trialling on part of a batch – results with smoke have been mixed.
Sowing depth & method	Sow thinly and cover lightly.

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Item	Details
After sowing	Keep moist through the cool months.
Germination time	About **4–8 weeks** (related species).
Pricking out	Early, at 1–2 leaves, while roots are small (estimate).
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Slow – seedlings die back to a small tuber in their first summer. Record your own.

Common problems

Problem	Likely cause and fix
Seedlings die back in summer	Normal dormancy. Keep pots barely moist until autumn – watering dormant plants rots the tubers.
Poor germination	Sown too warm, or seed too fresh. Sow in autumn with seed a few months old.
Seedlings collapse	Damping off. Sterile mix, airflow.

Notes

Other notes	Not divided – the tubers are replaced each year. Grow from seed.
Confidence	**Medium.** Nursery data recorded under the older name *B. umbellata* (6 tests) and a Perth study of seed release; growing notes from related plants.
Sources	See References: 3, 4, 49, 60, 110, 128, 129, 130, 136

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Calothamnus quadrifidus

One-sided Bottlebrush

Family **Myrtaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Easy	None	Mar – Apr, Sep – Oct	Surface	26%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	OK	Best	Best	OK	OK	OK	OK	Best	Best	OK	–

Best window · Workable – slower or less reliable · Avoid / low success

Plant snapshot

Habit	Erect, compact or spreading shrub, usually 0.9–2 m.
Flowering	Jun – Dec (red, one-sided brushes).
Habitat	Wide variety of soils and habitats.
Local note	Recorded in the Wanneroo LGA. Several subspecies exist – label stock by subspecies and provenance.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Good	Presumed to strike like related melaleucas (provisional).	→ Cuttings Guide
Seed	Main method	Easy from seed held in old capsules.	**This guide**
Division	Not used	Woody shrub.	–

Seed collection & storage

Item	Details
Seed collection	Woody capsules stay on the older stems for years. Collect capsules from **older wood** (at least a year old) at any time of year.
Extraction & cleaning	Place capsules in an open tray or paper bag in a warm spot; fine seed is released in a few days. No need to separate seed from chaff.
Seed quality	Very fine seed mixed with chaff – a cut-test and count gives seeds per gram.
Storage life	**Long** if kept dry and cool (typical of capsule-fruited Myrtaceae).
Typical germination	About **26%** (untreated, Alcoa nursery average). About **572 seedlings per gram** – a pinch goes a long way.

Sowing & growing on

Item	Details
Best sowing time	**Mar – Apr** and **Sep – Oct** (15–25°C). Feb, May – Aug and Nov workable. Avoid Dec – Jan.
Seed mix	Fine-textured, sterile native seed mix with a smooth surface.
Pre-treatment	**None needed.**
Sowing depth & method	Mix seed with dry, sterile yellow sand and sprinkle from a spice jar. **Sow on the surface** and press in lightly; don't bury.
After sowing	Water in with smoke water (Geraldton method) using a very fine spray, or water from below (bog method) so seed isn't washed away.
Germination time	About **7–10 days** (Geraldton nursery).

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Item	Details
Pricking out	When the **second pair of leaves** appears (about 2 cm). Seedlings are tiny – handle by the leaves.
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 4–6 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Seedlings too crowded	Sown too thickly. Dilute seed well with sand.
Seed washed into clumps	Overhead watering too coarse. Use a fine mist or bottom watering.
Damping off	Crowding and wet mix. Thin sowing, airflow, sterile mix.

Notes

Other notes	Cuttings are worth trialling if a particular form is wanted (see the Cuttings Guide).
Confidence	**High.** Large nursery data set (39 tests) and WA community-nursery method available.
Sources	See References: 3, 4, 5, 18

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Carpobrotus virescens

Coastal Pigface

Family **Aizoaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Easy–Moderate	Ferment fruit; smoke water	Mar – May	Surface	21%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	OK	Best	Best	Best	OK	–	–	OK	OK	–	–

Best window | Workable – slower or less reliable | Avoid / low success

Plant snapshot

Habit	Prostrate succulent groundcover, 0.1–0.3 m high and 0.5–3 m wide.
Flowering	Jun – Dec or Jan (purple-pink, sometimes white), followed by fleshy fruit.
Habitat	Coastal limestone and sand dunes.
Local note	Recorded in the Wanneroo LGA. Collect only from plants you are sure are native – the introduced Hottentot Fig (*C. edulis*) is naturalised on some coasts.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Main method	Stem pieces root very easily; rooted runners can be lifted.	→ Cuttings Guide
Seed	Good	Ferment fruit, sow fine seed on the surface; 6–8 days.	This guide
Division	Possible	Lift stems that have rooted – treat as cuttings.	–

Seed collection & storage

Item	Details
Seed collection	Collect the fleshy fruits when soft and fully ripe, after flowering.
Extraction & cleaning	**Ferment:** soak the crushed fruit in water for 4–5 days, then wash the tiny seeds out through a fine sieve and dry them.
Seed quality	Seed is very fine – each fruit holds many seeds.
Storage life	Not well documented. Store dry and cool.
Typical germination	About **21%** after a dry-heat treatment (Alcoa nursery average, 9 tests). About **454 seedlings per gram** – roughly 0.2 g of seed per 100 seedlings.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May**. Feb, Jun and Sep – Oct are workable. Avoid the coldest and hottest months.
Seed mix	Standard sterile native seed mix, extra free-draining.
Pre-treatment	Ferment and wash the seed (above). Alcoa used a dry-heat pulse instead. Water in with smoke water after sowing.
Sowing depth & method	**Surface-sow.** Mix the fine seed with dry yellow sand in a spice jar and shake it evenly over the tray (Geraldton method). Press in lightly.
After sowing	Water in with smoke water using a fine mist; keep just moist – succulent seedlings rot if wet.

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Item	Details
Germination time	About **6–8 days** (Geraldton nursery).
Pricking out	When seedlings have two pairs of leaves and can be handled.
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 3–4 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Seedlings rot	Too wet. Water less, improve drainage and airflow.
Seed washed into clumps	Use bottom watering or a fine mist.
Poor germination	Fruit not fully ripe, or seed not cleaned. Collect soft, ripe fruit and ferment fully.

Notes

Other notes	Stem cuttings are quicker and very reliable (see the Cuttings Guide).
Confidence	**Medium.** Nursery germination data (9 tests) and a WA community-nursery method are available.
Sources	See References: 3, 4, 5, 32

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Clematis linearifolia

Slender Clematis / Old Man's Beard

Family **Ranunculaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Moderate	Smoke-water soak 24 h	Apr – Jun	Just covered (1–2 mm)	0% untreated (1 test)

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	OK	Best	Best	Best	OK	–	–	–	–	–

Best window | Workable – slower or less reliable | Avoid / low success

Plant snapshot

Habit	Woody climber to about 5 m. **Male and female flowers on separate plants** (dioecious).
Flowering	Jul – Oct (white-cream), followed by feathery seed heads on female plants.
Habitat	Sand; coastal limestone and sand dunes.
Local note	Recorded in the Wanneroo LGA. Only female plants produce seed – note which plants you collect from.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Possible	Semi-hardened cuttings work for related clematis; lets you pick the plant's sex.	–
Seed	Main method	Smoke-water soak; 40–50 days in cool weather.	This guide
Division	Not used	Climber.	–

Seed collection & storage

Item	Details
Seed collection	Collect the feathery seed heads from female plants when **very dry** and the seeds come away easily (Geraldton nursery).
Extraction & cleaning	Rub off the feathery plumes over a sieve.
Seed quality	Cut-test a few seeds – look for plump, filled seed.
Storage life	Probably short: seed of the related *C. microphylla* dropped to about 50% viability after 6 months. Sow the same year.
Typical germination	Not recorded with treatment. The only Alcoa test, untreated, gave 0%.

Sowing & growing on

Item	Details
Best sowing time	**Apr – Jun** – it doesn't germinate until the cooler weather from late April (Geraldton). Mar and Jul workable.
Seed mix	Standard sterile native seed mix.
Pre-treatment	**Soak in smoke water (1:9) for 24 hours** before sowing (Geraldton method).
Sowing depth & method	Just cover the seed, about 1–2 mm. Sow in trays.
After sowing	Keep moist, not wet, in a cool shaded spot.
Germination time	About **40–50 days** (Geraldton nursery).

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Item	Details
Pricking out	When the second pair of leaves appears.
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 4–6 months; give each tube a small stake.

Common problems

Problem	Likely cause and fix
No germination	Sown too warm, or old seed. Sow fresh seed in cool weather after a smoke soak.
Empty seed	Heads from male plants or unpollinated females. Collect from female plants near males.
Tangled seedlings	Stake and space tubes.

Notes

Other notes	Cuttings of semi-hardened stems work for related clematis and let you choose male or female plants – worth trialling.
Confidence	**Medium.** WA community-nursery method available; the only nursery test (untreated) gave 0%.
Sources	See References: 3, 4, 5, 64, 65

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Conostylis aculeata

Prickly Conostylis

Family **Haemodoraceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Moderate	Smoke water	Mar – May	Just covered (1–2 mm)	19%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	Best	Best	Best	OK	OK	OK	–	–	–	–

Best window · Workable – slower or less reliable · Avoid / low success

Plant snapshot

Habit	Rhizomatous, tufted perennial, 0.06–0.5 m, with prickly-edged leaves.
Flowering	Aug – Nov (yellow).
Habitat	Sand, loam, clay, gravel, limestone or laterite; winter-wet areas and sand dunes.
Local note	Recorded in Joondalup and Wanneroo (subsp. *cygnorum* is the local coastal-plain form). There are 10 subspecies – label seed by subspecies. Dieback resistant.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Not used	Tufted herb.	–
Seed	Main method	Smoke water; about 19% in nursery tests.	This guide
Division	Good	Larger divisions, or rooted runners after flowering.	→ Division Guide

Seed collection & storage

Item	Details
Seed collection	Collect the dry capsules in late spring–early summer, after flowering, before they split.
Extraction & cleaning	Dry in paper bags; rub and sieve out the fine seed.
Seed quality	Very fine seed – about 700 seedlings per gram in nursery tests.
Storage life	Stores dry. *Conostylis* seed has cyclical dormancy – its response to smoke changes as it ages, so results vary between batches.
Typical germination	About **19%** with smoke (Alcoa nursery average, 12 tests). About **712 seedlings per gram** – roughly 0.14 g of seed per 100 seedlings.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May**. Jun – Aug workable. Avoid Sep – Feb.
Seed mix	Standard sterile native seed mix.
Pre-treatment	**Smoke water** – 24 h soak (1:9) or water trays with it after sowing.
Sowing depth & method	Mix the fine seed with dry sand and sow thinly; just cover.
After sowing	Keep moist, not wet.
Germination time	About **3–6 weeks** (record yours).
Pricking out	When seedlings have 2–3 leaves.

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Item	Details
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 5–6 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Patchy germination between batches	Normal – cyclical dormancy. Try seed of different ages; always use smoke.
Seedlings collapse	Damping off. Sow thinly; good airflow.
Seed washed away	Bottom water or use a fine mist.

Notes

Other notes	Also propagated by division, and by potting up rooted runners after flowering (see the Division Guide).
Confidence	**High.** Large nursery data set with smoke (12 tests); smoke response documented for the genus.
Sources	See References: 3, 4, 60, 72

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Conostylis candicans

Grey Cottonhead

Family **Haemodoraceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Moderate–Difficult	Smoke water; aged seed	Mar – May	Just covered	3%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	Best	Best	Best	OK	OK	OK	–	–	–	–

Best window · Workable – slower or less reliable · Avoid / low success

Plant snapshot

Habit	Rhizomatous tufted perennial, 0.05–0.4 m, with silvery-grey woolly leaves.
Flowering	Jul – Nov (yellow, in woolly heads).
Habitat	Sand, sandy loam and limestone; coastal and inland sand dunes.
Local note	Recorded in the Wanneroo LGA. Two former subspecies were raised to species in 2026 (*C. calcicola*, *C. procumbens*) – check labels.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Not used	Not grown from cuttings.	–
Seed	Main method	Smoke water; older seed; spasmodic.	This guide
Division	Possible	Less easy than kangaroo paws – large pieces (3+ big shoots), autumn.	–

Seed collection & storage

Item	Details
Seed collection	Collect dry capsules in late spring–early summer, after flowering, before they split.
Extraction & cleaning	Dry in paper bags; rub the woolly heads over a sieve.
Seed quality	Cut-test a sample.
Storage life	Stores dry. In eight *Conostylis* taxa, seed buried in soil for a year germinated better than fresh seed, so older seed may do better.
Typical germination	Low untreated: about **3%** (Alcoa, 7 tests), about 148 seedlings per gram. Smoke water and the smoke compounds glyceronitrile and karrikinolide significantly increased germination in research.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May**. Jun – Aug workable. Avoid Sep – Feb.
Seed mix	Standard sterile native seed mix, free-draining.
Pre-treatment	**Smoke water** – 24 h soak (1:9) or water trays with it. Use seed that is a year or more old if you have it.
Sowing depth & method	Sow thinly and just cover.
After sowing	Keep moist, not wet; keep trays going – germination is spasmodic.

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Item	Details
Germination time	Anywhere from about 3 weeks to 4 months.
Pricking out	At 2–3 leaves, gently.
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 5–6 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Very few seedlings	Normal – spasmodic germination. Use smoke, older seed, and larger sowings.
Seedlings rot	Too wet. Water less; free-draining mix.
Woolly leaves rot in the shade house	Too humid. More airflow, less overhead watering.

Notes

Other notes	Also propagated by division, though less easily than kangaroo paws (see the Division Guide).
Confidence	**High.** Nursery tests (7) plus research showing smoke compounds and soil burial lift germination.
Sources	See References: 3, 4, 60, 81, 82, 83, 90

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Conostylis setigera

Bristly Cottonhead

Family **Haemodoraceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Easy–Moderate	None; smoke water helps	Mar – May	Just covered	31%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	Best	Best	Best	OK	OK	OK	–	–	–	–

Best window · Workable – slower or less reliable · Avoid / low success

Plant snapshot

Habit	Rhizomatous, tufted perennial, 0.05–0.36 m, with bristly-edged leaves.
Flowering	Aug – Oct (yellow, sometimes red-tinged).
Habitat	White, grey or yellow sand, gravel, sandy loam and laterite.
Local note	Recorded in the Wanneroo LGA (subsp. *setigera*). Subsp. *dasys* is threatened and not local.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Not used	Not grown from cuttings.	–
Seed	Main method	About 31% untreated (Alcoa); smoke helps.	This guide
Division	Possible	Large pieces in autumn, as for other Conostylis.	–

Seed collection & storage

Item	Details
Seed collection	Collect dry capsules in late spring–summer, after flowering, before they split.
Extraction & cleaning	Dry in paper bags; rub and sieve out the fine seed.
Seed quality	Fine seed – about 410 seedlings per gram.
Storage life	Stores dry. In eight *Conostylis* taxa, seed buried in soil for a year germinated better than fresh seed.
Typical germination	About **31%** untreated (Alcoa nursery average, 4 tests), about 410 seedlings per gram – roughly 0.25 g of seed per 100 seedlings.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May**. Jun – Aug workable. Avoid Sep – Feb.
Seed mix	Standard sterile native seed mix.
Pre-treatment	None needed, but **smoke water** promotes germination in *Conostylis* – use it on the first watering.
Sowing depth & method	Mix the fine seed with dry sand, sow thinly and just cover.
After sowing	Keep moist, not wet.
Germination time	Not recorded for this species; *C. candicans* takes 3 weeks to 4 months. Check from 3 weeks and record yours.
Pricking out	At 2–3 leaves.

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Item	Details
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 5–6 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Patchy germination	Seed age affects dormancy. Always use smoke; try seed of different ages.
Seedlings collapse	Damping off. Sow thinly; airflow.
Seed washed away	Bottom water or use a fine mist.

Notes

Other notes	Clumps can be divided, but *Conostylis* divide less readily than kangaroo paws – use large pieces.
Confidence	**Medium.** Nursery data from 4 tests; smoke response documented for the genus.
Sources	See References: 3, 4, 60, 72, 82, 90

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Corymbia calophylla

Marri

Family **Myrtaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Easy	None	Mar – Apr, Sep – Oct	About 5 mm	60%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	OK	Best	Best	OK	OK	OK	OK	Best	Best	OK	–

Best window · Workable – slower or less reliable · Avoid / low success

Plant snapshot

Habit	Tree to 40 m (rarely a mallee), with rough, tessellated bark.
Flowering	Dec or Jan – May (white or pink).
Habitat	Wide range: clay loam, sandy clay, gravel, grey sand over limestone, granite and laterite.
Local note	Recorded in the Wanneroo LGA. Large urn-shaped gumnuts ('honky nuts').

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Not used	Eucalypts are very difficult from cuttings.	–
Seed	Main method	No pre-treatment; about 60% germination.	This guide
Division	Not used	Tree.	–

Seed collection & storage

Item	Details
Seed collection	Fruit takes about a year to mature. Collect brown, woody gumnuts when the valves look ready to open.
Extraction & cleaning	Keep gumnuts warm and dry in a tray or paper bag until the large winged seed is released.
Seed quality	Large seed – easy to check by cutting a few open.
Storage life	**Long** if stored cool and dry (eucalypt seed keeps its viability a long time).
Typical germination	About **60%** (untreated, Alcoa nursery average). About **19 seedlings per gram** – roughly 5 g of seed per 100 seedlings.

Sowing & growing on

Item	Details
Best sowing time	**Mar – Apr** and **Sep – Oct**. Feb, May – Aug and Nov workable. Avoid Dec – Jan.
Seed mix	Standard sterile native seed mix.
Pre-treatment	**None needed.** A dip in dilute bleach (10%) to surface-sterilise the seed can help.
Sowing depth & method	Large seed – cover about **5 mm**. Direct sow 1–2 seeds per tube (preferred) or space out in trays.
After sowing	Water in; keep moist.
Germination time	Usually within about **1–2 weeks**.
Pricking out	When seedlings have two pairs of true leaves.

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Item	Details
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 3–5 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Seedlings collapse	Damping off. Sterile mix, avoid crowding.
Leggy seedlings	Too much shade. Move to brighter light once up.
Roots circling	Kept too long in tubes. Plant out or pot on sooner.

Notes

Other notes	Fast-growing – time sowings so trees aren't held in tubes too long.
Confidence	**High.** Large nursery data set (66 tests) and WA eucalypt seed notes available.
Sources	See References: 3, 4, 5, 19

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Daviesia decurrens

Prickly Bitter-pea

Family **Fabaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Moderate	Dry heat or hot water	Mar – May	3–5 mm	25%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	OK	Best	Best	Best	OK	OK	OK	Best	OK	–	–

Best window · Workable – slower or less reliable · Avoid / low success

Plant snapshot

Habit	Erect, prickly shrub, usually to about 1 m (0.3–1.8 m).
Flowering	Jun – Sep (orange and red pea flowers).
Habitat	Red-brown loam over clay, gravel and laterite; well-drained slopes, often near rivers.
Local note	Recorded in the Wanneroo LGA (not listed for Joondalup). Two subspecies (*decurrens* and *hamata*) – keep collections separate. Sharp, pointed leaves – wear gloves. *Daviesia* is susceptible to dieback (*Phytophthora*) – keep nursery hygiene tight.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Difficult	Slow and unreliable in this genus.	–
Seed	Main method	Dry heat or hot water; about 25% (Alcoa).	This guide
Division	Not used	Shrub – not divided.	–

Seed collection & storage

Item	Details
Seed collection	Pods (8–13 mm, broadly triangular) ripen 2–3 months after flowering, in late spring–summer, and **release seed explosively**. Tie stocking or muslin bags over developing pods and check every few weeks. Keep ants out of the bags.
Extraction & cleaning	Rub the pods and sieve or winnow; leave the aril on the seed.
Seed quality	Hard-seeded. Float test in water with a drop of detergent: discard floaters and seed with insect holes.
Storage life	**Long.** Hard-coated pea seed keeps for many years when dry.
Typical germination	About **25%** after heat treatment (Alcoa nursery average, 9 tests). About **9 seedlings per gram** – roughly 11 g of seed per 100 seedlings.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May** and **Sep**. Feb, Jun – Aug and Oct are workable. Avoid Nov – Jan.
Seed mix	Standard sterile native seed mix.
Pre-treatment	**Dry heat** – the Alcoa nursery heat-treated this species. Seed Notes 15 gives 80–120°C in an oven for 10 minutes to 1 hour. Or pour near-boiling water over the seed, leave it to cool and soak up to 24 hours. Sow the swollen seed; re-treat the rest.
Sowing depth & method	Cover about **3–5 mm**. Direct sow 2–3 seeds per tube and thin, or sow in trays.

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Item	Details
After sowing	Water in; keep moist, not wet. Don't let swollen seed dry out.
Germination time	Not documented for this species – start checking about a week after treatment and record it.
Pricking out	At the cotyledon to first true leaf stage – early, before the taproot lengthens.
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 3–5 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Very low germination	Seed coat not broken. Re-treat unswollen seed, or try the other method.
Swollen seed rots	Mix too wet, or over-heated. Water less; keep oven times within the range.
Seedlings collapse	Damping off or dieback. Sterile mix, clean benches, good airflow.

Notes

Other notes	Cuttings are slow and unreliable in *Daviesia* – seed is the practical method. Boiling water scalds – use tongs and heat-proof containers.
Confidence	**High.** Nursery germination data from 9 tests; method follows WA hard-seed guidance (Seed Notes 15).
Sources	See References: 3, 4, 14, 120

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Daviesia divaricata

Marno

Family **Fabaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Moderate	Boiling water 30 s	Mar – May	3–5 mm	28%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	OK	Best	Best	Best	OK	OK	OK	Best	OK	–	–

Best window · Workable – slower or less reliable · Avoid / low success

Plant snapshot

Habit	Shrub, 0.4–2 m.
Flowering	Aug – Nov (pea flowers).
Habitat	Sand, limestone and laterite.
Local note	Recorded in Joondalup and Wanneroo.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Not used	Not documented.	–
Seed	Main method	Boiling water 30 s.	This guide
Division	Not used	–	–

Seed collection & storage

Item	Details
Seed collection	Pods ripen in late spring–summer and **release seed as soon as they are ripe**. Tie fine mesh or stocking bags over developing pods and collect once they split.
Extraction & cleaning	Rub pods and sieve or winnow to remove debris.
Seed quality	Hard-seeded: fresh seed stays dormant until the coat is treated. Float test: discard seed that floats.
Storage life	**Long.** Hard-coated pea seed keeps for many years when dry.
Typical germination	About **28%** after boiling-water treatment (Alcoa nursery average, 12 tests). About **11 seedlings per gram** – roughly 9 g of seed per 100 seedlings.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May** and **Sep**. Feb, Jun – Aug and Oct are workable. Avoid Nov – Jan.
Seed mix	Standard sterile native seed mix.
Pre-treatment	**Boiling water:** drop seed into boiling water for **30 seconds** (Alcoa method), then cool; or pour near-boiling water over the seed (10:1) and soak overnight. Sow the swollen seed; re-treat the rest.
Sowing depth & method	Cover about **3–5 mm**. Direct sow 2–3 seeds per tube and thin, or sow in trays.
After sowing	Water in; keep moist, not wet. Don't let swollen seed dry out.
Germination time	Can start about **one week** after treatment.
Pricking out	At the cotyledon to first true leaf stage – early, before the taproot lengthens.

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Item	Details
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 3–5 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Very low germination	Seed coat not broken. Re-treat unswollen seed.
Swollen seed rots	Mix too wet, or boiled too long. Keep to 30 s; water less.
Seedlings collapse	Damping off. Sterile mix, good airflow.

Notes

Other notes	Boiling water scalds – use tongs, a strainer and heat-proof containers.
Confidence	**High.** Nursery germination data from 12 tests; method follows WA hard-seed guidance.
Sources	See References: 3, 4, 14, 15

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Daviesia nudiflora

No widely used common name (a bitter-pea)

Family **Fabaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Moderate	Boiling water 30 s	Mar – May	3–5 mm	22%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	OK	Best	Best	Best	OK	OK	OK	Best	OK	–	–

Best window · Workable – slower or less reliable · Avoid / low success

Plant snapshot

Habit	Shrub, 0.3–1.5 m.
Flowering	Jun – Sep (orange-yellow and red-brown pea flowers).
Habitat	Sandy and clayey soils and laterite; low rises and flats.
Local note	Recorded in the Wanneroo LGA.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Not used	Not documented.	–
Seed	Main method	Boiling water 30 s.	This guide
Division	Not used	–	–

Seed collection & storage

Item	Details
Seed collection	Pods ripen in summer and **release seed as soon as they are ripe**. Tie nylon stockings or muslin bags over developing pods and collect once they split.
Extraction & cleaning	Rub pods and sieve or winnow to remove debris.
Seed quality	Hard-seeded: nearly all fresh seed is dormant until the seed coat is treated. Float test: discard seed that floats.
Storage life	**Long.** Hard-coated pea-family seed keeps for many years when dry (related *Acacia* seed has stayed viable 50+ years).
Typical germination	About **22%** after boiling-water treatment (Alcoa nursery average, few tests). About **57 seedlings per gram** – roughly 1.7 g of seed per 100 seedlings.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May** and **Sep**. Feb, Jun – Aug and Oct are workable. Avoid Nov – Jan (heat stress, damping off).
Seed mix	Standard sterile native seed mix.
Pre-treatment	**Boiling water:** drop seed into boiling water for **30 seconds** (Alcoa method), then cool; or pour near-boiling water over seed (at least 10 parts water to 1 part seed) and leave to soak overnight. **Sow seed that has swollen**; re-treat seed that hasn't.
Sowing depth & method	Cover to about twice the seed's thickness, roughly **3–5 mm**. Direct sow 2–3 seeds per tube and thin, or sow in trays.

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Item	Details
After sowing	Water in; keep moist, not wet. Sow soon after treatment – don't let swollen seed dry out.
Germination time	Can start about **one week** after treatment.
Pricking out	At the cotyledon to first true leaf stage – early, before the taproot lengthens.
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 3–5 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Very low germination	Seed coat not broken. Re-treat unswollen seed; try a dry-heat pulse (80–120°C) on the next batch.
Swollen seed rots	Mix too wet or seed damaged by overlong boiling. Keep boiling to 30 s; water less.
Seedlings collapse	Damping off. Sterile mix, good airflow, avoid crowding.

Notes

Other notes	Treat with care: boiling water scalds – use tongs, a strainer and heat-proof containers.
Confidence	**Medium.** Nursery germination data available (small number of tests); method follows WA hard-seed guidance.
Sources	See References: 1, 3, 5, 14, 15, 16

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Daviesia triflora

No widely used common name (a bitter-pea)

Family **Fabaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Moderate	Boiling water 30 s	Mar – May	3–5 mm	26%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	OK	Best	Best	Best	OK	OK	OK	Best	OK	–	–

Best window · Workable – slower or less reliable · Avoid / low success

Plant snapshot

Habit	Broom-like, almost leafless shrub, 0.25–0.6 m.
Flowering	May – Sep (pea flowers).
Habitat	White or grey sand and laterite.
Local note	Recorded in the Wanneroo LGA.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Not used	Not documented.	–
Seed	Main method	Boiling water 30 s.	This guide
Division	Not used	–	–

Seed collection & storage

Item	Details
Seed collection	Pods ripen in late spring–summer and shed seed as soon as ripe. Bag developing pods with fine mesh and collect once they split.
Extraction & cleaning	Rub pods and sieve or winnow.
Seed quality	Hard-seeded. Float test: discard seed that floats.
Storage life	**Long.** Many years when dry.
Typical germination	About **26%** after boiling-water treatment (Alcoa nursery average, 2 tests). About **11 seedlings per gram** – roughly 9 g of seed per 100 seedlings.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May** and **Sep**. Feb, Jun – Aug and Oct are workable. Avoid Nov – Jan.
Seed mix	Standard sterile native seed mix.
Pre-treatment	**Boiling water for 30 seconds** (Alcoa method), or a near-boiling overnight soak. Sow the swollen seed; re-treat the rest.
Sowing depth & method	Cover about **3–5 mm**. Direct sow 2–3 seeds per tube and thin.
After sowing	Water in; keep moist, not wet.
Germination time	Can start about **one week** after treatment.
Pricking out	Early, at the cotyledon to first true leaf stage.

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Item	Details
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 3–5 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Very low germination	Seed coat not broken. Re-treat unswollen seed.
Swollen seed rots	Too wet or over-boiled.
Seedlings collapse	Damping off. Sterile mix, good airflow.

Notes

Other notes	Small, slender plant – grow in deep tubes and avoid overpotting.
Confidence	**Medium.** Nursery germination data available (2 tests); method follows WA hard-seed guidance.
Sources	See References: 3, 4, 14, 15

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Dianella revoluta

Blueberry Lily / Flax Lily

Family **Asphodelaceae (Hemerocallidoideae)**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Difficult	**GA 24 h + smoke water 24 h**	**Mar – Apr**	**3–5 mm**	**<1% untreated**

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	OK	Best	Best	OK	–	–	–	–	–	–	–

Best window · Workable – slower or less reliable · Avoid / low success

Plant snapshot

Habit	Rhizomatous, tufted perennial herb, 0.3–1.5 m.
Flowering	Sep – Dec (blue-purple), followed by blue berries.
Habitat	White or yellow sand, sandy clay, laterite, loam and clay.
Local note	Recorded in Joondalup and Wanneroo (var. *divaricata* is common on the coast).

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Not used	Clumping herb.	–
Seed	Difficult	GA + smoke water; 8–11 weeks.	This guide
Division	Main method	The quickest way to produce plants.	→ Division Guide

Seed collection & storage

Item	Details
Seed collection	Collect ripe, deep-blue berries in summer after flowering.
Extraction & cleaning	Soak berries in water for a few days, then rub and wash off the pulp in a sieve. Dry the shiny black seed before storing.
Seed quality	Seed is dormant (morphophysiological dormancy): untreated seed gave under 1% in Alcoa tests.
Storage life	Not well documented. Store cleaned, dried seed sealed and cool.
Typical germination	Untreated: **under 1%** (Alcoa, 19 tests). Treated seed does much better – record your own results.

Sowing & growing on

Item	Details
Best sowing time	**Mar – Apr** – seed experiences warm conditions, then germinates as the weather cools (Geraldton: not until late April). Feb and May workable.
Seed mix	Standard sterile native seed mix.
Pre-treatment	**Gibberellic acid** (1 part per thousand) for 24 hours, then **smoke water** (1:9) for 24 hours (Geraldton method). Research shows warm stratification followed by smoke also breaks dormancy.
Sowing depth & method	Cover about **3–5 mm**. Sow in trays or direct into tubes.
After sowing	Keep moist and be patient – don't discard trays too early.
Germination time	About **58–75 days** (Geraldton nursery).

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Item	Details
Pricking out	When seedlings have 2–3 leaves and are large enough to handle.
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 5–7 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Nothing after 6 weeks	Normal – germination takes 8–11 weeks. Keep trays moist.
No germination at all	Seed still dormant. Use GA + smoke water, or keep trays for a second season.
Pulp left on seed goes mouldy	Clean seed thoroughly before drying and storing.

Notes

Other notes	Division of established clumps is a quicker way to produce plants.
Confidence	**Medium.** WA community-nursery method and dormancy research available; nursery results vary widely.
Sources	See References: 3, 4, 5, 23

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Diplolaena angustifolia

Yanchep Rose

Family **Rutaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Moderate–Difficult *	Smoke water (trial GA)	Mar – May	About 4 mm	Not recorded

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	Best	Best	Best	OK	OK	OK	OK	OK	–	–

Best window · Workable – slower or less reliable · Avoid / low success

Plant snapshot

Habit	Erect, compact or spreading shrub, 0.3–1.5 m.
Flowering	Jun – Oct (red or orange).
Habitat	White or yellow sand on limestone hills, rocky ridges and dunes.
Local note	Recorded in the Wanneroo LGA (not listed for Joondalup) – it grows from Wanneroo north to near Eneabba. Dislikes humid conditions and wet, boggy soils.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Main method	Firm current season's growth works in a relative (provisional).	→ Cuttings Guide
Seed	Possible	Slow – first seedlings at 4–7 weeks (Kings Park); smoke may help.	This guide
Division	Not used	Shrub – not divided.	–

Seed collection & storage

Item	Details
Seed collection	Not documented. Estimate Oct – Dec, after flowering. Bag a few fruiting heads with fine mesh until you know how quickly seed is shed.
Extraction & cleaning	Dry in paper bags, rub and sieve.
Seed quality	Oblong to bean-shaped seed about 6 mm long in the genus. A seed supplier lists about 85 seeds per gram. Cut-test a few.
Storage life	Not documented. Store dry and cool.
Typical germination	No germination % recorded. Kings Park germinated **13 seed lots**, with first seedlings at 28–50 days (mean 36). Two related species gave 0% in single Alcoa tests. A seed supplier says seed may lie dormant until conditions suit.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May** (estimate). Jun – Oct workable. Avoid Nov – Feb. A seed supplier gives 30–70 days when sown at a soil temperature of 22–26°C, so it may not need cold.
Seed mix	Standard sterile native seed mix, very free-draining.
Pre-treatment	**Smoke water** (24 h soak, 1:9) – suppliers say smoke may lift germination. For the related *D. grandiflora*, Geraldton soaks seed 24 h in gibberellic acid (1 part per thousand), then 24 h in smoke water – worth trialling on part of a batch.

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Item	Details
Sowing depth & method	Sow thinly and cover about **4 mm** (seed supplier).
After sowing	Keep moist, not wet. Keep trays into a second year – Geraldton gets further germination that way with the related species.
Germination time	First seedlings at **28–50 days** (mean 36) in 13 Kings Park lots; a seed supplier gives 30–70 days.
Pricking out	At 2–3 leaves, gently.
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 6–9 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Nothing comes up	Dormant seed. Keep trays a second year; trial smoke plus gibberellic acid.
Seedlings rot	Too wet or humid. Water less; free-draining mix and good airflow.
Seedlings collapse	Damping off. Sterile mix, airflow.

Notes

Other notes	Cuttings are likely to be more reliable – they work in a related species (see the Cuttings Guide).
Confidence	**Low.** No germination % recorded, but Kings Park germinated 13 seed lots. Handling is based on seed-supplier notes and a related species' method – treat as a trial.
Sources	See References: 3, 4, 5, 143, 154, 155, 158

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Dodonaea aptera

Coast Hop-bush

Family **Sapindaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Moderate	Hot water	Mar – May	Lightly covered	47%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	Best	Best	Best	OK	OK	OK	OK	OK	–	–

Best window | Workable – slower or less reliable | Avoid / low success

Plant snapshot

Habit	Erect to spreading shrub, 0.5–3.5 m. Male and female flowers are on separate plants.
Flowering	Mainly Apr – Jul (green).
Habitat	Sand in coastal limestone areas.
Local note	Recorded in the Wanneroo LGA (not listed for Joondalup). Grows along the coast from Shark Bay to Cape Freycinet.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Possible	Firm young growth, 2,000 ppm IBA in the genus; keeps the parent's sex.	–
Seed	Main method	Hot water; about 47% (Alcoa). Needs male plants nearby for seed.	This guide
Division	Not used	Shrub – not divided.	–

Seed collection & storage

Item	Details
Seed collection	Collect the four-angled capsules (5–6 mm, without wings) as they ripen in **early summer** and turn red-brown, with hard black seed inside. Only female plants fruit, and capsules may hold no viable seed unless a male plant grows nearby.
Extraction & cleaning	Dry capsules in paper bags for 1–2 weeks, then rub and sieve (related species).
Seed quality	The seed coat is waterproof (physical dormancy), so seed won't take up water until treated. Cut-test a few – capsules from isolated female plants are often empty.
Storage life	Not documented. Hard-coated seed is usually long-lived when kept dry and cool.
Typical germination	About **47%** after boiling water (Alcoa, 4 tests), about **66 seedlings per gram**. In a CSIRO trial with hop-bushes, boiling water gave roughly 80–90% (genus).

Sowing & growing on

Item	Details
Best sowing time	**Mar – May** (estimate). In WA research a related hop-bush (*D. petiolaris*) germinated about equally well from 15 to 25°C. Jun – Oct workable. Avoid Nov – Feb.
Seed mix	Standard sterile native seed mix.
Pre-treatment	**Hot water** – Alcoa dipped seed in boiling water for 30 seconds, and WA researchers used a 30-second hot-water dip on this species too. ANPSA's method for the genus is to pour boiling water over the seed and leave it to soak overnight. Sow the seeds that have swollen and re-treat any still hard. For a related species, Geraldton follows hot water with a 24 h soak in smoke water (1:9).

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Item	Details
Sowing depth & method	Cover lightly – about the depth of the seed.
After sowing	Keep moist, not wet.
Germination time	First seedlings at **14–37 days** (mean 22) in 6 Kings Park seed lots.
Pricking out	When the second pair of leaves appears.
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 4–6 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
No germination	Seed coat not broken. Re-treat with hot water; check seeds swell before sowing.
Empty seed	No male plant nearby. Collect from female plants in mixed stands of both sexes.
Seedlings collapse	Damping off. Thin sowing, sterile mix, airflow.

Notes

Other notes	Cuttings are possible in the genus – firm young growth, dipped for a few seconds in 2,000 ppm IBA (ANPSA). Cuttings keep the parent's sex, so grow both males and females if you want seed.
Confidence	**Medium.** Nursery data (4 tests, boiling water), Kings Park emergence records and WA research on the seed coat; some details from related hop-bushes.
Sources	See References: 3, 4, 5, 140, 141, 142, 143, 159

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Eremaea pauciflora

Orange-flowered Eremaea

Family **Myrtaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Easy–Moderate *	None	Mar – May	Surface – light cover	Not recorded

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	OK	Best	Best	OK	OK	OK	OK	Best	Best	OK	–

Best window · Workable – slower or less reliable · Avoid / low success

Plant snapshot

Habit	Erect to spreading shrub, to 4 m (var. *pauciflora* to about 2 m).
Flowering	Jul – Dec or Jan (orange, red or yellow).
Habitat	Sand, loam, gravel, laterite, limestone or granite; sandplains, ridges and moist depressions.
Local note	Recorded in the Wanneroo LGA (var. *pauciflora*). Label seed by variety.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Possible	Easy for the related E. beaufortioides – worth trialling.	–
Seed	Main method	No pre-treatment (related species).	This guide
Division	Not used	Shrub – not divided.	–

Seed collection & storage

Item	Details
Seed collection	Capsules sit on old stems and are only weakly serotinous. Collect mature, closed capsules from older wood.
Extraction & cleaning	Keep capsules warm and dry until they release the fine seed; sow with the chaff.
Seed quality	A seed supplier lists about 75 seeds per gram with chaff.
Storage life	Not well documented. Store dry and cool.
Typical germination	Not recorded.

Sowing & growing on

Item	Details
Best sowing time	**Mar – Apr** and **Sep – Oct**. Feb, May – Aug and Nov workable.
Seed mix	Standard sterile native seed mix.
Pre-treatment	**None needed** (as for the related *E. beaufortioides*).
Sowing depth & method	Scatter seed and chaff on the surface; cover very lightly.
After sowing	Water in with a fine spray; keep moist.
Germination time	Not recorded – check from 2 weeks.
Pricking out	When seedlings have two pairs of true leaves.

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Item	Details
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 4–6 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Seed washed into clumps	Bottom water or use a fine mist.
Damping off	Sow thinly; good airflow.
Few seedlings	Old or empty capsules. Collect from several plants.

Notes

Other notes	Cuttings are also reported to be easy for the related *E. beaufortioides* – worth trialling.
Confidence	**Low.** No species data. A related Eremaea grows easily from untreated seed – treat as a starting point.
Sources	See References: 4, 43, 93, 100

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Eryngium pinnatifidum

Blue Devils

Family **Apiaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Moderate *	None	Mar – May	Lightly covered	3%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	Best	Best	Best	OK	OK	OK	–	–	–	–

Best window | Workable – slower or less reliable | Avoid / low success

Plant snapshot

Habit	Perennial herb, 0.02–0.5 m.
Flowering	Aug – Nov (blue or white, in prickly heads).
Habitat	Clay, sandy clay, and grey or black peaty sand; winter-wet areas, claypans and sandy rises.
Local note	Recorded in the Wanneroo LGA on FloraBase, and listed as indigenous at Warwick Bushland, Joondalup. **Check the subspecies:** two informal subspecies are Priority flora (subsp. Palustre G.J.Keighery 13459, Priority Three; subsp. Umbraphilum G.J.Keighery 13967, Priority Two). Confirm what your source plants are, check the current DBCA list, and collect only under the right DBCA licence.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Not used	Not documented for Australian Eryngium.	–
Seed	Main method	No pre-treatment; 19–28 days (Kings Park). Check the subspecies.	This guide
Division	Not used	Not documented – grow from seed.	–

Seed collection & storage

Item	Details
Seed collection	Collect heads as they dry and turn straw-coloured (related species) – estimate Dec – Feb, after flowering. Wear gloves.
Extraction & cleaning	Rub dry heads with a rubber bung rather than your hands, sieve, and remove the lighter, empty fruit with an aspirator or by winnowing (related species). Each fruit splits into two halves.
Seed quality	Expect many empty fruits – cut-test a sample.
Storage life	Not documented. Store dry and cool.
Typical germination	Only one nursery test: about **3%** untreated (Alcoa), about 10 seedlings per gram. Kings Park raised it from autumn sowings with no special treatment, and a related species is non-dormant (88% in SA).

Sowing & growing on

Item	Details
Best sowing time	**Mar – May** (Kings Park sowed in autumn). Jun – Aug workable. Avoid Sep – Feb.
Seed mix	Standard sterile native seed mix. Keep it evenly moist – it grows naturally in winter-wet sites.
Pre-treatment	**None** – Kings Park used no pre-treatment.
Sowing depth & method	Sow thinly and cover lightly.
After sowing	Keep moist through the cool months.

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Item	Details
Germination time	First seedlings after about **19–28 days** (Kings Park, 7 seed lots).
Pricking out	At 2–3 leaves.
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 4–6 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Very few seedlings	Many empty fruits. Cut-test before sowing and sow generously.
Seedlings dry out	Keep trays evenly moist – it is a winter-wet plant.
Seedlings collapse	Damping off. Sterile mix, airflow.

Notes

Other notes	Cuttings and division are not documented for Australian *Eryngium* – grow from seed.
Confidence	**Low.** Only one nursery test (3%) plus Kings Park emergence records; handling follows a related species.
Sources	See References: 3, 4, 143, 144, 145, 149

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Eucalyptus decipiens

Redheart / Redheart Moit

Family **Myrtaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Easy	None	Mar – Apr, Sep – Oct	1–2 mm	46%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	OK	Best	Best	OK	OK	OK	OK	Best	Best	OK	–

Best window · Workable – slower or less reliable · Avoid / low success

Plant snapshot

Habit	Mallee or tree, 1.5–15 m, with rough, flaky or ribbony bark. Forms a lignotuber.
Flowering	Aug – Dec or Jan (white).
Habitat	White, yellow or grey sand, sandy clay, gravelly loam and laterite; sandplains, hills, swamp margins.
Local note	Recorded in Joondalup and Wanneroo.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Not used	Eucalypts are very difficult from cuttings.	–
Seed	Main method	No pre-treatment; about 46% germination.	This guide
Division	Not used	Tree.	–

Seed collection & storage

Item	Details
Seed collection	Collect unopened capsules once the valves look as if they are beginning to open. Ripe seed has a white, firm inside and a brown to black coat.
Extraction & cleaning	Keep capsules warm and dry for a day to several weeks until they release the seed; sieve out large chaff.
Seed quality	Seed comes mixed with chaff – that's normal. Cut-test a sample.
Storage life	**Long** if stored cool and dry.
Typical germination	About **46%** (untreated, Alcoa nursery average, 7 tests). About **481 seedlings per gram** of seed-and-chaff – roughly 0.2 g per 100 seedlings.

Sowing & growing on

Item	Details
Best sowing time	**Mar – Apr** and **Sep – Oct**. Feb, May – Aug and Nov workable. Avoid Dec – Jan.
Seed mix	Standard sterile native seed mix.
Pre-treatment	**None needed.** Surface-sterilising in dilute bleach can help (WA eucalypt seed notes).
Sowing depth & method	Mix the fine seed with dry sand, sow thinly and cover lightly, about **1–2 mm**. Or sow a pinch per tube and thin.
After sowing	Water in (smoke water optional); keep moist.
Germination time	About **1–2 weeks** (typical for WA eucalypts).

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Item	Details
Pricking out	When seedlings have two pairs of true leaves.
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 3–5 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Seedlings collapse	Damping off. Sow thinly; sterile mix; avoid crowding.
Trays too crowded	Seed is very fine. Dilute it with dry sand before sowing.
Roots circling	Held too long in tubes.

Notes

Other notes	Cuttings are very difficult for eucalypts – seed is the method.
Confidence	**High.** Nursery germination data (7 tests) and WA eucalypt seed notes available.
Sources	See References: 3, 4, 5, 19

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Eucalyptus gomphocephala

Tuart

Family **Myrtaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Easy	None	Mar – Apr, Sep – Oct	1–2 mm	42%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	OK	Best	Best	OK	OK	OK	OK	Best	Best	OK	–

Best window · Workable – slower or less reliable · Avoid / low success

Plant snapshot

Habit	Tree, 10–40 m.
Flowering	Jan – Apr (white).
Habitat	Sand over limestone on the coastal plain.
Local note	Recorded in Joondalup and Wanneroo.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Not used	Eucalypts are very difficult from cuttings.	–
Seed	Main method	No pre-treatment; 1–2 weeks.	This guide
Division	Not used	Tree.	–

Seed collection & storage

Item	Details
Seed collection	Fruit takes about a year to mature. Collect brown capsules when the valves are just starting to open; many trees hold seed for a year.
Extraction & cleaning	Keep capsules warm and dry from a day to several weeks until seed drops. Seed is mixed with chaff.
Seed quality	A cut-test and count on seed and chaff estimates seeds per gram.
Storage life	**Long** if stored cool and dry.
Typical germination	About **42%** (untreated, Alcoa nursery average). About **228 seedlings per gram** – under 0.5 g per 100 seedlings.

Sowing & growing on

Item	Details
Best sowing time	**Mar – Apr** and **Sep – Oct**. Feb, May – Aug and Nov workable. Avoid Dec – Jan.
Seed mix	Standard sterile native seed mix.
Pre-treatment	**None needed.** Surface-sterilising in dilute bleach (10%) can help.
Sowing depth & method	Mix with dry sand and sprinkle thinly; cover very lightly (about **1–2 mm**).
After sowing	Water in with a fine spray (smoke water optional); keep moist.
Germination time	About **1–2 weeks**.
Pricking out	When seedlings have two pairs of true leaves.

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Item	Details
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 3–5 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Seedlings collapse	Damping off. Thin sowing, sterile mix, airflow.
Uneven germination	Chaff clumps. Mix seed well with sand.
Roots circling	Held too long in tubes. Time sowings to planting.

Notes

Other notes	Local coastal-plain provenance matters for revegetation – record seed source.
Confidence	**High.** Nursery data (16 tests) and WA eucalypt seed notes available.
Sources	See References: 3, 4, 5, 19

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Eucalyptus marginata

Jarrah

Family **Myrtaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Easy	None	Mar – Apr, Sep – Oct	2–3 mm	31%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	OK	Best	Best	OK	OK	OK	OK	Best	Best	OK	–

Best window · Workable – slower or less reliable · Avoid / low success

Plant snapshot

Habit	Tree to 40 m.
Flowering	Jun – Dec or Jan (white-cream to pink).
Habitat	Grey sand, clay or sandy loam and laterite; hills and rises.
Local note	Recorded in Joondalup and Wanneroo.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Not used	Eucalypts are very difficult from cuttings.	–
Seed	Main method	No pre-treatment; 1–2 weeks.	This guide
Division	Not used	Tree.	–

Seed collection & storage

Item	Details
Seed collection	Fruit takes about a year to mature. Collect brown capsules when the valves look ready to open.
Extraction & cleaning	Keep capsules warm and dry until seed is released; sieve out large debris.
Seed quality	Cut-test a sample; seed is mixed with chaff.
Storage life	**Long** if stored cool and dry.
Typical germination	About **31%** (untreated, Alcoa nursery average). About **33 seedlings per gram** – roughly 3 g of seed per 100 seedlings.

Sowing & growing on

Item	Details
Best sowing time	**Mar – Apr** and **Sep – Oct**. Feb, May – Aug and Nov workable. Avoid Dec – Jan.
Seed mix	Standard sterile native seed mix.
Pre-treatment	**None needed.** Surface-sterilising in dilute bleach (10%) can help.
Sowing depth & method	Cover lightly, about **2–3 mm**. Direct sow a few seeds per tube and thin, or sow thinly in trays.
After sowing	Water in; keep moist.
Germination time	About **1–2 weeks**.
Pricking out	When seedlings have two pairs of true leaves.

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Item	Details
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 3–5 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Seedlings collapse	Damping off. Sterile mix, avoid crowding.
Low germination	Old or immature seed. Collect mature capsules; cut-test.
Roots circling	Held too long in tubes.

Notes

Other notes	Jarrah seedlings develop a lignotuber early – a good sign of healthy stock.
Confidence	**High.** Large nursery data set (67 tests) and WA eucalypt seed notes available.
Sources	See References: 3, 4, 5, 19

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Exocarpos sparteus

Broom Ballart

Family **Santalaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Difficult *	Gibberellic acid 48 h	Apr – Jun	About 5 mm	36% with GA (SA trial)

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	OK	Best	Best	Best	OK	–	–	–	–	–

Best window · Workable – slower or less reliable · Avoid / low success

Plant snapshot

Habit	Weeping, broom-like shrub, 1–4 m. A **root hemiparasite** – it taps the roots of nearby plants.
Flowering	Feb – Oct (tiny, yellow-green), followed by a small nut on a swollen pink-red stalk.
Habitat	White sand over limestone, red sand and laterite; coastal and inland dunes, sandplains.
Local note	Recorded in Joondalup and Wanneroo. Hosts in the wild include sheoaks, wattles, myrtles and proteas.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Not used	Not documented.	–
Seed	Difficult	GA gave 36% in one trial; needs a host plant.	This guide
Division	Not used	Root hemiparasite.	–

Seed collection & storage

Item	Details
Seed collection	Collect ripe fruits; the edible red stalk falls away. Keep the **shiny brown nuts filled with white endosperm** – cut some open to check. Viable seed has even been collected from emu droppings.
Extraction & cleaning	Remove the fleshy stalk and any pulp; dry the nuts.
Seed quality	Often poor – only 14% of one SA seed lot was viable (about 45 seeds per gram).
Storage life	Not well documented. Store dry and cool; sow within the year.
Typical germination	**36%** after 48 h in gibberellic acid (GA, 400 mg/L) in an SA trial; leaching alone gave 0% and GA plus smoke 18%. Alcoa's single untreated test gave 0%.

Sowing & growing on

Item	Details
Best sowing time	**Apr – Jun** – seed germinates in winter. Mar and Jul workable.
Seed mix	Free-draining, sterile mix.
Pre-treatment	**Soak in gibberellic acid (400 mg/L) for 48 hours** (SA Seed Conservation Centre trial).
Sowing depth & method	Cover about 5 mm. Sow in deep tubes – ideally with a host seedling already growing in the tube.
After sowing	Keep moist, not wet, in a cool spot over winter.
Germination time	Slow; expect winter emergence and record the timing.
Pricking out	Avoid – sow direct into deep tubes with a host.

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Item	Details
Shade house & hardening	Grow on with its host in light shade, then harden off gradually.
Time to planting size	Slow and unpredictable – record your own.

Common problems

Problem	Likely cause and fix
No germination	Empty seed or no GA. Cut-test; use GA; sow in winter.
Seedlings stall and die	No host. Pot with a host plant (e.g. a small wattle or sheoak) early.
Low fill	Normal for this species – collect plenty.

Notes

Other notes	Growers find establishment difficult because of its semi-parasitic habit. Plant out next to suitable hosts.
Confidence	**Medium.** One published seed trial (SA, 2016) found gibberellic acid gave 36%. Root hemiparasite – needs a host to grow on.
Sources	See References: 3, 4, 66

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Ficinia nodosa

Knotted Club Rush

Family
Cyperaceae (sedges)

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Easy	None; smoke water to water in	Mar – May; Sep – Oct	Surface	55%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	OK	Best	Best	Best	OK	OK	OK	Best	Best	OK	–

Best window | Workable – slower or less reliable | Avoid / low success

Plant snapshot

Habit	Erect, clumping, rhizomatous sedge to 1 m high and 0.8 m wide, with round, rush-like stems.
Flowering	Oct – Dec or Jan (brown knot-like heads).
Habitat	Calcareous sand and limestone; coastal dunes, flats, seasonally wet swamps and salt-lake shores.
Local note	Recorded in Joondalup and Wanneroo. Salt tolerant and fast growing – an excellent sand binder.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Not used	Sedge.	–
Seed	Main method	No pre-treatment; about 55% germination.	This guide
Division	Main method	Divides readily; plantable in a few months.	→ Division Guide

Seed collection & storage

Item	Details
Seed collection	Hand-collect the dry brown seed heads in late summer–autumn.
Extraction & cleaning	Little cleaning needed – rub heads to free the tiny seed.
Seed quality	Very fine seed – about 2,300 seedlings per gram in nursery tests.
Storage life	Not well documented. Store dry and cool.
Typical germination	About **55%** (untreated, Alcoa nursery average, 7 tests). A pinch goes a long way – under 0.1 g of seed for 100 seedlings.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May** and **Sep – Oct**. Feb, Jun – Aug and Nov workable. Avoid Dec – Jan.
Seed mix	Standard sterile native seed mix; stays a little moister than most.
Pre-treatment	**None needed.** Water trays in with smoke water (Geraldton method).
Sowing depth & method	**Surface-sow:** dilute the seed with sterile yellow sand and shake it from a spice jar (Geraldton method).
After sowing	Water in with smoke water; keep moist.
Germination time	About **15–19 days** (Geraldton nursery).
Pricking out	Prick out small clumps of seedlings together once 2–3 cm tall.

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Item	Details
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 3–4 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Trays too crowded	Seed is very fine – dilute with sand before sowing.
Seedlings dry out	Tiny seedlings dry quickly. Bottom water or mist often.
Damping off	Sow thinly; airflow.

Notes

Other notes	Also divides easily – the quickest way to multiply existing plants (see the Division Guide).
Confidence	**High.** Nursery data from 7 tests plus a WA community-nursery method.
Sources	See References: 3, 4, 5, 79

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Frankenia pauciflora

Seaheath

Family **Frankeniaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Easy *	None	Apr – Jun	Surface – light cover	96% (SA seed bank)

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	OK	Best	Best	Best	OK	OK	OK	–	–	–

Best window · Workable – slower or less reliable · Avoid / low success

Plant snapshot

Habit	Prostrate to ascending small shrub, 0.08–0.5 m, with tiny heath-like leaves.
Flowering	Jun – Dec or Jan – Feb (pink or white).
Habitat	Sandy, often saline soils; coastal limestone, saline flats and salt lakes.
Local note	Recorded in the City of Joondalup. Very salt tolerant (the genus has salt-excreting leaf glands).

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Trial	A related Frankenia is grown from cuttings; untested.	–
Seed	Main method	Non-dormant; 96% in SA seed-bank tests.	This guide
Division	Not used	Not recorded.	–

Seed collection & storage

Item	Details
Seed collection	Collect the brown, cylindrical fruiting heads that are slightly swollen at the base.
Extraction & cleaning	Dry heads for about a week, then rub and sieve.
Seed quality	Tiny seed – cut-test under a hand lens.
Storage life	Store with a desiccant in an airtight container, cool and dry.
Typical germination	Seed is **non-dormant**: South Australian seed germinated **96%** without treatment in winter conditions. No WA nursery data.

Sowing & growing on

Item	Details
Best sowing time	**Apr – Jun** (cool). Mar and Jul – Sep workable.
Seed mix	Free-draining, sandy seed mix.
Pre-treatment	**None needed.**
Sowing depth & method	Mix the tiny seed with dry sand and surface-sow; cover very lightly.
After sowing	Keep moist, not wet.
Germination time	First seedlings from about **7 days**; half up by about 14 days (SA).
Pricking out	Prick out small clumps once seedlings can be handled.

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Item	Details
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 3–4 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Seedlings rot	Too wet. Sandy mix; water less.
Trays too crowded	Seed is tiny – dilute with sand.
Weak growth	Too much shade. Full sun once established.

Notes

Other notes	A related Frankenia is grown from cuttings – worth trialling for this species.
Confidence	**Medium.** Seed data are from South Australian material (var. gunnii); WA plants may differ.
Sources	See References: 4, 88, 101

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Gastrolobium capitatum

Bacon and Eggs

Family **Fabaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Moderate *	Hot water or dry heat	Mar – May	2–3 mm	Not recorded

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	OK	Best	Best	Best	OK	OK	OK	Best	OK	–	–

Best window | Workable – slower or less reliable | Avoid / low success

Plant snapshot

Habit	Prostrate to low, bushy shrub, to 1 m.
Flowering	Jun – Sep (orange-yellow).
Habitat	Sandy to loamy soils, laterite, granite; slopes, outcrops, swampy areas and plains.
Local note	Recorded in the Wanneroo LGA. Many *Gastrolobium* (poison peas) contain fluoroacetate, which is toxic to stock and people. No data were found for this species – wear gloves and keep seed away from pets and stock.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Not used	Not documented.	–
Seed	Main method	Hot water or dry heat (trial).	This guide
Division	Not used	–	–

Seed collection & storage

Item	Details
Seed collection	Seed is usually shed in summer. Tie nylon stockings or muslin bags over developing pods and collect once they split.
Extraction & cleaning	Rub pods and sieve or winnow.
Seed quality	Hard-seeded. Float test: discard seed that floats. A seed supplier lists about 250 seeds per gram.
Storage life	**Long.** Hard-coated pea seed keeps many years when dry.
Typical germination	Not recorded for this species.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May** and **Sep**. Feb, Jun – Aug and Oct are workable. Avoid Nov – Jan.
Seed mix	Standard sterile native seed mix.
Pre-treatment	Choose one (WA *Gastrolobium* seed notes): • **Hot water:** soak for up to 24 hours in warm to hot water (pour near-boiling water over and let it cool). • **Dry heat:** 80–120°C in an oven for 10 minutes to 1 hour. • **Nick** or file a small piece of the seed coat. Sow the seed that has swollen.
Sowing depth & method	Cover about **2–3 mm**. Direct sow 2–3 seeds per tube and thin.

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Item	Details
After sowing	Water in; keep moist, not wet.
Germination time	Not recorded – check from about one week after treatment.
Pricking out	Early, at the cotyledon to first true leaf stage.
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 3–5 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Low germination	Seed coat not broken. Re-treat, or try the dry-heat method.
Swollen seed rots	Mix too wet. Water less.
Seedlings collapse	Damping off. Sterile mix, good airflow.

Notes

Other notes	Treat as a trial and record germination so this page can be updated.
Confidence	**Medium.** WA genus seed notes available; no nursery germination data for this species.
Sources	See References: 4, 14, 43, 49

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Gastrolobium nervosum

No widely used common name

Family **Fabaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Moderate *	Hot water or dry heat	Mar – May	2–3 mm	Not recorded

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	OK	Best	Best	Best	OK	OK	OK	Best	OK	–	–

Best window | Workable – slower or less reliable | Avoid / low success

Plant snapshot

Habit	Small shrub, to 0.5 m.
Flowering	Aug – Oct (yellow, red and purple).
Habitat	Yellow, white or grey sand, sandy clay, laterite gravel and limestone; coastal plains and sandplains.
Local note	Recorded in Joondalup and Wanneroo. Many *Gastrolobium* (poison peas) contain fluoroacetate, which is toxic to stock and people. No data were found for this species – wear gloves and keep seed away from pets and stock.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Not used	Not documented.	–
Seed	Main method	Hot water or dry heat (trial).	This guide
Division	Not used	–	–

Seed collection & storage

Item	Details
Seed collection	Seed is usually shed in summer. Tie nylon stockings or muslin bags over developing pods and collect once they split.
Extraction & cleaning	Rub pods and sieve or winnow.
Seed quality	Hard-seeded. Float test: discard seed that floats. A seed supplier lists about 175 seeds per gram.
Storage life	**Long.** Hard-coated pea seed keeps many years when dry.
Typical germination	Not recorded for this species.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May** and **Sep**. Feb, Jun – Aug and Oct are workable. Avoid Nov – Jan.
Seed mix	Standard sterile native seed mix.
Pre-treatment	Choose one (WA *Gastrolobium* seed notes): • **Hot water:** soak for up to 24 hours in warm to hot water. • **Dry heat:** 80–120°C in an oven for 10 minutes to 1 hour. • **Nick** or file a small piece of the seed coat. Sow the seed that has swollen.
Sowing depth & method	Cover about **2–3 mm**. Direct sow 2–3 seeds per tube and thin.

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Item	Details
After sowing	Water in; keep moist, not wet.
Germination time	Not recorded – check from about one week after treatment.
Pricking out	Early, at the cotyledon to first true leaf stage.
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 3–5 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Low germination	Seed coat not broken. Re-treat, or try the dry-heat method.
Swollen seed rots	Mix too wet. Water less.
Seedlings collapse	Damping off. Sterile mix, good airflow.

Notes

Other notes	Treat as a trial and record germination so this page can be updated.
Confidence	**Medium.** WA genus seed notes available; no nursery germination data for this species.
Sources	See References: 4, 14, 43

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Gompholobium tomentosum

Hairy Yellow Pea

Family **Fabaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Easy–Moderate	Dry heat or hot water	Mar – May	3–4 mm	34%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	OK	Best	Best	Best	OK	OK	OK	Best	OK	–	–

Best window · Workable – slower or less reliable · Avoid / low success

Plant snapshot

Habit	Erect, often spindly shrub, 0.15–1 m.
Flowering	Jul – Jan (yellow pea flowers).
Habitat	White, grey or yellow sand, sandy clay and lateritic sand; coastal limestone, dunes and plains.
Local note	Recorded in Joondalup and Wanneroo.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Not used	Not documented.	–
Seed	Main method	Dry heat or hot water.	This guide
Division	Not used	–	–

Seed collection & storage

Item	Details
Seed collection	Pods ripen in summer and shed seed when ripe. **Bag developing pods** with stockings or muslin and collect once split.
Extraction & cleaning	Rub and sieve out the small seed.
Seed quality	Hard-seeded – almost all fresh seed is dormant until the coat is treated.
Storage life	**Long.** Keeps for many years when dry (typical of hard-coated pea-family seed).
Typical germination	About **34%** after heat treatment (Alcoa nursery average). About **125 seedlings per gram** – roughly 0.8 g of seed per 100 seedlings.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May** and **Sep**. Feb, Jun – Aug and Oct workable. Avoid Nov – Jan.
Seed mix	Standard sterile native seed mix.
Pre-treatment	Either: • **Dry heat shock:** oven at 80–120°C for 10 min to 1 hour (Alcoa used a dry-heat pulse for this species); or • **Hot water:** pour near-boiling water over the seed and let it cool in the water (easiest method). Sow seed that has swollen; re-treat the rest.
Sowing depth & method	Small seed – cover about **3–4 mm**. Direct sow a few seeds per tube and thin, or sow thinly in trays.
After sowing	Water in; keep moist, not wet. Sow promptly after hot-water treatment.

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Item	Details
Germination time	Can start about **one week** after treatment.
Pricking out	At the cotyledon to first true leaf stage, before the taproot lengthens.
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 3–5 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Low germination	Seed coat not broken. Re-treat unswollen seed or switch between heat and hot-water methods.
Seed rots after sowing	Too wet. Water less and improve drainage.
Seedlings collapse	Damping off. Sterile mix, airflow, avoid crowding.

Notes

Other notes	Spindly habit – tip-pruning young plants can give a fuller tube.
Confidence	**High.** Nursery germination data and WA hard-seed guidance available for this species.
Sources	See References: 1, 3, 5, 14, 15, 16

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Grevillea vestita

No widely accepted common name

Family **Proteaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Moderate	Nick seed coat; smoke water	Mar – May	3–5 mm	Not recorded

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	Best	Best	Best	OK	OK	OK	–	–	–	–

Best window | Workable – slower or less reliable | Avoid / low success

Plant snapshot

Habit	Erect, spreading, prickly shrub, usually 0.5–3 m.
Flowering	Mainly Jun – Nov (white to cream, often pink-tinged, scented).
Habitat	Sandy soils over limestone or granite on the Swan Coastal Plain.
Local note	Two subspecies exist – label seed by subspecies and provenance. Prickly – wear gloves.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Main method	Firm current season's growth strikes reliably.	→ Cuttings Guide
Seed	Good	Grows reasonably easily from seed; seedlings vary from the parent.	This guide
Division	Possible	Can sucker – lift rooted suckers.	–

Seed collection & storage

Item	Details
Seed collection	Follicles split and drop seed soon after ripening. **Tie fine mesh or stocking bags over maturing follicles** and collect once they open.
Extraction & cleaning	Seed usually falls into the bag; remove follicle debris.
Seed quality	Cut-test a few seeds – good seed is firm and white inside.
Storage life	Not well documented. Store dry, sealed and cool.
Typical germination	Not recorded. Untreated seed can take 1–2 years to germinate.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May** (cool autumn). Jun – Aug workable. Avoid Sep – Feb.
Seed mix	Free-draining, sterile, **low-phosphorus** mix.
Pre-treatment	**Nick the seed coat** with a scalpel (or peel it off after a short soak), then soak in smoke water for 24 hours. Gibberellic acid (GA) can also help (WA *Grevillea* seed notes).
Sowing depth & method	Cover about **3–5 mm**. Direct sow into tubes – seedlings have a taproot.
After sowing	Keep moist, not wet; strict hygiene.
Germination time	A few weeks for treated seed; untreated seed can take **1–2 years**.
Pricking out	If tray-sown, prick out early at the cotyledon stage.

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Item	Details
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 4–6 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
No germination after months	Seed coat not treated. Nick or peel, then smoke water.
Seedlings yellow or die	Phosphorus toxicity or root rot. Low-P mix and feed; good hygiene.
Seedlings differ from parent	Normal for seed. Use cuttings to keep a particular form.

Notes

Other notes	Cuttings are the main method and keep the parent's form (see the Cuttings Guide).
Confidence	**Medium.** ANPSA says it grows reasonably easily from seed; treatment follows WA *Grevillea* seed notes.
Sources	See References: 4, 34, 35

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Guichenotia ledifolia

No widely used common name

Family **Malvaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Moderate–Difficult	**Boiling water over seed; smoke**	**Mar – May**	**2–3 mm**	**Variable**

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	Best	Best	Best	OK	OK	OK	Best	OK	–	–

Best window | Workable – slower or less reliable | Avoid / low success

Plant snapshot

Habit	Spreading shrub, 0.4–1.5 (–2) m.
Flowering	Jul – Oct (blue-purple or pink).
Habitat	Sandy or rocky clay; coastal limestone, sandplains and granite rocks.
Local note	Recorded in Joondalup and Wanneroo. Tolerates drought and alkaline (limestone) soils.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Main method	Often grown from cuttings; semi-hardwood.	→ Cuttings Guide
Seed	Good	Hard coat: boiling water over seed; variable.	**This guide**
Division	Not used	Shrub – not divided.	–

Seed collection & storage

Item	Details
Seed collection	Fruit develops in late spring–summer. Collect the **swollen fruits once the calyx turns papery**; each capsule holds up to 15 seeds.
Extraction & cleaning	Dry in paper bags and sieve out the seed.
Seed quality	Hard seed coat (physical dormancy). A seed supplier lists about 550 seeds per gram.
Storage life	Not documented; hard-coated seed is usually long-lived when dry.
Typical germination	Variable, as reported in WA seed notes. No nursery figures.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May** and **Sep**. Jun – Aug and Oct workable.
Seed mix	Standard sterile native seed mix.
Pre-treatment	Break the hard coat: **pour boiling water over the sown seed** (Geraldton method), or soak in water just off the boil for 24 hours, or chip the seed coat. Water in with smoke water.
Sowing depth & method	Cover about **2–3 mm**.
After sowing	Keep moist, not wet.
Germination time	**30+ days** (Geraldton nursery).
Pricking out	When the second pair of leaves appears.

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Item	Details
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 4–5 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Low, patchy germination	Seed coat not broken. Try chipping, or the 24 h hot-water soak.
Seed killed	Water too hot for too long. Use water just off the boil.
Damping off	Sow thinly; airflow.

Notes

Other notes	Seed is rarely available commercially – it is usually grown from cuttings (see the Cuttings Guide).
Confidence	**Medium.** WA seed notes and a WA community-nursery method available; germination is reported to be variable.
Sources	See References: 4, 5, 43, 92, 99

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Haemodorum laxum

Bloodroot

Family **Haemodoraceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Moderate	None (smoke optional)	Mar – May	Lightly covered	73%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	Best	Best	Best	OK	OK	OK	–	–	–	–

Best window | Workable – slower or less reliable | Avoid / low success

Plant snapshot

Habit	Bulbous perennial herb with a red underground bulb; flowering stems 0.6–1.4 m.
Flowering	Oct – Nov (black, brown or green-brown).
Habitat	Grey or yellow sand, clay, gravel and laterite; dry or seasonally damp sites.
Local note	Recorded in Joondalup and Wanneroo. Rarely grown in nurseries – germinates well, but the bulb is slow to develop.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Not used	Not grown from cuttings.	–
Seed	Main method	No pre-treatment; about 73% (Alcoa). Slow to form a bulb.	This guide
Division	Not used	Not documented – grow from seed.	–

Seed collection & storage

Item	Details
Seed collection	Collect the flowering stems as the capsules dry and brown, in early summer after the Oct – Nov flowering (estimate). Capsules dry hard and split, and the small winged seeds are shaken out – don't wait too long.
Extraction & cleaning	Dry in paper bags, shake out the seed and sieve off the capsule pieces.
Seed quality	Flat, black, disc-shaped seed with a thin wing (genus). Cut-test a few.
Storage life	Stores dry. The Alcoa nursery tests used seed that was usually 1–2 years old.
Typical germination	About **73%** untreated (Alcoa nursery average, 33 tests) – high for a WA monocot. About **150 seedlings per gram**.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May**. Jun – Aug workable. Avoid Sep – Feb – the nursery tests were run cool, at about 15–20°C.
Seed mix	Standard sterile native seed mix, free-draining.
Pre-treatment	**None needed.** Smoke water is optional – *Haemodorum* is listed as a smoke-responsive genus.
Sowing depth & method	Sow thinly and cover lightly.
After sowing	Keep moist, not wet, through the cool months.

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Item	Details
Germination time	Most within about **6 weeks** – the nursery tests were read after 40 days.
Pricking out	Early and gently, at 1–2 leaves, so the developing bulb isn't damaged.
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Slow – a related species takes a couple of seasons to form a decent bulb, and is summer-dormant. Record your own.

Common problems

Problem	Likely cause and fix
Seedlings die back in summer	Probably normal summer dormancy (related species). Keep the pots, water sparingly and wait for autumn.
Bulbs rot	Too wet while dormant. Free-draining mix; water less over summer.
Seedlings collapse	Damping off. Sterile mix, airflow.

Notes

Other notes	Not divided – grow from seed. The red bulbs of *Haemodorum* were eaten and used as a dye, and survive summer and autumn fires.
Confidence	**Medium.** Strong nursery germination data (33 tests); growing-on notes come from related species.
Sources	See References: 3, 4, 72, 110, 111, 112

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Hakea lissocarpha

Honey Bush

Family **Proteaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Easy	None	Mar – May	3–5 mm	66%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	Best	Best	Best	OK	OK	OK	OK	–	–	–

Best window | Workable – slower or less reliable | Avoid / low success

Plant snapshot

Habit	Erect to sprawling, prickly shrub, 0.4–1.5 (–3) m.
Flowering	May – Sep (scented white-cream, yellow or pink).
Habitat	White, grey or yellow sand, sandy loam, granitic soils and laterite.
Local note	Recorded in the Wanneroo LGA. Prickly – wear gloves.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Difficult	Hakeas are harder from cuttings than seed.	–
Seed	Main method	No pre-treatment; about 66% germination.	This guide
Division	Not used	Woody shrub.	–

Seed collection & storage

Item	Details
Seed collection	Woody fruits usually hold their seed for at least a year. Collect mature, closed fruits.
Extraction & cleaning	Keep fruits in a paper bag in a warm, dry place – they open after several days to several weeks. If they won't open, oven at about **125°C for 1 hour**. Each fruit holds two winged seeds.
Seed quality	Cut-test a few seeds.
Storage life	**Long** – viable for many years stored dry and cool.
Typical germination	About **66%** (untreated, Alcoa nursery average, 41 tests). About **50 seedlings per gram** – roughly 2 g of seed per 100 seedlings.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May**. Jun – Sep workable. Avoid Oct – Feb.
Seed mix	Free-draining, pathogen-free, **low-phosphorus** mix.
Pre-treatment	**None needed.**
Sowing depth & method	Direct sow 1–2 seeds per tube, covered about **3–5 mm**.
After sowing	Water in; keep moist, not wet. Watch for damping off.
Germination time	About **3–6 weeks**.
Pricking out	If tray-sown, prick out at the cotyledon stage.

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Item	Details
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 4–6 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Seedlings collapse	Damping off. Sterile mix, good airflow, bottom watering.
Yellowing seedlings	Phosphorus toxicity. Very low-P feed only.
Fruits won't open	Warm them longer, or use the oven method.

Notes

Other notes	Hakea cuttings are harder than seed; seed is the method.
Confidence	**High.** Large nursery data set (41 tests) and WA *Hakea* seed notes available.
Sources	See References: 3, 4, 51, 52

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Hakea prostrata

Harsh Hakea

Family **Proteaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Easy	None	Mar – May	3–5 mm	79%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	Best	Best	Best	OK	OK	OK	OK	–	–	–

Best window | Workable – slower or less reliable | Avoid / low success

Plant snapshot

Habit	Erect to spreading or prostrate shrub, 1–3 m, with stiff, prickly-edged leaves. No lignotuber.
Flowering	Jul – Oct (white-cream).
Habitat	Sandy soils, often over laterite; hillslopes, granite outcrops and coastal dunes.
Local note	Recorded in the Wanneroo LGA. Extremely phosphorus-sensitive. Prostrate coastal forms exist – label by form.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Difficult	Hakeas are harder from cuttings than seed.	–
Seed	Main method	No pre-treatment; bag fruit – it sheds seed when ripe.	This guide
Division	Not used	Woody shrub.	–

Seed collection & storage

Item	Details
Seed collection	**Unlike most hakeas, it releases seed when the fruit ripens** (WA Hakea seed notes). Bag maturing fruits with mesh, or pick them just before they split.
Extraction & cleaning	Keep fruits in a paper bag in a warm, dry place until they open; each holds two winged seeds.
Seed quality	Cut-test a few seeds.
Storage life	**Long** – viable for many years stored dry and cool.
Typical germination	About **79%** (untreated, Alcoa nursery average, 30 tests). About **21 seedlings per gram** – roughly 5 g of seed per 100 seedlings.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May**. Jun – Sep workable. Avoid Oct – Feb.
Seed mix	Free-draining, pathogen-free mix with **no added phosphorus**.
Pre-treatment	**None needed.**
Sowing depth & method	Direct sow 1–2 seeds per tube, covered about **3–5 mm**.
After sowing	Water in; keep moist, not wet. Watch for damping off.
Germination time	About **3–6 weeks**.
Pricking out	If tray-sown, prick out at the cotyledon stage.

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Item	Details
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 4–6 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Seedlings yellow and die	Phosphorus toxicity – this species is extremely sensitive. No P in mix or feed.
Seedlings collapse	Damping off. Sterile mix, good airflow.
Fruits empty when collected	Seed already shed. Bag fruits earlier.

Notes

Other notes	Bird and bee attracting. Hakea cuttings are harder than seed.
Confidence	**High.** Nursery data from 30 tests and WA *Hakea* seed notes available.
Sources	See References: 3, 4, 43, 51, 52, 58

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Hakea trifurcata

Two-leaf Hakea

Family **Proteaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Easy	None	Mar – May	3–5 mm	69%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	Best	Best	Best	OK	OK	OK	OK	–	–	–

Best window · Workable – slower or less reliable · Avoid / low success

Plant snapshot

Habit	Rounded or open shrub, 1.5–3 m high and to 3.5 m wide. Two leaf shapes: needle-like and flat.
Flowering	Apr – Oct (white to cream-pink).
Habitat	White, grey or brown sand over limestone or laterite, loam and gravel.
Local note	Recorded in Joondalup and Wanneroo. Doesn't resprout after fire.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Difficult	Hakeas are harder from cuttings than seed.	–
Seed	Main method	No pre-treatment; about 69% germination.	This guide
Division	Not used	Woody shrub.	–

Seed collection & storage

Item	Details
Seed collection	Fruits **stay green when mature** and look like the flat leaves, so they're easy to miss. Collect full-sized, closed fruits.
Extraction & cleaning	Keep fruits in a paper bag in a warm, dry place until they open (several days to weeks), or oven at about 125°C for 1 hour.
Seed quality	Cut-test a few seeds.
Storage life	**Long** – viable for many years stored dry and cool.
Typical germination	About **69%** (untreated, Alcoa nursery average, 26 tests). About **58 seedlings per gram** – roughly 1.7 g of seed per 100 seedlings.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May**. Jun – Sep workable. Avoid Oct – Feb.
Seed mix	Free-draining, pathogen-free, **low-phosphorus** mix.
Pre-treatment	**None needed.**
Sowing depth & method	Direct sow 1–2 seeds per tube, covered about **3–5 mm**.
After sowing	Water in; keep moist, not wet. Watch for damping off.
Germination time	About **3–6 weeks**.
Pricking out	If tray-sown, prick out at the cotyledon stage.

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Item	Details
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 4–6 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Seedlings collapse	Damping off. Sterile mix, good airflow.
Yellowing seedlings	Phosphorus toxicity. Very low-P feed only.
Can't find fruit	They're green and leaf-like – check along the stems among the flat leaves.

Notes

Other notes	Bird and bee attracting. Hakea cuttings are harder than seed.
Confidence	**High.** Nursery data from 26 tests and WA *Hakea* seed notes available.
Sources	See References: 3, 4, 51, 52, 59

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Hardenbergia comptoniana

Native Wisteria

Family **Fabaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Moderate	Hot water (or boil 30 s); smoke	Mar – May	3–5 mm	31%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	OK	Best	Best	Best	OK	OK	OK	Best	OK	–	–

Best window · Workable – slower or less reliable · Avoid / low success

Plant snapshot

Habit	Vigorous twining climber or scrambling shrub.
Flowering	Jul – Oct (blue-purple, occasionally white).
Habitat	Sandy soils; coastal limestone, sandplains and dunes.
Local note	Recorded in Joondalup and Wanneroo. More vigorous than *H. violacea* – can smother small shrubs.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Good	Firm current season's growth strikes well.	→ Cuttings Guide
Seed	Main method	Hot water + smoke water; about 10 days.	This guide
Division	Not used	–	–

Seed collection & storage

Item	Details
Seed collection	Pods ripen and split in late spring–summer. Bag developing pods and collect once they split.
Extraction & cleaning	Rub pods and sieve or winnow.
Seed quality	Hard-seeded – the impervious coat must be broken. Float test: discard seed that floats.
Storage life	**Long** – seed keeps its viability for many years (ANPSA).
Typical germination	About **31%** after boiling-water treatment (Alcoa nursery average, 34 tests). About **19 seedlings per gram** – roughly 5 g of seed per 100 seedlings.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May** and **Sep**. Feb, Jun – Aug and Oct workable. Avoid Nov – Jan.
Seed mix	Standard sterile native seed mix.
Pre-treatment	**Hot water:** simmer at 95°C for 2 minutes, pour over the seed and soak 24 hours, then **soak 24 hours in smoke water** (Geraldton method). Alcoa used a 30 s boil instead. Sow the swollen seed.
Sowing depth & method	Cover about **3–5 mm**. Direct sow 2 seeds per tube and thin.
After sowing	Water in; keep moist, not wet.
Germination time	About **10 days** (Geraldton nursery).
Pricking out	Early, at the cotyledon to first true leaf stage.

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Item	Details
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 3–4 months. Give tubes a short stake or trim runners so they don't tangle.

Common problems

Problem	Likely cause and fix
Low germination	Seed coat not broken. Re-treat unswollen seed.
Swollen seed rots	Mix too wet. Water less.
Tangled seedlings	Space tubes and trim or stake runners.

Notes

Other notes	Cuttings of firm current season's growth also strike well (see the Cuttings Guide).
Confidence	**High.** Large nursery data set (34 tests) plus a WA community-nursery method.
Sources	See References: 3, 4, 5, 40

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Hovea pungens

Devil's Pins

Family **Fabaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Easy–Moderate	Boiling water 30 s	Mar – May	4–6 mm	41%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	OK	Best	Best	Best	OK	OK	OK	Best	OK	–	–

Best window · Workable – slower or less reliable · Avoid / low success

Plant snapshot

Habit	Erect, prickly, broom-like shrub, 0.2–1.8 m.
Flowering	May – Nov (blue-purple).
Habitat	Wide range: shallow soils on rock, sand, laterite, gravel, clay and loam; granite outcrops, coastal limestone and sandplains.
Local note	Recorded in Joondalup and Wanneroo. Sharp leaves – wear gloves when collecting.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Not used	Not documented.	–
Seed	Main method	Boiling water 30 s.	This guide
Division	Not used	–	–

Seed collection & storage

Item	Details
Seed collection	Pods ripen in late spring to summer and **eject seed when ripe**. Bag developing pods with stockings or muslin.
Extraction & cleaning	Collect seed from the bags; sieve out pod fragments.
Seed quality	Hard-seeded – fresh seed is dormant until the coat is treated. Float test to remove damaged seed.
Storage life	**Long.** Keeps for many years when dry (typical of hard-coated pea-family seed).
Typical germination	About **41%** after boiling-water treatment (Alcoa nursery average). About **33 seedlings per gram** – roughly 3 g of seed per 100 seedlings.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May** and **Sep**. Feb, Jun – Aug and Oct workable. Avoid Nov – Jan.
Seed mix	Standard sterile native seed mix.
Pre-treatment	**Boiling water:** drop into boiling water for **30 seconds** (Alcoa method), then cool; or pour near-boiling water over the seed and soak overnight. Sow swollen seed; re-treat any that haven't swollen.
Sowing depth & method	Larger seed – cover about **4–6 mm** (roughly twice its thickness). Direct sow 2–3 seeds per tube and thin.
After sowing	Water in; keep moist, not wet.

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Item	Details
Germination time	Can start about **one week** after treatment.
Pricking out	If tray-sown, prick out at the cotyledon stage – early, as the taproot grows fast.
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 3–5 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Low germination	Seed coat not broken. Re-treat unswollen seed.
Swollen seed rots	Over-boiled or mix too wet. Keep to 30 s; water less.
Taproot damage at pricking out	Pricked out too late. Direct sow into tubes instead.

Notes

Other notes	Wear gloves – the pungent (sharp) leaves make handling seedlings and stock plants prickly.
Confidence	**High.** Nursery germination data available for this species; method follows WA hard-seed guidance.
Sources	See References: 1, 3, 5, 14, 15, 16

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Hovea trisperma

Common Hovea

Family **Fabaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Moderate	Boiling water 30 s	Mar – May	3–5 mm	31%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	OK	Best	Best	Best	OK	OK	OK	Best	OK	–	–

Best window · Workable – slower or less reliable · Avoid / low success

Plant snapshot

Habit	Straggling, weak to ascending shrub, 0.1–0.7 m.
Flowering	May – Nov (purple-blue).
Habitat	Sandy soils, laterite, gravel and clay loam.
Local note	Recorded in the Wanneroo LGA.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Not used	Not documented.	–
Seed	Main method	Boiling water 30 s.	This guide
Division	Not used	–	–

Seed collection & storage

Item	Details
Seed collection	Pods shed seed as soon as they are ripe (late spring–summer). Bag developing pods with fine mesh.
Extraction & cleaning	Rub pods and sieve or winnow.
Seed quality	Hard-seeded. Float test: discard seed that floats.
Storage life	**Long.** Hard-coated pea seed keeps many years when dry.
Typical germination	About **31%** after boiling-water treatment (Alcoa nursery average, 25 tests). About **23 seedlings per gram** – roughly 4.4 g of seed per 100 seedlings.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May** and **Sep**. Feb, Jun – Aug and Oct workable. Avoid Nov – Jan.
Seed mix	Standard sterile native seed mix.
Pre-treatment	**Boiling water for 30 seconds** (Alcoa method), or a near-boiling overnight soak. Sow the swollen seed; re-treat the rest.
Sowing depth & method	Cover about **3–5 mm**. Direct sow 2–3 seeds per tube and thin.
After sowing	Water in; keep moist, not wet.
Germination time	Can start about **one week** after treatment.
Pricking out	Early, before the taproot lengthens.

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Item	Details
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 3–5 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Low germination	Seed coat not broken. Re-treat unswollen seed.
Swollen seed rots	Over-boiled or too wet.
Seedlings collapse	Damping off. Sterile mix, good airflow.

Notes

Other notes	Grows well under taller shrubs – a good understorey plant.
Confidence	**High.** Nursery germination data from 25 tests; method follows WA hard-seed guidance.
Sources	See References: 3, 4, 39

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Hyalosperma cotula

No widely used common name (an everlasting daisy)

Family **Asteraceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Moderate	None, or smoke water	Apr – May	Surface – light cover	27%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	OK	Best	Best	OK	–	–	–	–	–	–

Best window · Workable – slower or less reliable · Avoid / low success

Plant snapshot

Habit	Slender, erect annual herb, 0.05–0.25 m.
Flowering	Jul – Dec (white and yellow).
Habitat	A variety of soils, in periodically damp places.
Local note	Recorded in the Wanneroo LGA (not listed for Joondalup); a Friends of Warwick Bushland trail page shows a photo. An annual – it flowers the spring after sowing, then dies.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Not used	Annual.	–
Seed	Main method	Annual everlasting – untreated or smoke; compare both.	This guide
Division	Not used	Annual.	–

Seed collection & storage

Item	Details
Seed collection	Collect heads or whole plants as they dry off and turn brown – **Oct – Dec** (genus, SA).
Extraction & cleaning	Dry heads in a tray for about a week, then rub them through a fine sieve (genus).
Seed quality	Tiny, glossy pale brown seed about 1 mm long with a ring of feathery bristles – about 10,000 seeds per gram (seed supplier).
Storage life	Store with a desiccant in an airtight container, somewhere cool and dry (genus); sow the following autumn.
Typical germination	With smoke about **27%** (Alcoa, 15 tests) and about **3,180 seedlings per gram** (19 tests). Two untreated tests under the name *Helipterum cotula* averaged 57% (about 1,100 per gram). A 2025 lab study got no germination from fully viable seed, untreated or with gibberellic acid or a smoke chemical – so some lots may be dormant.

Sowing & growing on

Item	Details
Best sowing time	**Apr – May**, after the first autumn rains. Mar and Jun workable. Related everlastings germinate best at 10–20°C.
Seed mix	Standard sterile native seed mix.
Pre-treatment	**None, or smoke water – sow both and compare.** At Alcoa, untreated seed gave 57% (2 tests) and smoked seed 27% (15 tests; aerosol smoke or a 5% smoke water solution). Smoke suppresses germination in some paper daisies, and high doses of smoke water can stop daisies germinating (Dixon), so keep to a weak solution.

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Item	Details
Sowing depth & method	Mix the fine seed with dry sand and sprinkle it on the surface; cover very lightly or not at all – light helps most everlastings germinate.
After sowing	Water in with a fine spray; keep moist.
Germination time	First seedlings at **7–28 days** (mean 14) in 8 Kings Park seed lots, recorded as *Helipterum cotula*.
Pricking out	Prick out small, at the second-leaf stage – or sow a pinch into each cell to skip pricking out.
Shade house & hardening	Short – a week or so in the shade house, then full sun.
Time to planting size	Plant out small, about **6–10 weeks** after sowing (estimate), so plants establish in winter and flower in spring.

Common problems

Problem	Likely cause and fix
Very few seedlings	Seed lots vary, and smoke may suppress it. Compare smoked and untreated trays; try another lot.
Seed washed away	Seed is very fine. Bottom-water or use a very fine mist.
Leggy seedlings	Too much shade. Move to full sun early.

Notes

Other notes	A good pot plant – it ranked seventh for horticultural potential in Redlands trials of everlastings (ANPSA).
Confidence	**Medium.** Nursery data (15 tests with smoke, 2 untreated under the old name) and Kings Park emergence records; a 2025 lab study got no germination, so results vary between seed lots.
Sources	See References: 3, 4, 30, 43, 60, 143, 146, 147, 148, 149, 152

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Jacksonia furcellata

Grey Stinkwood

Family **Fabaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Moderate	Boiling water 30 s	Mar – May	2–3 mm	34%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	OK	Best	Best	Best	OK	OK	OK	Best	OK	–	–

Best window · Workable – slower or less reliable · Avoid / low success

Plant snapshot

Habit	Prostrate to weeping, erect shrub, 0.4–4 (–6) m, with grey, leafless-looking stems.
Flowering	Oct – Dec or Jan – Mar (yellow and orange-red).
Habitat	Sandy soils; sandplains, rises, swampy depressions and river banks.
Local note	Recorded in Joondalup and Wanneroo.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Not used	Not documented for WA species.	–
Seed	Main method	Boiling water 30 s.	This guide
Division	Not used	–	–

Seed collection & storage

Item	Details
Seed collection	Pea pods can burst open when ripe, leaving only a short window. Bag developing pods with fine mesh and collect once they split.
Extraction & cleaning	Rub pods and sieve or winnow.
Seed quality	Hard-seeded. Float test: discard seed that floats.
Storage life	**Long.** Hard-coated pea seed keeps many years when dry.
Typical germination	About **34%** after boiling-water treatment (Alcoa nursery average, 11 tests). About **48 seedlings per gram** – roughly 2 g of seed per 100 seedlings.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May** and **Sep**. Feb, Jun – Aug and Oct workable. Avoid Nov – Jan.
Seed mix	Standard sterile native seed mix.
Pre-treatment	**Boiling water for 30 seconds** (Alcoa method), or a near-boiling overnight soak. A 24 h smoke-water soak afterwards is used for another WA *Jacksonia* at Geraldton. Sow the swollen seed.
Sowing depth & method	Cover about **2–3 mm**. Direct sow 2–3 seeds per tube and thin.
After sowing	Water in; keep moist, not wet.
Germination time	Can start about **one week** after treatment.
Pricking out	Early, at the cotyledon to first true leaf stage.

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Item	Details
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 3–5 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Low germination	Seed coat not broken. Re-treat unswollen seed.
Swollen seed rots	Over-boiled or too wet.
Seedlings collapse	Damping off. Sterile mix, good airflow.

Notes

Other notes	No cutting data found for WA Jacksonia – seed is the method.
Confidence	**High.** Nursery germination data from 11 tests; method follows WA hard-seed guidance.
Sources	See References: 3, 4, 5, 43

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Jacksonia sericea

Waldjumi

Family **Fabaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Moderate	Boiling water 30 s	Mar – May	2–3 mm	26%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	OK	Best	Best	Best	OK	OK	OK	Best	OK	–	–

Best window · Workable – slower or less reliable · Avoid / low success

Plant snapshot

Habit	Low, spreading shrub to 0.6 m. Mature plants are usually leafless, with flattened grey stems.
Flowering	Usually Dec or Jan – Feb (orange).
Habitat	Calcareous and sandy soils.
Local note	Recorded in Joondalup and Wanneroo. **Priority Four** conservation species – collect seed only under the right DBCA licence.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Not used	Not documented for WA species.	–
Seed	Main method	Boiling water 30 s.	This guide
Division	Not used	–	–

Seed collection & storage

Item	Details
Seed collection	Pods ripen after the summer flowering and can burst open. Bag developing pods with fine mesh.
Extraction & cleaning	Rub pods and sieve or winnow.
Seed quality	Hard-seeded. Float test: discard seed that floats.
Storage life	**Long.** Hard-coated pea seed keeps many years when dry.
Typical germination	About **26%** after boiling-water treatment (Alcoa, only 1 test). About **84 seedlings per gram** – roughly 1.2 g of seed per 100 seedlings.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May** and **Sep**. Feb, Jun – Aug and Oct workable. Avoid Nov – Jan.
Seed mix	Standard sterile native seed mix, extra free-draining.
Pre-treatment	**Boiling water for 30 seconds** (Alcoa method), or a near-boiling overnight soak. Sow the swollen seed.
Sowing depth & method	Cover about **2–3 mm**. Direct sow 2–3 seeds per tube and thin.
After sowing	Water in; keep moist, not wet.
Germination time	Can start about **one week** after treatment.
Pricking out	Early, at the cotyledon to first true leaf stage.

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Item	Details
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 3–5 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Low germination	Seed coat not broken. Re-treat unswollen seed.
Swollen seed rots	Over-boiled or too wet.
Seedlings collapse	Damping off. Sterile mix, good airflow.

Notes

Other notes	Record every batch – the germination figure is from a single test.
Confidence	**Medium.** Nursery germination data from a single test; method follows WA hard-seed guidance.
Sources	See References: 3, 4, 37, 43

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Jacksonia sternbergiana

Stinkwood / Green Stinkwood

Family **Fabaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Moderate	Boiling water 30 s	Mar – May	3–5 mm	33%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	OK	Best	Best	Best	OK	OK	OK	Best	OK	–	–

Best window · Workable – slower or less reliable · Avoid / low success

Plant snapshot

Habit	Erect, weeping shrub or small tree, 1.5–5 m.
Flowering	Throughout the year (yellow-orange).
Habitat	Sandy soils along rivers and creeks, near swamps, flats and dunes.
Local note	Recorded in the Wanneroo LGA. Food plant for several butterfly larvae.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Not used	Not documented for WA species.	–
Seed	Main method	Boiling water 30 s.	This guide
Division	Not used	–	–

Seed collection & storage

Item	Details
Seed collection	Pods can burst open when ripe. Bag developing pods with fine mesh and collect once they split. Plants flower through the year, so check for ripe pods often.
Extraction & cleaning	Rub pods and sieve or winnow.
Seed quality	Hard-seeded. Float test: discard seed that floats.
Storage life	**Long.** Hard-coated pea seed keeps many years when dry.
Typical germination	About **33%** after boiling-water treatment (Alcoa nursery average, 9 tests). About **33 seedlings per gram** – roughly 3 g of seed per 100 seedlings.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May** and **Sep**. Feb, Jun – Aug and Oct workable. Avoid Nov – Jan.
Seed mix	Standard sterile native seed mix.
Pre-treatment	**Boiling water for 30 seconds** (Alcoa method), or a near-boiling overnight soak. Sow the swollen seed.
Sowing depth & method	Cover about **3–5 mm**. Direct sow 2–3 seeds per tube and thin.
After sowing	Water in; keep moist, not wet.
Germination time	Can start about **one week** after treatment.
Pricking out	Early, at the cotyledon to first true leaf stage.

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Item	Details
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 3–5 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Low germination	Seed coat not broken. Re-treat unswollen seed.
Swollen seed rots	Over-boiled or too wet.
Seedlings collapse	Damping off. Sterile mix, good airflow.

Notes

Other notes	No cutting data found for WA Jacksonia – seed is the method.
Confidence	**High.** Nursery germination data from 9 tests; method follows WA hard-seed guidance.
Sources	See References: 3, 4, 43

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Kennedia prostrata

Scarlet Runner / Running Postman

Family **Fabaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Easy	Hot water + smoke water	Mar – May	About 5 mm	30%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	OK	Best	Best	Best	OK	OK	OK	Best	OK	–	–

Best window · Workable – slower or less reliable · Avoid / low success

Plant snapshot

Habit	Prostrate or twining groundcover, spreading to about 2.5 m.
Flowering	Apr – Nov (red pea flowers).
Habitat	Usually sandy, gravelly soils.
Local note	Recorded in Joondalup and Wanneroo.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Good	Firm current season's growth strikes well.	→ Cuttings Guide
Seed	Main method	Hot water + smoke water; about 8 days.	This guide
Division	Not used	–	–

Seed collection & storage

Item	Details
Seed collection	Pods (about 50 mm) ripen in late spring to summer and split when ripe – bag them or collect as they turn brown.
Extraction & cleaning	Dry pods in paper bags and collect the seed.
Seed quality	Impervious seed coat – dormant until treated. Float test to remove damaged seed.
Storage life	**Long.** Seed maintains viability for extended periods.
Typical germination	About **30%** after dry heat (100°C, 3 h; Alcoa). About **12 seedlings per gram** – roughly 8 g of seed per 100 seedlings.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May** and **Sep**. Feb, Jun – Aug and Oct workable. Avoid Nov – Jan.
Seed mix	Standard sterile native seed mix.
Pre-treatment	**Hot water:** simmer water at 95°C for 2 minutes, pour over the seed (at least 10:1 water to seed) and soak 24 hours, then soak 24 hours in smoke water (Geraldton method). Alternatives: boiling water, abrasion, or dry heat 100°C for 3 hours.
Sowing depth & method	Larger seed – cover about **5 mm**. Direct sow 1–2 seeds per tube.
After sowing	Water in; keep moist, not wet.
Germination time	About **8 days** (Geraldton nursery).

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Item	Details
Pricking out	If tray-sown, prick out at the cotyledon stage.
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 3–4 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Low germination	Seed coat not broken. Re-treat unswollen seed.
Runners tangle in the nursery	Space tubes out; trim runners before dispatch.
Seedlings collapse	Damping off. Sterile mix, airflow.

Notes

Other notes	Cuttings of firm current season's growth also strike well.
Confidence	**High.** Nursery data, WA community-nursery method and ANPSA notes agree.
Sources	See References: 3, 4, 5, 14, 24

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Laxmannia squarrosa

Paper Lily

Family **Asparagaceae (Lomandroideae)**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Moderate–Difficult	Smoke water	Mar – May	Lightly covered	5%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	Best	Best	Best	OK	OK	OK	–	–	–	–

Best window | Workable – slower or less reliable | Avoid / low success

Plant snapshot

Habit	Small tufted perennial herb, 0.03–0.1 m.
Flowering	Sep – Nov (white).
Habitat	Lateritic sand and gravel.
Local note	Recorded in the Wanneroo LGA (not listed for Joondalup). A good container plant (Friends of Kings Park).

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Not used	Not grown from cuttings.	–
Seed	Main method	Smoke water; about 5% untreated (Alcoa).	This guide
Division	Trial	A related species is divided in Tasmania; untested.	–

Seed collection & storage

Item	Details
Seed collection	All western *Laxmannia* species are in mature seed in **January** (Keighery 2002). Collect the small heads as the capsules dry.
Extraction & cleaning	Dry in paper bags, rub and sieve.
Seed quality	Small, black, triangular seed with a finely bumpy surface (genus). Cut-test a few.
Storage life	Not documented. In a related species, seed buried in soil for a year fell from 97% to 19% viable (RIRDC) – don't rely on old seed.
Typical germination	Low untreated: about **5%** (Alcoa, 10 tests), about **176 seedlings per gram**. *Laxmannia* is on the list of smoke-responsive genera in WA research (RIRDC 1998).

Sowing & growing on

Item	Details
Best sowing time	**Mar – May** (estimate). Jun – Aug workable. Avoid Sep – Feb.
Seed mix	Standard sterile native seed mix, free-draining.
Pre-treatment	**Smoke water** – 24 h soak (1:9) or water the trays with it.
Sowing depth & method	Sow thinly and cover lightly.
After sowing	Keep moist, not wet. Seedlings are slow (related species) – keep trays going.
Germination time	Not documented.
Pricking out	At 2–3 leaves, gently.

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Item	Details
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Slow – est. 6–9 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Very few seedlings	Normal untreated. Use smoke water and fresh seed, and sow generously.
Seedlings slow to size up	Normal for the genus. Keep them in the shade house longer.
Seedlings collapse	Damping off. Sterile mix, airflow.

Notes

Other notes	Grown from seed in Perth restoration work (Gnangara). A related species is divided in Tasmania; untested here.
Confidence	**Medium.** Nursery data from 10 tests; smoke response documented for the genus.
Sources	See References: 3, 4, 110, 133, 134, 135

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Lepidosperma gladiatum

Coast Sword-sedge

Family
Cyperaceae (sedges)

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Moderate–Difficult *	**None, or heat + smoke (trial)**	**Apr – May**	**3–5 mm**	**41%**

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	OK	Best	Best	OK	OK	–	–	–	–	–

Best window Workable – slower or less reliable Avoid / low success

Plant snapshot

Habit	Robust, rhizomatous, tufted sedge, 0.5–1.5 (–3) m, clumps to 1.5 m wide. Flat leaves with **knife-sharp edges**.
Flowering	Nov – Dec or Jan – May (brown).
Habitat	White, grey or calcareous sand, limestone and loam; dunes and creeklines.
Local note	Recorded in the Wanneroo LGA. A strong dune stabiliser. Wear gloves – the leaf edges cut.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Not used	Sedge.	–
Seed	Difficult	Variable; 41% in one test, 53–72 days.	This guide
Division	Possible	Crown division is used commercially (trial here).	→ Division Guide

Seed collection & storage

Item	Details
Seed collection	Seed can only be harvested for a **few weeks after it ripens**. Check heads often and collect as the nutlets harden and darken.
Extraction & cleaning	Rub the heads over a sieve.
Seed quality	In *Lepidosperma*, 21–77% of nutlets can be empty – cut-test a sample and sow generously.
Storage life	Not well documented. Store dry and cool.
Typical germination	About **41%** untreated (Alcoa, 1 test), about 296 seedlings per gram. Most *Lepidosperma* seed needs dormancy broken first, so expect big differences between seed lots.

Sowing & growing on

Item	Details
Best sowing time	**Apr – May**. Mar, Jun – Jul workable.
Seed mix	Free-draining, sterile mix.
Pre-treatment	Sow untreated first. If a lot won't germinate, **trial** a heat shock (100°C for 10 minutes) followed by smoke water – the best treatment for fresh seed of the related *L. scabrum*.
Sowing depth & method	Cover about **3–5 mm**, in deep trays.
After sowing	Keep moist, not wet. Soil disturbance helps germination in the wild.

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Item	Details
Germination time	About **53–72 days** in one study – keep trays going.
Pricking out	At the first leaf, into deep tubes.
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Slow – est. 6–9 months (record your own).

Common problems

Problem	Likely cause and fix
No germination	Dormant or empty seed. Cut-test; trial heat plus smoke.
Harvested heads are empty	Collected too late – seed had already dropped. Check heads weekly as they ripen.
Slow growth	Normal. Deep tubes, full sun once established.

Notes

Other notes	Division of the crown is also used, and commercial stock is increasingly tissue-cultured (see the Division Guide).
Confidence	**Medium.** One nursery test (41% untreated) and genus research on dormancy; results vary widely between seed lots.
Sources	See References: 3, 4, 73, 74, 75

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Leucophyta brownii

Cushion Bush

Family **Asteraceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Easy–Moderate	None	Mar – May	Surface – light cover	Not recorded

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	Best	Best	Best	OK	OK	OK	OK	–	–	–

Best window · Workable – slower or less reliable · Avoid / low success

Plant snapshot

Habit	Compact, rounded shrub to about 1 m, densely covered in silvery-white woolly hairs.
Flowering	Dec or Jan – Feb (small yellow button heads).
Habitat	Coastal foredunes and cliffs.
Local note	Recorded in the Wanneroo LGA. Grows poorly in high humidity and dies easily if waterlogged.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Main method	Semi-hardened cuttings are the preferred method.	→ Cuttings Guide
Seed	Good	No pre-treatment; 10–30 days.	**This guide**
Division	Not used	–	–

Seed collection & storage

Item	Details
Seed collection	Collect the dry flower heads in late summer, once they turn brown and papery.
Extraction & cleaning	Rub heads gently over a sieve to free the small seeds.
Seed quality	Many florets may be empty – sow generously.
Storage life	Not well documented. Sow in the first autumn after collection.
Typical germination	Not recorded. ANBG notes seed germinates 10–30 days after sowing.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May**. Jun – Sep workable. Avoid Oct – Feb.
Seed mix	Standard sterile native seed mix, extra free-draining.
Pre-treatment	**None needed** (ANBG).
Sowing depth & method	Surface-sow and cover very lightly with fine sand – daisy-family seed often needs light.
After sowing	Water in with a fine spray; keep just moist. Good airflow.
Germination time	About **10–30 days**.
Pricking out	When the second pair of leaves appears.
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.

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Item	Details
Time to planting size	Est. 3–4 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Seedlings collapse	Damping off or too humid. Sow thinly, water less, more airflow.
Seed buried too deep	Needs light. Surface-sow.
Plants die in pots	Waterlogging. Very free-draining mix; don't overwater.

Notes

Other notes	Semi-hardened cuttings are the preferred method (see the Cuttings Guide).
Confidence	**Medium.** ANBG growing notes give treatment and germination time; no nursery test data.
Sources	See References: 4, 30, 31

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Leucopogon parviflorus

Coast Beard-heath

Family **Ericaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Difficult *	Peroxide + smoke water	Mar – May	Just covered	Not recorded

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	Best	Best	Best	OK	OK	OK	–	–	–	–

Best window | Workable – slower or less reliable | Avoid / low success

Plant snapshot

Habit	Erect, densely branched shrub or small tree, usually 0.3–3 m (to 5 m).
Flowering	Feb – Mar or Jun – Oct (white).
Habitat	Sandy soils over limestone or granite; coastal dunes and limestone.
Local note	Recorded in Joondalup and Wanneroo. Birds and lizards eat the fruit (State Flora SA).

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Difficult	Low strike and fine roots – trial protocol (provisional).	→ Cuttings Guide
Seed	Difficult	Dormant stones; peroxide + smoke water worked best in SA trials.	This guide
Division	Not used	Shrub – not divided.	–

Seed collection & storage

Item	Details
Seed collection	Pick fruit that has turned **translucent white** with a hard stone inside – green fruit is unripe. In SA this is Nov – Feb; watch local plants after the spring flowering.
Extraction & cleaning	Soak overnight, rub off the flesh by hand, rinse, and dry the stones on paper towel (Seeds of SA). The stone – pale yellow, ridged, about 4 mm – is what you sow.
Seed quality	Viability varies a lot (10–90% in SA seed lots). Cut-test a few stones before a big sowing.
Storage life	Store dry and cool, in an airtight container with desiccant.
Typical germination	Not tested in WA nurseries. SA lab trials: **100%** after 15 minutes in 30% hydrogen peroxide then 24 hours in 10% smoke water; **80%** after 2 minutes at 100°C, then 15 minutes in 30% hydrogen peroxide, a rinse and 72 hours in gibberellic acid (1000 mg/L). Cold stratification or peroxide alone gave 0%.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May** (cool, moist conditions; the SA trials were incubated at spring/autumn temperatures). Jun – Aug workable.
Seed mix	Standard sterile native seed mix, very free-draining.
Pre-treatment	Trial one of the SA treatments on part of the batch: **hydrogen peroxide then smoke water** (the best result), or **dry heat, hydrogen peroxide, then gibberellic acid**. 30% hydrogen peroxide is corrosive – wear gloves and eye protection for both. Untreated seed can still come up slowly, especially sown in soil from under the parent plants (Victoria).

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Item	Details
Sowing depth & method	Sow the stones and just cover them.
After sowing	Keep moist, not wet. **Keep trays for at least a year** – don't tip them out early.
Germination time	First seedlings about **7 weeks** and half by 11 weeks in the SA trials; untreated seed 3–18 months (Tasmania).
Pricking out	Early and gently – the roots are very fine and easily damaged.
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Slow – est. 9–12 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Nothing comes up for months	Normal for epacrids. Keep the trays going; try the other pre-treatment on the next batch.
Few seedlings from a big sowing	Low-viability seed lot. Cut-test before sowing; collect only white, ripe fruit.
Seedlings collapse	Damping off. Sterile mix, airflow, careful watering.

Notes

Other notes	Bird-passed seed germinates much better (APS Warrnambool). Cuttings are the other option – see the Cuttings Guide.
Confidence	**Low.** No WA nursery data. Method based on South Australian germination trials (Seeds of SA); eastern growers report slow, erratic results.
Sources	See References: 4, 23, 102, 103, 104, 105, 121

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Macrozamia fraseri

Sandplain Zamia

Family **Zamiaceae (cycads)**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Moderate	Remove fleshy coat; float test	Feb – Mar (fresh seed)	Surface	~34% (related M. riedlei)

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	OK	Best	Best	OK	OK	–	–	–	–	–	–	–

Best window · Workable – slower or less reliable · Avoid / low success

Plant snapshot

Habit	Cycad with a trunk that can exceed 1 m. Male and female cones on separate plants (dioecious).
Flowering	Cones, not flowers. Seeds with a bright red fleshy coat are shed in late summer to early autumn.
Habitat	Sand, in heath. Swan Coastal Plain and Geraldton Sandplains.
Local note	Recorded in the Wanneroo LGA. **All parts, especially the seeds, are poisonous** – wear gloves, wash hands, and keep seed away from pets and stock.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Not used	Cycad – no cuttings.	–
Seed	Main method	Sow fresh seed; takes over a year to germinate.	This guide
Division	Not used	Not documented.	–

Seed collection & storage

Item	Details
Seed collection	Collect fallen seed, or seed from cones as they break apart, in **late summer to early autumn** (February in the 2023 study).
Extraction & cleaning	Remove the red fleshy coat (sarcotesta). Researchers used a cement mixer with water and 10–30 mm aggregate; small lots can be scrubbed by hand in gloves.
Seed quality	**Float test:** soak in water for about 5 minutes – seeds that sink are viable; floaters are dead.
Storage life	**Short – sow fresh.** Seeds are shed moist (about 46% moisture) with an unfinished embryo. Standard cold, dry seed-bank storage (5°C, ~20% RH) damages them: viability was still over 50% at about 6 months but fell to 0–29% after 30+ months.
Typical germination	No figure for this species. The related *M. riedlei* averaged about **34%** untreated in Alcoa tests (10 tests). Seeds are large – sow individually.

Sowing & growing on

Item	Details
Best sowing time	**As soon as seed is cleaned** – usually Feb – Mar. Jan and Apr – May are workable.
Seed mix	Coarse, free-draining, sand-based mix in **deep containers** – cycads have a long root.
Pre-treatment	None beyond removing the fleshy coat and float testing.
Sowing depth & method	**Surface-sow:** lay seeds on the mix and press them in slightly; don't bury. Sow one seed per deep tube or pot, or in deep trays.

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Item	Details
After sowing	Keep outdoors under natural temperatures and rainfall, as the researchers recommend. Water lightly in dry spells – avoid heavy irrigation.
Germination time	Slow. The embryo keeps developing after the seed falls; the first root appears after about 5 months and full germination takes **more than 12 months**.
Pricking out	Avoid if possible – sow direct into deep containers. If tray-sown, pot each seed on once its root appears, keeping the root intact.
Shade house & hardening	Grow on in light shade and harden off to full sun over several weeks once the first leaf is firm.
Time to planting size	Very slow – allow 1–2 years or more from sowing (record your own).

Common problems

Problem	Likely cause and fix
Seed rots	Too wet, or damaged when the fleshy coat was removed. Water lightly; handle gently.
No sign of life after 6 months	Normal – germination takes over a year. Keep trays going.
Poor results from stored seed	Seed dried or cold-stored. Sow fresh, within a few months of collection.

Notes

Other notes	Poisonous: label trays clearly and keep seed and seedlings out of reach of children, pets and stock.
Confidence	**High.** Species-specific seed storage and germination research published in 2023. Germination figure is from the related *M. riedlei*.
Sources	See References: 3, 4, 26, 27

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Melaleuca cardiophylla

Tangling Melaleuca

Family **Myrtaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Easy–Moderate	None (smoke water to water in)	Mar – Apr, Sep – Oct	Surface	19%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	OK	Best	Best	OK	OK	OK	OK	Best	Best	OK	–

Best window | Workable – slower or less reliable | Avoid / low success

Plant snapshot

Habit	Dense, prickly, erect to spreading shrub, 0.2–2 m (to 4 m), with tangled branches.
Flowering	Aug – Dec or Jan (white-cream).
Habitat	White, grey or brown sand on limestone ridges, outcrops and dunes.
Local note	Recorded in Joondalup and Wanneroo.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Good	Reliable in this genus; keeps a chosen form.	→ Cuttings Guide
Seed	Main method	Surface-sow fine seed from old capsules; no pre-treatment.	This guide
Division	Not used	Shrub – not divided.	–

Seed collection & storage

Item	Details
Seed collection	Seed stays in the woody capsules (about 10 mm – large for a melaleuca), so collect at any time. Use capsules **at least 12 months old** from older stems. Wear gloves.
Extraction & cleaning	Place capsules in an open container somewhere warm; fine seed is released in 2–3 days.
Seed quality	Fine seed with chaff; no need to separate.
Storage life	**Long** if kept dry and cool.
Typical germination	About **19%** (untreated, Alcoa average of only 2 tests). About **222 seedlings per gram**.

Sowing & growing on

Item	Details
Best sowing time	**Mar – Apr** and **Sep – Oct**. Feb, May – Aug and Nov workable. Avoid Dec – Jan.
Seed mix	Fine-textured, sterile native seed mix.
Pre-treatment	**None needed.**
Sowing depth & method	Dilute with dry, sterile yellow sand and sprinkle from a spice jar (Geraldton). **Sow on the surface**; don't cover.
After sowing	Water in with smoke water using a fine mist (Geraldton), or use the bog method.
Germination time	About **10 days** at Geraldton; 14–30 days is typical for the genus.
Pricking out	When the second pair of leaves appears. Handle tiny seedlings by the leaves.

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Item	Details
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 4–6 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Low germination	Young or empty capsules. Use capsules 12+ months old; sow more.
Seed washed away	Use bottom watering or a very fine mist.
Damping off	Crowding and wet mix. Thin sowing, airflow.

Notes

Other notes	Cuttings are also reliable in the genus (see the Cuttings Guide).
Confidence	**Medium.** Few nursery tests (2); WA community-nursery method and ANPSA genus notes agree.
Sources	See References: 3, 4, 5, 18

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Melaleuca huegelii

Chenille Honeymyrtle

Family **Myrtaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Easy	None	Mar – Apr, Sep – Oct	Surface	58%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	OK	Best	Best	OK	OK	OK	OK	Best	Best	OK	–

Best window · Workable – slower or less reliable · Avoid / low success

Plant snapshot

Habit	Shrub or small tree, 0.5–5 m.
Flowering	Sep – Dec or Jan (pink, white or pink-purple).
Habitat	Sand; limestone cliffs, coastal plains and dunes.
Local note	Recorded in Joondalup and Wanneroo.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Good	Reliable in this genus; keeps a chosen form.	→ Cuttings Guide
Seed	Main method	Surface-sow fine seed; no pre-treatment.	**This guide**
Division	Not used	Woody shrub.	–

Seed collection & storage

Item	Details
Seed collection	Seed stays in the woody capsules indefinitely, so capsules can be collected at any time. Use capsules **at least 12 months old**.
Extraction & cleaning	Place capsules in an open container somewhere warm; fine seed is released in 2–3 days.
Seed quality	Extremely fine seed with chaff; no need to separate.
Storage life	**Long** if kept dry and cool; seed also keeps on the plant in closed capsules.
Typical germination	About **58%** (untreated, Alcoa nursery average). About **775 seedlings per gram** – a pinch is plenty.

Sowing & growing on

Item	Details
Best sowing time	**Mar – Apr** and **Sep – Oct**. Feb, May – Aug and Nov workable. Avoid Dec – Jan.
Seed mix	Fine-textured, sterile native seed mix.
Pre-treatment	**None needed.**
Sowing depth & method	Mix with dry, sterile yellow sand and sprinkle from a spice jar. **Sow on the surface**; don't cover.
After sowing	Water in with smoke water using a fine mist (Geraldton), or stand pots in a saucer of water until germination (bog method).
Germination time	About **9 days** at Geraldton; 14–30 days is typical for the genus.
Pricking out	When the second pair of leaves appears. Handle tiny seedlings by the leaves.

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Item	Details
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 4–6 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Seedlings too crowded	Dilute seed more with sand.
Seed washed away	Use bottom watering or a very fine mist.
Damping off	Crowding and wet mix. Thin sowing, airflow.

Notes

Other notes	Cuttings are also reliable if a particular form is wanted.
Confidence	**High.** Nursery data, WA community-nursery method and ANPSA genus notes agree.
Sources	See References: 3, 4, 5, 18

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Melaleuca lanceolata

Rottnest Teatree / Moonah

Family **Myrtaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Easy	None (smoke water to water in)	Mar – Apr, Sep – Oct	Surface	55%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	OK	Best	Best	OK	OK	OK	OK	Best	Best	OK	–

Best window · Workable – slower or less reliable · Avoid / low success

Plant snapshot

Habit	Shrub or tree, 1–8 m.
Flowering	Mainly in summer – FloraBase's calendar shows Nov – Mar (white-cream).
Habitat	Limestone ridges, coastal cliffs and dunes, salt flats and the edges of salt lakes.
Local note	Recorded in the Wanneroo LGA (not listed for Joondalup). Resists salt spray – a good windbreak or shelterbelt plant.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Good	Reliable in this genus (provisional).	→ Cuttings Guide
Seed	Main method	Surface-sow; no pre-treatment; about 55% (Alcoa).	This guide
Division	Not used	Shrub – not divided.	–

Seed collection & storage

Item	Details
Seed collection	Seed is held in the capsules, so collect at any time. Pick **large, hard capsules with closed valves** – the mature ones are usually lower down the stems (SA). The genus rule is capsules at least 12 months old.
Extraction & cleaning	Dry capsules in a paper bag or open tray for about 2 weeks, until the valves open, then shake and sieve. Seed is very hard to separate from the chaff – sow them together.
Seed quality	Tiny (to about 1.5 × 0.5 mm), dark brown, pyramid-shaped seed mixed with chaff of similar size. SA seed lots were 85–95% viable.
Storage life	Store dry, with a desiccant, in an airtight container somewhere cool (SA).
Typical germination	About **55%** with no pre-treatment (Alcoa, 7 tests), about **1,350 seedlings per gram** (10 tests). Seed is non-dormant.

Sowing & growing on

Item	Details
Best sowing time	**Mar – Apr** and **Sep – Oct**. Feb, May – Aug and Nov workable. Avoid Dec – Jan.
Seed mix	Fine-textured, sterile native seed mix.
Pre-treatment	**None needed.**
Sowing depth & method	Mix with dry, sterile sand and sprinkle from a spice jar (Geraldton). **Sow on the surface**; don't cover.

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Item	Details
After sowing	Water in with smoke water (Geraldton) – use a fine mist, or the bog method.
Germination time	About **11 days** at Geraldton; first seedlings at 15–64 days (mean 24) in 10 Kings Park lots.
Pricking out	When the second pair of leaves appears. Handle tiny seedlings by the leaves.
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Seedlings grow slowly at first (Geraldton). Est. 5–7 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Low germination	Young or empty capsules. Use large, hard capsules from lower on the stems.
Seed washed away	Use bottom watering or a very fine mist.
Damping off	Crowding and wet mix. Thin sowing, airflow.

Notes

Other notes	Cuttings are also reliable in the genus (see the Cuttings Guide).
Confidence	**High.** Nursery data (7 tests), Kings Park emergence records, a WA community-nursery method and SA seed-bank notes agree.
Sources	See References: 3, 4, 5, 18, 139, 143, 156

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Melaleuca systena

Coastal Honeymyrtle

Family **Myrtaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Easy–Moderate	None	Mar – Apr, Sep – Oct	Surface	14%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	OK	Best	Best	OK	OK	OK	OK	Best	Best	OK	–

Best window | Workable – slower or less reliable | Avoid / low success

Plant snapshot

Habit	Erect to spreading shrub, usually 0.5–2 m.
Flowering	Feb – Mar or Aug – Dec (yellow-cream).
Habitat	Sand over laterite or limestone; stabilised coastal dunes and rocky limestone.
Local note	Recorded in Joondalup and Wanneroo.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Good	Reliable in this genus; a good option given low seed germination.	→ Cuttings Guide
Seed	Main method	Surface-sow fine seed; no pre-treatment.	This guide
Division	Not used	Woody shrub.	–

Seed collection & storage

Item	Details
Seed collection	Collect capsules at least 12 months old from older stems at any time; seed is retained in closed capsules.
Extraction & cleaning	Place capsules somewhere warm in an open container; seed is released in 2–3 days.
Seed quality	Fine seed with chaff. Germination in nursery tests was low – sow generously.
Storage life	**Long** if kept dry and cool.
Typical germination	About **14%** (untreated, Alcoa average of only 3 tests). About **301 seedlings per gram**.

Sowing & growing on

Item	Details
Best sowing time	**Mar – Apr** and **Sep – Oct**. Feb, May – Aug and Nov workable. Avoid Dec – Jan.
Seed mix	Fine-textured, sterile native seed mix.
Pre-treatment	**None needed.**
Sowing depth & method	Mix with dry sand and sprinkle from a spice jar. **Sow on the surface**; don't cover.
After sowing	Water in with smoke water using a fine mist, or use the bog method.
Germination time	About **7 days** at Geraldton; allow up to 4 weeks.
Pricking out	When the second pair of leaves appears.

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Item	Details
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 4–6 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Low germination	Empty or old seed. Use older capsules (12+ months) from vigorous plants; sow more.
Seed washed away	Use bottom watering or a very fine mist.
Damping off	Crowding and wet mix.

Notes

Other notes	Cuttings are a reliable alternative for this genus.
Confidence	**Medium.** Few nursery tests (3); WA community-nursery method and ANPSA genus notes available.
Sources	See References: 3, 4, 5, 18

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Mesomelaena pseudostygia

Semaphore Sedge

Family
Cyperaceae (sedges)

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Difficult *	**Heat + smoke (trial)**	**Apr – May**	**2–3 mm**	**Not recorded (relatives 0%)**

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	OK	Best	Best	OK	OK	–	–	–	–	–

Best window | Workable – slower or less reliable | Avoid / low success

Plant snapshot

Habit	Tufted perennial sedge, 0.2–0.75 (–1) m, clumps to 0.5 m wide.
Flowering	Mainly Apr – Jun (brown-black heads held out like signal flags).
Habitat	White, grey or yellow sand.
Local note	Recorded in Joondalup and Wanneroo. Described in research as a difficult-to-propagate dryland sedge.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Not used	Sedge.	–
Seed	Difficult	No data; relatives 0%. Heat + smoke is a trial.	This guide
Division	Trial	Not documented. Tube stock is usually tissue-cultured.	–

Seed collection & storage

Item	Details
Seed collection	Collect ripe heads in late spring–summer when they are dry and the nutlets come away easily.
Extraction & cleaning	Rub heads over a sieve.
Seed quality	Cut-test a sample – expect many empty nutlets.
Storage life	For the related *M. tetragona*, seed kept best very dry (about 11% relative humidity) at 23°C or cooler, and declined in the freezer.
Typical germination	Not recorded. The related *M. tetragona* and *M. graciliceps* gave 0% untreated in Alcoa tests.

Sowing & growing on

Item	Details
Best sowing time	**Apr – May** (trial). Mar, Jun – Jul workable.
Seed mix	Free-draining, sterile mix.
Pre-treatment	**Trial only:** no treatment has been published. A heat shock (100°C for 10 minutes) plus smoke water, which helped a related sedge, is a reasonable trial.
Sowing depth & method	Cover about **2–3 mm**. Sow generously in deep trays.
After sowing	Keep moist, not wet. Keep trays going for months.
Germination time	Not recorded.
Pricking out	At the first leaf, into deep tubes.

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Item	Details
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Slow – record your own.

Common problems

Problem	Likely cause and fix
No germination	Expected. Try heat plus smoke, older seed, and larger sowings.
Empty seed	Cut-test before sowing.
Seedlings rot	Too wet. Water less.

Notes

Other notes	Tissue culture has given over 80% rooting and 95% survival in research, and is how most tube stock is produced. Record any seed trial results.
Confidence	**Low.** No seed data for this species; related species gave 0%. Commercial stock is tissue-cultured. Trial protocol.
Sources	See References: 3, 4, 74, 76, 77, 80

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Myoporum insulare

Blueberry Tree / Common Boobialla

Family **Scrophulariaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Moderate	Ferment berries; smoke-water soak	Mar – May	2–3 mm	Not recorded

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	Best	Best	Best	OK	OK	OK	OK	–	–	–

Best window | Workable – slower or less reliable | Avoid / low success

Plant snapshot

Habit	Dense, low spreading to erect shrub or small tree, 0.25–5 m, with thick, fleshy leaves.
Flowering	Jul – Dec, also Jan – Feb (white, purple-spotted), followed by purple berries.
Habitat	White or grey sand on coastal dunes and limestone.
Local note	Recorded in Joondalup and Wanneroo. Forms range from prostrate to tree-like; seedlings vary.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Main method	Firm young growth roots readily.	→ Cuttings Guide
Seed	Good	Ferment berries, smoke water; 15–30 days. Seedling form varies.	This guide
Division	Not used	Woody shrub.	–

Seed collection & storage

Item	Details
Seed collection	Pick the **darkest purple**, fully ripe berries.
Extraction & cleaning	**Ferment:** soak the berries in water for 4–5 days, then wash off the pulp through a sieve and dry the stones.
Seed quality	Hard stones; cut-test a few to check they are filled.
Storage life	Not well documented. Store dry and cool.
Typical germination	Not recorded.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May**. Jun – Sep workable. Avoid Oct – Feb.
Seed mix	Standard sterile native seed mix.
Pre-treatment	After fermenting, **soak in smoke water (1:9) for 24 hours** (Geraldton method).
Sowing depth & method	Cover about **2–3 mm**. Sow in trays or 2–3 stones per tube.
After sowing	Water in; keep moist, not wet.
Germination time	About **15–30 days** (Geraldton nursery).
Pricking out	When the second pair of leaves appears.

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Item	Details
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 3–5 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Poor germination	Unripe berries or pulp left on. Use the darkest berries and ferment fully.
Mould on trays	Pulp residue. Wash stones well before sowing.
Plants not the form you wanted	Seedlings vary. Use cuttings for a particular form.

Notes

Other notes	Cuttings are the main method and keep the parent's form (see the Cuttings Guide).
Confidence	**Medium.** WA community-nursery method available; no nursery test data.
Sources	See References: 4, 5

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Nuytsia floribunda

WA Christmas Tree / Moodjar

Family **Loranthaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Moderate–Difficult	None – sow fresh	Feb – Apr (fresh seed)	Surface, upright	29%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	OK	Best	Best	Best	OK	–	–	–	–	–	–	OK

Best window · Workable – slower or less reliable · Avoid / low success

Plant snapshot

Habit	Tree or shrub to 10 m. A **root parasite** – it taps the roots of nearby plants for water and nutrients.
Flowering	Oct – Dec or Jan (brilliant yellow-orange).
Habitat	Sand, sandy loam, gravel over clay, granite, laterite and limestone; sandplains and slopes.
Local note	Recorded in the Wanneroo LGA. Takes about 7–10 years to flower from seed.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Possible	Reported to strike, but plants rarely survive without a host.	–
Seed	Main method	Fresh seed germinates readily; seedlings need a host plant.	This guide
Division	Not used	Old trees sucker from the roots; not a nursery method.	–

Seed collection & storage

Item	Details
Seed collection	Collect the papery winged seed as it matures on the tree, within a few weeks of flowering.
Extraction & cleaning	Little cleaning needed – remove twigs and old flower parts.
Seed quality	Use freshly collected seed.
Storage life	Not well documented. Growers rely on fresh seed – sow as soon as it is collected.
Typical germination	About **29%** (untreated, Alcoa nursery average, 6 tests). About **41 seedlings per gram** – roughly 2.4 g of seed per 100 seedlings. Most growers find fresh seed easy to germinate.

Sowing & growing on

Item	Details
Best sowing time	**As soon as seed is collected** – usually Feb – Apr. Jan, May and Dec are workable.
Seed mix	Standard sterile native seed mix. **No fertiliser** – it is adapted to poor soils.
Pre-treatment	None.
Sowing depth & method	Stand the papery seed **upright on the surface** of the mix and keep moist.
After sowing	Keep moist, not wet (about 18–22°C suits it).
Germination time	About **10–35 days**.
Pricking out	Move seedlings into forestry tubes to grow on, then pot each into a pot **with a host plant** once big enough. Hosts that have worked: dichondra, native grasses, kangaroo paw, even carrot seedlings; research found *Acacia* hosts passed nitrogen to the seedlings.

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Item	Details
Shade house & hardening	Grow on in light shade with the host, then harden off gradually to full sun.
Time to planting size	Slow and unpredictable – allow 12 months or more (record your own).

Common problems

Problem	Likely cause and fix
Seedlings stall or die after a year or two	No host, or a poor one. Pot up with a host plant early.
Poor germination	Old seed. Sow fresh.
Yellowing after feeding	Fertiliser. Don't feed.

Notes

Other notes	Growers warn tube stock is hard to produce – set more than you need and plant out with a nearby host. It can also parasitise underground cables and pipes, so choose planting sites with care.
Confidence	**Medium.** Nursery germination data (6 tests) and grower methods published; host-plant research available. Germinating seed is easy – keeping seedlings alive is the hard part.
Sources	See References: 3, 4, 45, 46, 47, 48, 50

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Olearia axillaris

Coastal Daisybush

Family **Asteraceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Easy–Moderate	Air-dry 3 months; smoke water	Aug – Sep	Surface	21%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	OK	OK	OK	OK	OK	Best	Best	OK	–	–

Best window | Workable – slower or less reliable | Avoid / low success

Plant snapshot

Habit	Erect, much-branched grey shrub, 0.2–3 m.
Flowering	Jan – Jul or Nov – Dec (small white to yellow daisies).
Habitat	White, grey or red sand and loam; coastal limestone, sand dunes and rocky hillsides.
Local note	Recorded in Joondalup and Wanneroo.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Good	Perennial daisies strike quickly from vigorous growth.	→ Cuttings Guide
Seed	Main method	Fresh seed, aired 3 months; surface-sow.	This guide
Division	Not used	Woody shrub.	–

Seed collection & storage

Item	Details
Seed collection	Seed usually ripens around **April – May**, but plants can set seed about **6 weeks after a big rain**, so watch for opportunistic crops (e.g. after summer storms).
Extraction & cleaning	Collect the fluffy seed heads into paper bags.
Seed quality	Germination in nursery tests was moderate (about 21%).
Storage life	**Short – needs fresh seed each year.** Seed must be **aired for about 3 months** before storing or sowing.
Typical germination	About **21%** (untreated, Alcoa nursery average). About **73 seedlings per gram** – roughly 1.4 g of seed per 100 seedlings.

Sowing & growing on

Item	Details
Best sowing time	**Aug – Sep** – about 3 months after an autumn collection. Mar – Jul and Oct workable depending on when seed was collected. Avoid Nov – Feb.
Seed mix	Standard sterile native seed mix.
Pre-treatment	Air the seed for about 3 months after collection. Water trays with smoke water after sowing (Geraldton method).
Sowing depth & method	**Sow on the surface** and press in lightly – daisy-family seed often needs light to germinate.
After sowing	Water in with smoke water; keep moist.

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Item	Details
Germination time	About **7–16 days** (Geraldton nursery).
Pricking out	When the second pair of leaves appears.
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 3–4 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Poor germination	Old seed or not aired. Use this season's seed, aired for 3 months.
Seed buried too deep	Needs light. Surface sow.
Leggy seedlings	Too much shade. Move to full sun sooner.

Notes

Other notes	Also strikes readily from cuttings.
Confidence	**High.** Nursery data and a detailed WA community-nursery method available.
Sources	See References: 3, 4, 5, 23

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Opercularia vaginata

Dog Weed

Family **Rubiaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Difficult *	Smoke water (trial)	Apr – May	1–2 mm	<1% untreated

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	OK	Best	Best	OK	OK	–	–	–	–	–

Best window · Workable – slower or less reliable · Avoid / low success

Plant snapshot

Habit	Low, decumbent to erect perennial herb or subshrub, 0.1–0.45 m.
Flowering	Jul – Dec (green to green-yellow heads).
Habitat	Sandy, lateritic or granitic soils and coastal limestone.
Local note	Recorded in Joondalup and Wanneroo. Many *Opercularia* have an unpleasant smell. May sucker.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Not used	Not documented.	–
Seed	Difficult	Under 1% in nursery tests; trial smoke and aged seed.	This guide
Division	Possible	May sucker; untested.	–

Seed collection & storage

Item	Details
Seed collection	Collect the fruit heads when dry; the capsules shed a lid to release the seed.
Extraction & cleaning	Dry heads in a paper bag and sieve out the seed.
Seed quality	Cut-test a sample.
Storage life	Not well documented. Store dry and cool.
Typical germination	Under 1% untreated (0.17%, Alcoa, 4 tests), about 2.6 seedlings per gram.

Sowing & growing on

Item	Details
Best sowing time	**Apr – May** (trial). Mar, Jun – Jul workable.
Seed mix	Standard sterile native seed mix.
Pre-treatment	**Trial only:** no treatment has been published. Smoke water (24 h soak) and seed aged for a year or more are low-cost things to try – record the results.
Sowing depth & method	Cover lightly, about 1–2 mm.
After sowing	Keep moist, not wet.
Germination time	Not recorded.
Pricking out	At the second leaf stage.

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Item	Details
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Record your own.

Common problems

Problem	Likely cause and fix
No germination	Expected untreated. Try smoke and aged seed; sow generously.
Seedlings collapse	Damping off. Sterile mix; airflow.
Weak seedlings	Too much shade.

Notes

Other notes	Suckering plants may be divided – untested. Record any trial so this page can be updated.
Confidence	**Low.** Nursery tests show under 1% germination untreated; no published treatment. Trial protocol.
Sources	See References: 3, 4

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Orthrosanthus laxus

Morning Iris

Family **Iridaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Moderate–Difficult	Smoke water	Mar – May	Lightly covered	7%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	Best	Best	Best	OK	OK	OK	–	–	–	–

Best window | Workable – slower or less reliable | Avoid / low success

Plant snapshot

Habit	Rhizomatous, tufted perennial; grass-like leaves 0.1–0.45 m, flower stems to about 0.7 m.
Flowering	Aug – Nov (pale to bright blue, rarely white). Flowers are best in the morning and fade through the day.
Where it grows	Grows locally in the Perth region; suits full sun or light shade and is drought hardy (ReWild Perth).
Local note	Increasingly used for revegetation and landscaping. The variety *gramineus* (Grass-leaved Orthrosanthus) is a separate form – keep provenances separate.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Not used	Not grown from cuttings.	–
Seed	Main method	Smoke water; germination varies between seed lots.	This guide
Division	Possible	Clumps can be divided in spring (RHS); untested here.	–

Seed collection & storage

Item	Details
Seed collection	Collect capsules (12–15 mm, pointed) as they dry and brown after the spring flowering – est. Nov – Dec, as for the related *O. multiflorus*.
Extraction & cleaning	Dry 1–2 weeks in paper bags, rub and sieve (genus).
Seed quality	Brown, wrinkled, rounded-angular seed about 2 mm. Fresh WA seed tested 93% viable.
Storage life	Not well documented. Seed buried in soil for a year fell to 34% viable, although the survivors germinated better with smoke – so don't rely on old seed.
Typical germination	Low untreated: about **7%** (Alcoa, 8 tests; the inland var. *gramineus* 18%). About **18 seedlings per gram**. WA research lists it as **smoke-responsive**.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May**. Jun – Aug workable. Avoid Sep – Feb (germinates at 13–18°C, RHS).
Seed mix	Standard sterile native seed mix, free-draining.
Pre-treatment	**Smoke water** – 24 h soak (1:9) or water the trays with it.
Sowing depth & method	Lightly cover. Because germination varies a lot between seed lots, sow **about 4 seeds per cell** (as in WA research) and thin.
After sowing	Keep moist, not wet. Keep trays for a second season if few come up.
Germination time	Not documented – expect it to be uneven.

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Item	Details
Pricking out	At 2–3 leaves, gently.
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 5–6 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Very few seedlings	Normal for some seed lots. Use smoke water, sow more per cell, and record germination by lot.
Seedlings rot	Too wet. Water less; free-draining mix.
Seedlings collapse	Damping off. Sterile mix, airflow.

Notes

Other notes	Can also be divided in spring (RHS); not tested here – see the Division Guide's list of other species.
Confidence	**Medium.** Nursery tests (8) and WA smoke research; germination varies between seed lots and years.
Sources	See References: 3, 4, 60, 110, 116, 117, 118, 119

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Patersonia occidentalis

Purple Flag

Family **Iridaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Difficult	**Smoke; aged seed**	**May – Jun**	**Lightly covered**	**2%**

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	OK	OK	Best	Best	OK	OK	–	–	–	–

Best window · Workable – slower or less reliable · Avoid / low success

Plant snapshot

Habit	Rhizomatous, tufted perennial herb, to 1.5 m in flower.
Flowering	Aug – Dec or Jan (purple). Each flower lasts less than a day, but each stem carries many.
Habitat	Wide range of soils – sand, sandy clay, loam, gravel, laterite, granite and limestone; winter-wet areas, dunes and granite outcrops.
Local note	Recorded in the Wanneroo LGA (not listed for Joondalup). Several varieties – keep provenances separate.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Not used	Not grown from cuttings.	–
Seed	Main method	Smoke and aged seed; low in nursery tests.	This guide
Division	Possible	Large pieces in autumn; ANBG calls division unreliable.	–

Seed collection & storage

Item	Details
Seed collection	Collect capsules as they dry and turn brown, with hard brown seed inside – Dec – Mar in SA, after the spring flowering.
Extraction & cleaning	Dry 1–2 weeks, rub the capsules gently and sieve.
Seed quality	Dark brown, flattened oval seed 2–3 mm, with no aril. Cut-test a few.
Storage life	Airtight with a desiccant, cool and dry. In WA research, seed stored in soil for a year then smoked germinated best (viability 67% fresh, 58% after a year).
Typical germination	Low in WA nursery tests: about **2%** even with smoke (Alcoa, 7 tests); about **7 seedlings per gram**. WA research found it needs **both soil storage and smoke** for its best germination.

Sowing & growing on

Item	Details
Best sowing time	**May – Jun** (Kings Park). Mar – Apr and Jul – Aug workable. Avoid Sep – Feb.
Seed mix	Standard sterile native seed mix, free-draining.
Pre-treatment	**Smoke water** – soak 12 hours in 10% smoke water (Dixon, ANPSA). Seed a year or more old is likely to do better.
Sowing depth & method	Sow thinly and cover lightly.
After sowing	Keep moist, not wet. If germination is poor, **keep the trays over summer** and water with smoke water again next autumn – this mimics the soil storage that worked in research.

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Item	Details
Germination time	Not documented – expect slow, spasmodic germination.
Pricking out	At 2–3 leaves, gently.
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 5–6 months from germination (record your own).

Common problems

Problem	Likely cause and fix
Very few seedlings	Normal in WA tests. Use smoke, older seed, and keep trays a second season.
Seedlings rot	Too wet. Water less; free-draining mix.
Seedlings collapse	Damping off. Sterile mix, airflow.

Notes

Other notes	Clumps can be divided, but ANBG calls it unreliable; if you try, use large pieces in autumn (see the Division Guide's list of other species).
Confidence	**Medium.** Nursery tests (7, with smoke) and WA smoke research agree germination is low; South Australian seed is reported to germinate readily.
Sources	See References: 3, 4, 60, 110, 113, 114, 115

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Petrophile axillaris

No widely used common name

Family **Proteaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Difficult *	Remove hairs, nick coat; smoke	Mar – May	3–5 mm	Not recorded

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	Best	Best	Best	OK	OK	OK	–	–	–	–

Best window · Workable – slower or less reliable · Avoid / low success

Plant snapshot

Habit	Erect, multi-stemmed shrub to 3 m, with sharp-pointed leaves.
Flowering	Sep – Nov (pink, purple and grey).
Habitat	Red, yellow or grey sand over limestone, sandy loam and gravel; sandplains, slopes and ridges.
Local note	Recorded in Joondalup and Wanneroo. The pink form once sold as *P. serruriae* is this species. Prickly – gloves.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Difficult	Petrophile cuttings take 3–10 months; 10–67% strike.	–
Seed	Main method	Remove hairs and nick the seed coat; slow.	This guide
Division	Not used	Shrub – not divided.	–

Seed collection & storage

Item	Details
Seed collection	Cones are usually ripe about six months after flowering, when they turn grey to grey-brown. Cones over a year old hold mature fruit.
Extraction & cleaning	Keep cones warm and dry – after a few days the scales open and the hairy nuts can be pulled out with tweezers.
Seed quality	Many 'false' (empty) fruits are produced – cut-test a sample.
Storage life	Not well documented. Store dry and cool.
Typical germination	Not recorded. Fresh seed is reported to germinate well once treated, but times vary.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May** (cool autumn). Jun – Aug workable. Avoid Sep – Feb.
Seed mix	Free-draining, sterile, **low-phosphorus** mix.
Pre-treatment	**Remove the hairs and nick a small section of the seed coat** to reveal the white seed, without damaging it. Smoke water may also help.
Sowing depth & method	Cover about **3–5 mm**. Direct sow into tubes.
After sowing	Keep moist, not wet; strict hygiene.
Germination time	Can take **several months**.

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Item	Details
Pricking out	If tray-sown, prick out at the cotyledon stage.
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 6–8 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
No germination	Seed coat not nicked, or empty nuts. Nick carefully; cut-test.
Seedlings die in pots	Phosphorus toxicity or dieback – petrophiles are very sensitive to both.
Seedlings collapse	Damping off. Sterile mix; bottom watering.

Notes

Other notes	Cuttings are hard in *Petrophile* (3–10 months to strike, 10–67% success in study-group trials).
Confidence	**Medium.** No species data; follows WA Petrophile seed notes, as for *P. linearis*.
Sources	See References: 4, 21, 94, 95, 100

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Petrophile brevifolia

No widely used common name

Family **Proteaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Difficult *	Remove hairs, nick coat; smoke	Mar – May	3–5 mm	Not recorded

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	Best	Best	Best	OK	OK	OK	–	–	–	–

Best window | Workable – slower or less reliable | Avoid / low success

Plant snapshot

Habit	Erect, multi-stemmed shrub, 0.2–1 (–2) m, with short needle-like leaves.
Flowering	May – Dec (yellow to cream-white).
Habitat	Grey, yellow or brown sand over laterite or limestone; sandplains, sometimes swampy areas.
Local note	Recorded in Joondalup and Wanneroo.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Difficult	Petrophile cuttings take 3–10 months; 10–67% strike.	–
Seed	Main method	Remove hairs and nick the seed coat; slow.	This guide
Division	Not used	Shrub – not divided.	–

Seed collection & storage

Item	Details
Seed collection	Cones are usually ripe about six months after flowering, when grey to grey-brown. Cones over a year old hold mature fruit.
Extraction & cleaning	Keep cones warm and dry until the scales open; pull the hairy nuts out with tweezers.
Seed quality	Expect many empty 'false' fruits – cut-test.
Storage life	Not well documented. Store dry and cool.
Typical germination	Not recorded.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May**. Jun – Aug workable. Avoid Sep – Feb.
Seed mix	Free-draining, sterile, **low-phosphorus** mix.
Pre-treatment	**Remove the hairs and nick the seed coat** to show the white seed; smoke water may also help.
Sowing depth & method	Cover about **3–5 mm**. Direct sow into tubes.
After sowing	Keep moist, not wet; strict hygiene.
Germination time	Can take **several months**.
Pricking out	If tray-sown, prick out at the cotyledon stage.

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Item	Details
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 6–8 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
No germination	Seed coat not nicked, or empty nuts.
Seedlings die in pots	Phosphorus toxicity or dieback.
Seed damaged by nicking	Cut less deeply.

Notes

Other notes	Cuttings are hard in *Petrophile*. Record any trials so this page can be updated.
Confidence	**Medium.** No species data; follows WA Petrophile seed notes, as for *P. linearis*.
Sources	See References: 4, 21, 94, 95, 100

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Petrophile linearis

Pixie Mops

Family **Proteaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Difficult	**Remove hairs, nick coat; smoke**	**Mar – May**	**3–5 mm**	**<1% untreated**

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	Best	Best	Best	OK	OK	OK	–	–	–	–

Best window · Workable – slower or less reliable · Avoid / low success

Plant snapshot

Habit	Erect shrub, 0.2–1 m.
Flowering	Aug – Dec (pink and white ‘mops’).
Habitat	White, grey or yellow sand, often over laterite; sandplains.
Local note	Recorded in Joondalup and Wanneroo.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Not used	Not documented.	–
Seed	Difficult	Nick the seed coat; slow.	This guide
Division	Not used	–	–

Seed collection & storage

Item	Details
Seed collection	Cones are usually ripe about **six months after flowering**, when they turn grey to grey-brown and lose their old floral parts.
Extraction & cleaning	Seed does not come out easily in *Petrophile* – you may need to pull the hairy nuts out of the cone with forceps.
Seed quality	Very low germination untreated (0.17% in Alcoa tests). Many nuts may be empty – cut-test.
Storage life	Not well documented. Store dry and cool.
Typical germination	Untreated: **under 1%** (Alcoa, 6 tests). Treated seed should do much better – record your results.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May** (cool autumn). Jun – Aug workable. Avoid Sep – Feb.
Seed mix	Free-draining, sterile, **low-phosphorus** mix.
Pre-treatment	**Remove the hairs and nick a small section of the seed coat** to reveal the white seed (without damaging it). Smoke water (24 h soak) may also help.
Sowing depth & method	Cover about **3–5 mm**. Direct sow into tubes to avoid root damage.
After sowing	Keep moist, not wet; strict hygiene.
Germination time	Can take **several months** – keep trays going.

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Item	Details
Pricking out	If tray-sown, prick out at the cotyledon stage.
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 6–8 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
No germination	Seed coat not nicked, or empty seed. Nick carefully; cut-test.
Seed damaged by nicking	Cut too deep. Remove just enough coat to see white seed.
Seedlings die in pots	Phosphorus toxicity or root rot. Low-P mix and fertiliser.

Notes

Other notes	A slow, fiddly species from seed – set larger batches and keep good records.
Confidence	**Medium.** Nursery tests show very low untreated germination; method follows WA Petrophile seed notes.
Sources	See References: 3, 4, 21

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Petrophile macrostachya

No widely used common name

Family **Proteaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Difficult *	Remove hairs, nick coat; smoke	Mar – May	3–5 mm	Not recorded

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	Best	Best	Best	OK	OK	OK	–	–	–	–

Best window | Workable – slower or less reliable | Avoid / low success

Plant snapshot

Habit	Erect, compact or spreading, prickly shrub, 0.3–1.2 m.
Flowering	Jul – Nov (yellow-cream).
Habitat	White or yellow-brown sand and gravel; sandplains.
Local note	Recorded in the Wanneroo LGA. Prickly – wear gloves.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Not used	Not documented.	–
Seed	Difficult	Nick the seed coat; slow (trial).	This guide
Division	Not used	–	–

Seed collection & storage

Item	Details
Seed collection	Cones are usually ripe about six months after flowering, when grey to grey-brown and the old floral parts have gone.
Extraction & cleaning	Pull the hairy nuts from the cones with forceps.
Seed quality	Cut-test a sample – expect many empty nuts.
Storage life	Not well documented. Store dry and cool.
Typical germination	Not recorded. Expect low germination unless the seed coat is treated.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May**. Jun – Aug workable. Avoid Sep – Feb.
Seed mix	Free-draining, sterile, low-phosphorus mix.
Pre-treatment	Remove the hairs and nick a small section of the seed coat; smoke water may also help.
Sowing depth & method	Cover about **3–5 mm**. Direct sow into tubes.
After sowing	Keep moist, not wet; strict hygiene.
Germination time	Can take **several months**.
Pricking out	If tray-sown, prick out at the cotyledon stage.

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Item	Details
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 6–8 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
No germination	Seed coat not nicked, or empty seed.
Seed damaged by nicking	Cut less deeply.
Seedlings die in pots	Phosphorus toxicity or root rot.

Notes

Other notes	Treat as a trial – record results so this page can be updated.
Confidence	**Low.** No species data found. Method based on WA Petrophile seed notes and *P. linearis* results – trial protocol.
Sources	See References: 4, 21

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Philotheca spicata

Pepper and Salt

Family **Rutaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Moderate	Smoke water	Mar – May	Lightly covered	35%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	Best	Best	Best	OK	OK	OK	–	–	–	–

Best window Workable – slower or less reliable Avoid / low success

Plant snapshot

Habit	Slender, erect shrub, 0.2–0.6 m (occasionally to 1.2 m).
Flowering	Jun – Nov (pink-purple, blue or white).
Habitat	A variety of soils.
Local note	Recorded in Joondalup and Wanneroo.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Trial	Varies in the genus – some strike readily, some are very difficult.	–
Seed	Main method	Smoke water; about 35% (Alcoa). Bag fruit – seed is shed quickly.	This guide
Division	Not used	Shrub – not divided.	–

Seed collection & storage

Item	Details
Seed collection	Seed is released as soon as it ripens, and all of it within 1–2 weeks (genus). **Bag fruiting stems** after flowering (estimate: Nov – Dec) and check often.
Extraction & cleaning	Dry the bags, then sieve out the seed.
Seed quality	Smooth, black seed (genus), roughly kidney-shaped, from fruit segments about 3 mm long. Cut-test a few.
Storage life	Not documented. Store dry and cool.
Typical germination	With smoke about **35%** (Alcoa, 5 tests), about **21 seedlings per gram**. Untreated, only about 6% (5 tests).

Sowing & growing on

Item	Details
Best sowing time	**Mar – May**. Jun – Aug workable. Avoid Sep – Feb (nursery tests ran cool, at about 15–20°C).
Seed mix	Standard sterile native seed mix, free-draining.
Pre-treatment	**Smoke water** – 24 h soak (1:9) or water the trays with it. Across 112 NSW Rutaceae species, gibberellic acid plus smoke averaged 62% (GA alone 40%, untreated 13%, smoke alone 9%) – GA is worth trialling on part of a batch.
Sowing depth & method	Sow thinly and cover lightly.
After sowing	Keep moist, not wet.
Germination time	Not documented – keep trays at least 3 months.

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Item	Details
Pricking out	At 2–3 leaves, gently.
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 5–6 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Empty bags	Seed released before bagging. Bag earlier, as soon as flowering finishes.
Low germination	No smoke. Use smoke water on every batch.
Seedlings collapse	Damping off. Sterile mix, airflow.

Notes

Other notes	Cuttings vary in the genus – some philothecas strike readily from new growth, others are very difficult. Worth a small trial.
Confidence	**Medium.** Nursery data (5 tests with smoke); smoke response confirmed in WA research.
Sources	See References: 3, 4, 60, 124, 125

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Pigea calycina

Wild Violet

Family **Violaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Difficult *	Smoke water	Apr – May	2–3 mm	0% untreated (1 test)

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	OK	Best	Best	OK	OK	–	–	–	–	–

Best window | Workable – slower or less reliable | Avoid / low success

Plant snapshot

Habit	Straggly perennial herb, 0.1–0.6 m.
Flowering	Mainly Jul – Oct (blue-purple violet-like flowers).
Habitat	Sandy soils.
Local note	Recorded in Joondalup and Wanneroo. Renamed from *Hybanthus calycinus* in 2021.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Trial	An eastern relative grows from cuttings.	–
Seed	Difficult	0% untreated; smoke helps in the genus (trial).	This guide
Division	Not used	–	–

Seed collection & storage

Item	Details
Seed collection	Capsules split when ripe (late spring–summer). Bag developing capsules with fine mesh.
Extraction & cleaning	Collect the bags once capsules have opened; sieve the seed.
Seed quality	Cut-test a sample.
Storage life	Not well documented. Store dry and cool.
Typical germination	0% untreated in the only Alcoa test. Smoke promotes germination in *Hybanthus* species with low germination.

Sowing & growing on

Item	Details
Best sowing time	**Apr – May**. Mar, Jun – Jul workable.
Seed mix	Standard sterile native seed mix.
Pre-treatment	**Smoke water** – 24 h soak (1:9), or water trays in with it.
Sowing depth & method	Cover about **2–3 mm**.
After sowing	Keep moist, not wet.
Germination time	Not recorded.
Pricking out	At the second leaf stage.

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Item	Details
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Record your own.

Common problems

Problem	Likely cause and fix
No germination	Try smoke water and aged seed; sow generously.
Capsules empty when collected	Seed already shed. Bag capsules earlier.
Seedlings collapse	Damping off. Sterile mix; airflow.

Notes

Other notes	Soft tip cuttings are worth a trial – an eastern relative grows from cuttings.
Confidence	**Low.** The only nursery test (untreated) gave 0%; smoke is reported to promote germination in the genus. Trial protocol.
Sources	See References: 3, 4, 60

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Poa poiformis

Coastal Poa

Family **Poaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Easy–Moderate *	Chill 1–2 weeks	Mar – May	Just covered	Not recorded

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	Best	Best	Best	OK	OK	OK	OK	OK	–	–

Best window | Workable – slower or less reliable | Avoid / low success

Plant snapshot

Habit	Densely tufted perennial tussock grass, 0.15–0.9 m, with fine blue-green leaves.
Flowering	Oct – Nov (green to yellow).
Habitat	Sand, granite and sandy loam; coastal areas, estuaries and dunes.
Local note	Recorded in the Perth area (Cottesloe, Stirling); FloraBase has no Joondalup or Wanneroo specimens. Tolerates salt spray – good for dune stabilisation.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Not used	Grasses are not grown from cuttings.	–
Seed	Main method	Chill 1–2 weeks; sow at 15°C or more.	This guide
Division	Good	Divide mature clumps in early spring.	→ Division Guide

Seed collection & storage

Item	Details
Seed collection	Not documented. Seed ripens after the Oct – Nov flowering: collect heads in early summer as they turn straw-coloured and seed falls easily.
Extraction & cleaning	Rub heads and sieve out the seed.
Seed quality	Cut-test a sample.
Storage life	Not well documented. Store dry and cool.
Typical germination	Not recorded.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May** (autumn). Jun – Oct workable. Needs at least 15°C to germinate well.
Seed mix	Standard sterile native seed mix.
Pre-treatment	**Chill the seed in the fridge for 1–2 weeks** before sowing (seed supplier's advice).
Sowing depth & method	Just cover, in trays or a pinch per tube.
After sowing	Keep moist, not wet.
Germination time	About **2–8 weeks**.
Pricking out	At 2–3 leaves, in small clumps.

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Item	Details
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 3–4 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Slow, patchy germination	Too cold. Sow when it's at least 15°C.
Seedlings collapse	Damping off. Sow thinly.
Leggy seedlings	Too much shade.

Notes

Other notes	Mature clumps can also be divided (see the Division Guide).
Confidence	**Low.** Only a seed supplier's method found; no WA nursery or research data.
Sources	See References: 4, 91, 98

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Podotheca angustifolia

Sticky Longheads

Family **Asteraceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Easy–Moderate	None	Apr – May	Surface – light cover	52–96% in SA lab tests

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	OK	Best	Best	OK	–	–	–	–	–	–

Best window · Workable – slower or less reliable · Avoid / low success

Plant snapshot

Habit	Low, often sticky annual herb, 0.03–0.2 m (to 0.3 m) – sprawling to upright.
Flowering	Aug – Dec (yellow, long narrow heads).
Habitat	Sand or sandy gravel over limestone, granite or laterite; coastal and inland.
Local note	Recorded in the Wanneroo LGA on FloraBase, and listed as indigenous at Warwick Bushland, Joondalup. An annual – it flowers the spring after sowing, then dies.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Not used	Annual.	–
Seed	Main method	Annual everlasting – no pre-treatment; sow after autumn rains.	This guide
Division	Not used	Annual.	–

Seed collection & storage

Item	Details
Seed collection	Cut mature fruiting heads as they dry and put them somewhere warm to finish drying – collected in November in SA; estimate Oct – Dec here.
Extraction & cleaning	Separate the seed from the dried heads and remove unwanted material (SA). Sow the seed with its bristles.
Seed quality	Grey-brown, finely hairy seed 2–4 mm long with five feathery bristles. SA seed lots were 95% viable.
Storage life	Store dry and cool; sow the following autumn. SA seed-bank lots are held at −18°C.
Typical germination	No nursery germination % recorded. Alcoa records only 145.92 from 3 tests – too high to be a %, so probably about **146 seedlings per gram**. In SA lab trials with no pre-treatment, 96% germinated under winter conditions and 52% at 10/22°C. A 2025 lab study got none (seed viability not tested) – expect variable results.

Sowing & growing on

Item	Details
Best sowing time	**Apr – May**, after the first autumn rains. Mar and Jun workable. Germination was best under winter conditions in SA trials.
Seed mix	Standard sterile native seed mix.
Pre-treatment	**None needed.**
Sowing depth & method	Surface-sow and cover very lightly. Sowing a pinch straight into each cell avoids pricking out.

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Item	Details
After sowing	Water in with a fine spray; keep moist.
Germination time	First seedlings in about **8 days**, half up by about 17 days (SA lab); 7–14 days (mean 9) in 7 Kings Park lots.
Pricking out	If tray-sown, prick out small, at the second-leaf stage. Thin direct-sown cells to 1–3 seedlings.
Shade house & hardening	Short – a week or so in the shade house, then full sun.
Time to planting size	Plant out small, about **6–10 weeks** after sowing (estimate), so plants establish in winter and flower in spring.

Common problems

Problem	Likely cause and fix
Very low germination	Seed lots vary. Sow generously and record each lot.
Leggy seedlings	Too much shade. Move to full sun early.
Plants flower in the tube	Held too long. Plant out while small.

Notes

Other notes	Handle like *Podotheca gnaphalioides*. Annual *Podotheca* was among the most common plants coming up from transferred topsoil in Perth Banksia woodland restoration. Common after fire (SA).
Confidence	**Medium.** SA seed-bank germination trials and nursery seedlings-per-gram data (3 tests); a 2025 lab study got no germination, so results vary.
Sources	See References: 3, 4, 143, 146, 149, 150, 151

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Podotheca gnaphalioides

Golden Long-heads

Family **Asteraceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Moderate	Smoke water (optional)	Apr – May	Surface – light cover	7%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	OK	Best	Best	OK	–	–	–	–	–	–

Best window | Workable – slower or less reliable | Avoid / low success

Plant snapshot

Habit	Annual herb, 0.02–0.6 m.
Flowering	Aug – Nov (golden-yellow, long narrow heads).
Habitat	Sandy soils.
Local note	Recorded in Joondalup and Wanneroo. An annual – it flowers the spring after sowing, then dies.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Not used	Annual.	–
Seed	Main method	Annual everlasting – sow after the first autumn rains.	This guide
Division	Not used	Annual.	–

Seed collection & storage

Item	Details
Seed collection	Collect heads in late spring–early summer once they dry and the fluffy seed starts to lift.
Extraction & cleaning	Rub heads gently over a sieve. The seed can be sown with its fluff.
Seed quality	Germination in nursery tests was low (about 7%) – sow generously.
Storage life	Store dry and cool and sow the following autumn.
Typical germination	About **7%** (untreated, Alcoa nursery average, 8 tests). About **84 seedlings per gram** – roughly 1.2 g of seed per 100 seedlings.

Sowing & growing on

Item	Details
Best sowing time	**Apr – May**, after the first autumn rains. Mar and Jun are workable. Later sowings flower poorly.
Seed mix	Standard sterile native seed mix.
Pre-treatment	None required. Smoke water in the first watering may help (it helps many annual daisies).
Sowing depth & method	Surface-sow and cover very lightly – daisy seed often needs light. Sowing a pinch straight into each tube or cell avoids pricking out.
After sowing	Water in with a fine spray; keep moist.
Germination time	About **2–4 weeks**.
Pricking out	If tray-sown, prick out small, at the second-leaf stage. Thin direct-sown cells to 1–3 seedlings.

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Item	Details
Shade house & hardening	Short – a week or so in the shade house, then full sun.
Time to planting size	Plant out small, about **6–10 weeks** after sowing, so plants establish in winter and flower in spring.

Common problems

Problem	Likely cause and fix
Very low germination	Normal for this species. Sow thickly; try smoke water.
Leggy seedlings	Too much shade. Move to full sun early.
Plants flower in the tube	Held too long. Plant out while small.

Notes

Other notes	Often direct-seeded into revegetation sites after the first autumn rains.
Confidence	**Medium.** Nursery germination data (8 tests); sowing follows standard practice for annual everlastings.
Sources	See References: 3, 4, 30

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Ptilotus drummondii

Narrowleaf Mulla Mulla

Family **Amaranthaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Moderate–Difficult	Remove woolly floral parts; smoke	Mar – May	1–2 mm	4%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	OK	Best	Best	Best	OK	OK	OK	OK	–	–	–

Best window · Workable – slower or less reliable · Avoid / low success

Plant snapshot

Habit	Perennial herb, 0.05–0.8 m.
Flowering	Mar, May and Jul – Dec (white, cream or pink woolly spikes).
Habitat	Sandy soils.
Local note	Recorded in the Wanneroo LGA.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Possible	Cuttings have worked in the genus; not documented for this species.	–
Seed	Main method	Remove the woolly floral parts to lift germination.	This guide
Division	Not used	–	–

Seed collection & storage

Item	Details
Seed collection	Seed is held inside the woolly flowers. Collect spikes when the flowers are **papery and very dry** and rub off easily.
Extraction & cleaning	Rub spikes over a sieve to separate the individual flowers – each holds at most one seed.
Seed quality	Many flowers are empty. Check a few by rubbing off the wool.
Storage life	Not well documented. Store dry and cool.
Typical germination	Untreated: about **4%** (Alcoa, 29 tests), about 12 seedlings per gram of whole flowers. In the related *P. exaltatus*, removing the woolly floral parts raised germination to 60–80%.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May**. Feb and Jun – Sep are workable. Avoid Oct – Jan.
Seed mix	Standard sterile seed mix, extra free-draining.
Pre-treatment	**Remove the woolly floral parts** by rubbing the flowers gently on a sieve or between rough boards to free the seed. Heat and cold treatments did not help in *P. exaltatus*. Water in with smoke water (Geraldton method for *Ptilotus*).
Sowing depth & method	Cover about **1–2 mm**. Sow cleaned seed thinly, or 2–3 seeds direct into tubes.
After sowing	Keep just moist – *Ptilotus* rot easily if wet or humid.
Germination time	About **6–10 days** for other *Ptilotus* at Geraldton; untreated seed is slow and erratic.

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Item	Details
Pricking out	Early and carefully, at the first true leaves – or sow direct into tubes.
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 3–4 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Almost no germination	Wool left on, or empty flowers. Clean the seed; sow more.
Seedlings collapse	Too wet. Water less; good drainage and airflow.
Plants short-lived in pots	Humidity and wet mix. Full sun, very free-draining mix.

Notes

Other notes	Cuttings have been successful in the genus (ANPSA) – worth trialling from non-flowering shoots.
Confidence	**Medium.** Nursery test data (29 tests) show low untreated germination; treatment is based on research with *P. exaltatus*.
Sources	See References: 3, 4, 5, 28, 29

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Ptilotus manglesii

Pom Poms

Family **Amaranthaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Moderate–Difficult	**Remove woolly floral parts; smoke**	**Mar – May**	**1–2 mm**	**3%**

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	OK	Best	Best	Best	OK	OK	OK	OK	–	–	–

Best window | Workable – slower or less reliable | Avoid / low success

Plant snapshot

Habit	Low perennial herb, 0.05–0.3 m.
Flowering	Sep – Dec or Jan (pink to magenta pom-pom heads).
Habitat	Sandy and gravelly soils.
Local note	Recorded in Joondalup and Wanneroo. Showy, but often short-lived in cultivation.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Possible	Cuttings have worked in the genus; not documented for this species.	–
Seed	Main method	Remove the woolly floral parts to lift germination.	This guide
Division	Not used	–	–

Seed collection & storage

Item	Details
Seed collection	Collect heads when the flowers are **papery and very dry** (early summer).
Extraction & cleaning	Rub heads over a sieve to separate the individual flowers.
Seed quality	Many flowers are empty. Check a few by rubbing off the wool.
Storage life	Not well documented. Store dry and cool.
Typical germination	Untreated: about **3%** (Alcoa, 11 tests), about 6 seedlings per gram of whole flowers. Removing the woolly floral parts raised germination to 60–80% in the related *P. exaltatus*.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May**. Feb and Jun – Sep are workable. Avoid Oct – Jan.
Seed mix	Standard sterile seed mix, extra free-draining.
Pre-treatment	**Remove the woolly floral parts** to free the seed, then water in with smoke water after sowing.
Sowing depth & method	Cover about **1–2 mm**. Sow cleaned seed thinly, or direct into tubes.
After sowing	Keep just moist; good airflow.
Germination time	About **6–10 days** for other *Ptilotus* at Geraldton; untreated seed is slow and erratic.
Pricking out	Early and carefully, at the first true leaves – or sow direct into tubes.

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Item	Details
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 3–4 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Almost no germination	Wool left on, or empty flowers. Clean the seed; sow more.
Seedlings collapse	Too wet. Water less; improve drainage.
Plants die after potting on	Overwatering or humidity. Full sun, free-draining mix.

Notes

Other notes	Cuttings have been successful in the genus (ANPSA) – worth trialling.
Confidence	**Medium.** Nursery test data (11 tests) show low untreated germination; treatment is based on research with *P. exaltatus*.
Sources	See References: 3, 4, 5, 28, 29

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Ptilotus polystachyus

Prince of Wales Feather

Family **Amaranthaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Moderate *	Remove woolly floral parts; smoke	Mar – May	1–2 mm	Not recorded

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	OK	Best	Best	Best	OK	OK	OK	OK	–	–	–

Best window | Workable – slower or less reliable | Avoid / low success

Plant snapshot

Habit	Annual or short-lived perennial herb, 0.15–1.5 m.
Flowering	May – Nov or Jan (green, white or yellow feathery spikes).
Habitat	Sandy soils.
Local note	Recorded in the Perth area (Queens Park). Short-lived – sow each year.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Possible	Cuttings have worked in the genus; not documented for this species.	–
Seed	Main method	Remove the woolly floral parts to lift germination.	This guide
Division	Not used	Annual or short-lived.	–

Seed collection & storage

Item	Details
Seed collection	Collect spikes when the flowers are papery and very dry and rub off easily.
Extraction & cleaning	Rub spikes over a sieve to separate the individual flowers.
Seed quality	Expect many empty flowers – check a few.
Storage life	Not well documented. Store dry and cool.
Typical germination	Not recorded. Untreated *Ptilotus* seed usually germinates poorly.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May**. Feb and Jun – Sep are workable.
Seed mix	Standard sterile seed mix, extra free-draining.
Pre-treatment	**Remove the woolly floral parts** to free the seed; water in with smoke water.
Sowing depth & method	Cover about **1–2 mm**, or sow direct into tubes.
After sowing	Keep just moist; good airflow.
Germination time	About **6–10 days** for other *Ptilotus*; record your own.
Pricking out	Early and carefully, or sow direct into tubes.

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Item	Details
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 3–4 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Almost no germination	Wool left on, or empty flowers.
Seedlings collapse	Too wet. Water less.
Weak, floppy plants	Too much shade. Full sun.

Notes

Other notes	Treat as a trial and record results so this page can be updated.
Confidence	**Low.** No species data found. Method follows other *Ptilotus* and research with *P. exaltatus* – trial protocol.
Sources	See References: 4, 5, 28, 29

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Rhagodia baccata

Berry Saltbush

Family **Amaranthaceae (chenopods)**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Moderate	Ferment berries; smoke-water soak	Mar – May	2–3 mm	11%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	Best	Best	Best	OK	OK	OK	OK	–	–	–

Best window | Workable – slower or less reliable | Avoid / low success

Plant snapshot

Habit	Spreading shrub, 0.3–2 m, with grey-green leaves.
Flowering	Variable – Feb, Apr – May or Oct – Dec, followed by red berries.
Habitat	Sand dunes and coastal rock.
Local note	Recorded in Joondalup and Wanneroo. FloraBase uses *Rhagodia baccata*; national lists use *Chenopodium baccatum*.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Good	Chenopod shrubs generally strike well (provisional).	→ Cuttings Guide
Seed	Main method	Ferment berries, then smoke water.	This guide
Division	Not used	Woody shrub.	–

Seed collection & storage

Item	Details
Seed collection	Pick fully ripe, deep red berries.
Extraction & cleaning	**Ferment:** soak the berries in water for 4–5 days, wash off the pulp through a sieve and dry the seed.
Seed quality	Small, dark seeds; discard any that float after washing.
Storage life	Not well documented. Store dry and cool.
Typical germination	About **11%** (untreated, Alcoa nursery average, 7 tests). About **75 seedlings per gram** – roughly 1.3 g of seed per 100 seedlings.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May**. Jun – Sep workable. Avoid Oct – Feb.
Seed mix	Standard sterile native seed mix.
Pre-treatment	Ferment and clean (above), then **soak in smoke water (1:9) for 24 hours** (Geraldton method for *R. preissii*).
Sowing depth & method	Cover about **2–3 mm**. Sow thinly in trays.
After sowing	Water in; keep moist, not wet.
Germination time	About **7–14 days** (Geraldton, *R. preissii*).

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Item	Details
Pricking out	When the second pair of leaves appears.
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 3–4 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Poor germination	Pulp left on, or unripe berries. Ferment fully; use ripe fruit.
Mould on trays	Pulp residue. Wash seed well.
Seedlings collapse	Damping off. Sow thinly; good airflow.

Notes

Other notes	Tip cuttings are worth trialling (see the Cuttings Guide).
Confidence	**Medium.** Nursery test data (7 tests); treatment follows the WA community-nursery method for *Rhagodia*.
Sources	See References: 3, 4, 5, 37

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Ricinocarpos undulatus

Wedding Bush

Family **Euphorbiaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Difficult *	Smoke water (trial GA)	Mar – May	Covered ~5 mm (dark)	Not recorded

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	OK	Best	Best	Best	OK	OK	OK	OK	OK	–	–

Best window | Workable – slower or less reliable | Avoid / low success

Plant snapshot

Habit	Erect, woody shrub, 0.5–1 m (to 1.5 m).
Flowering	Sep – Oct (white).
Habitat	White sand over limestone, sandy loam, sandy clay, gravel and laterite; plains, ridges and road verges.
Local note	Recorded in Joondalup and Wanneroo. The similar *R. glaucus* also grows on the Swan Coastal Plain – check labels.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Trial	Eastern relatives: half-ripened growth, can be difficult to strike.	–
Seed	Difficult	Dormant; smoke water, trial GA (related R. brevis).	This guide
Division	Not used	Shrub – not divided.	–

Seed collection & storage

Item	Details
Seed collection	Fruits mature Sep – Nov. The capsules (8–10 mm) split into three when ripe – collect them as they turn brown, or bag branches.
Extraction & cleaning	Dry in a paper bag to release the seed, then sieve (genus). Leave the cream caruncle on.
Seed quality	Good seed is plump and shiny, 5–6 × about 3.8 mm, with the caruncle intact. Cut-test a few. The caruncle attracts ants – keep seed covered.
Storage life	Not documented. Store dry and cool and record how old each lot is.
Typical germination	Not tested in WA nurseries. In the related *R. brevis* the seed has physiological dormancy and germinated better with smoke water, karrikinolide and gibberellic acid. Priming – wetting the seed for a few days, then drying it back before sowing – also helped (Kings Park).

Sowing & growing on

Item	Details
Best sowing time	**Mar – May** (estimate). Feb and Jun – Oct workable. Avoid Nov – Jan.
Seed mix	Standard sterile native seed mix, free-draining.
Pre-treatment	Soak in **smoke water** for 24 hours (1:9) before sowing. As a trial, treat part of the batch with gibberellic acid instead and compare.
Sowing depth & method	**Cover the seed** (about 5 mm) – light suppressed germination in *R. brevis*, which germinated across 15–30°C.

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Item	Details
After sowing	Keep moist, not wet, and keep trays at least 3 months.
Germination time	Not documented. An eastern species germinates in 4–5 weeks from fresh seed.
Pricking out	At the first true leaves, gently.
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 5–6 months from germination (record your own).

Common problems

Problem	Likely cause and fix
Nothing comes up	Dormant seed. Keep the trays; try smoke water plus gibberellic acid on the next batch and record what worked.
Seed missing from trays	Ants carry off seed with a caruncle. Stand trays where ants can't reach them.
Seedlings collapse	Damping off. Sterile mix, airflow.

Notes

Other notes	Cuttings are worth a small trial – eastern relatives are grown from half-ripened current season's growth, but it can be difficult to strike.
Confidence	**Low.** No germination data for this species. Method based on Kings Park research on the related *R. brevis* – treat as a trial.
Sources	See References: 4, 106, 107, 108, 109

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Santalum acuminatum

Quandong

Family **Santalaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Moderate–Difficult	Aged seed; crack shell; GA	May – Jul	About the stone's thickness	53%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	–	OK	Best	Best	Best	OK	–	–	–	–

Best window · Workable – slower or less reliable · Avoid / low success

Plant snapshot

Habit	Small tree or shrub, 1–7 m. A **root hemiparasite** – needs host plants.
Flowering	Variable (green-white to red-brown); shiny red fruits ripen in late spring–summer.
Habitat	Sand and clay loam; coastal dunes, gravelly plains, granite outcrops and creek beds.
Local note	Recorded in the City of Joondalup. The fruit is a well-known bush food.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Not used	Grafting onto seedling stock is used instead.	–
Seed	Main method	Aged seed, cracked shell, gibberellin; sow with a host.	This guide
Division	Not used	Root hemiparasite.	–

Seed collection & storage

Item	Details
Seed collection	Collect fruits once they turn shiny red, in late spring–summer.
Extraction & cleaning	Remove the flesh and dry the hard, pitted stones.
Seed quality	Check stones are heavy and sound; discard cracked or light ones.
Storage life	Needs **after-ripening**: germination rose from about 5–30% after one year's storage to 35% after three years (ANBG). Don't let treated seed dry out again – drying then rewetting inhibits germination.
Typical germination	About **53%** after a silica + gibberellin treatment (Alcoa, 1 test). Stones are large – about 0.13 seedlings per gram, roughly 770 g of stones per 100 seedlings.

Sowing & growing on

Item	Details
Best sowing time	**May – Jul** – germination is best at about 15°C, so winter suits it. Apr and Aug workable.
Seed mix	Free-draining, sterile mix in deep pots.
Pre-treatment	• Use stored (after-ripened) seed. • **Crack the shell** gently in a vice, or saw a nick into it (sow with the nick facing down). • Gibberellins improve germination – research found GA4 far more effective than GA3.
Sowing depth & method	Cover the stone to about its own thickness. **Sow with a host plant** in the same pot, e.g. a native grass, legume or small shrub.

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Item	Details
After sowing	Keep moist, not wet, through winter.
Germination time	Usually within a month but can take up to 3–4 months.
Pricking out	Avoid – sow direct into the pot with its host.
Shade house & hardening	Grow on with the host in light shade, then harden off to full sun.
Time to planting size	Est. 6–12 months (record your own). Plant out with its host, or next to a suitable host.

Common problems

Problem	Likely cause and fix
No germination	Fresh seed or intact shell. Use seed stored 1+ years; crack or nick the shell.
Seedlings stall after a few months	No host. Pot up with a host plant (e.g. *Acacia saligna*, a native grass).
Stones rot	Too wet. Free-draining mix; water less.

Notes

Other notes	Grafting onto seedling stock is also used for fruit production. Acacia saligna was the best host in a Jurien Bay trial.
Confidence	**High.** Well researched: after-ripening, seed-coat cracking, gibberellins and host plants are all published.
Sources	See References: 3, 4, 67, 68, 69, 70

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Spinifex hirsutus

Hairy Spinifex

Family **Poaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Moderate *	Cut off spines/awns; smoke water	Mar – May	10 mm (needs dark)	Not recorded

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	OK	OK	Best	Best	Best	–	–	–	OK	OK	OK	OK

Best window | Workable – slower or less reliable | Avoid / low success

Plant snapshot

Habit	Rhizomatous, tussock-forming grass, 0.3–0.9 m, with long runners and silvery, hairy leaves.
Flowering	Oct – Jan. Male and female flowers on separate plants; female heads are large spiky balls.
Habitat	White sand on coastal dunes.
Local note	Recorded in the City of Joondalup.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Good	Runner (stolon) sections with 1–2 nodes.	→ Cuttings Guide
Seed	Main method	The usual nursery method; cover seed (needs dark).	**This guide**
Division	Possible	Tussock divisions work but survived less well than rooted runners.	→ Division Guide

Seed collection & storage

Item	Details
Seed collection	Collect the spiky female heads in summer as they ripen and start to break away. **Collect from female plants growing near male plants** – isolated females set little seed.
Extraction & cleaning	Pull the seed-bearing spikelets from the head and **cut off the long spines (awns)**.
Seed quality	Many spikelets are empty – squeeze-test a few.
Storage life	Not well documented. Store dry and cool.
Typical germination	Not recorded for this species. In *S. sericeus* trials, seed germinated far better in darkness (56%) than in light (14%).

Sowing & growing on

Item	Details
Best sowing time	**Mar – May** (provisional). Sep – Feb workable. Avoid Jun – Aug.
Seed mix	Coarse, sandy, free-draining mix.
Pre-treatment	Cut off the awns; water in with smoke water after sowing (Geraldton method).
Sowing depth & method	**Cover well (about 10 mm)** – seed germinates best in the dark. Sow en masse in deep trays.
After sowing	Water in with smoke water; keep moist, not wet.
Germination time	About **10 days** (Geraldton nursery).
Pricking out	When seedlings have 2–3 leaves, into deep tubes.

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Item	Details
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Slow – est. 6–9 months from sowing (record your own). Plant out in late autumn or winter.

Common problems

Problem	Likely cause and fix
Little germination	Seed from isolated females, or sown too shallow. Collect near males; cover seed.
Seedlings rot	Mix too wet. Use coarse sand and water less.
Slow growth	Normal for spinifex. Full sun, deep tubes, light feeding.

Notes

Other notes	Rooted runner sections (Cuttings Guide) and tussock division (Division Guide) are alternatives.
Confidence	**Medium.** WA community-nursery method for spinifex plus trial results for the closely related *S. sericeus*.
Sources	See References: 4, 5, 36

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Spinifex longifolius

Beach Spinifex

Family **Poaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Moderate *	Cut off spines/awns; smoke water	Mar – May	10 mm (needs dark)	Not recorded

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	OK	OK	Best	Best	Best	–	–	–	OK	OK	OK	OK

Best window · Workable – slower or less reliable · Avoid / low success

Plant snapshot

Habit	Rhizomatous, tussock-forming grass, 0.3–1 m high and up to 2 m wide, with long runners.
Flowering	Apr – Dec or Jan. Male and female plants are separate; female heads up to 250 mm across.
Habitat	White sand on coastal dunes; a key foredune stabiliser.
Local note	Recorded in the Wanneroo LGA.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Good	Runner (stolon) sections with 1–2 nodes.	→ Cuttings Guide
Seed	Main method	The usual nursery method; smoke water, about 10 days.	**This guide**
Division	Possible	Tussock divisions work but survived less well than rooted runners.	→ Division Guide

Seed collection & storage

Item	Details
Seed collection	Collect ripe female heads as they begin to break away. **Choose female plants growing close to males**.
Extraction & cleaning	Separate the seed from the head and **cut off the awns**.
Seed quality	Many spikelets are empty – squeeze-test a few.
Storage life	Not well documented. Store dry and cool.
Typical germination	Not recorded for this species. In *S. sericeus* trials, seed germinated far better in darkness (56%) than in light (14%).

Sowing & growing on

Item	Details
Best sowing time	**Mar – May** (provisional). Sep – Feb workable. Avoid Jun – Aug.
Seed mix	Coarse, sandy, free-draining mix.
Pre-treatment	Cut off the awns and **sow en masse with smoke water** (Geraldton method).
Sowing depth & method	**Cover well (about 10 mm)** – seed germinates best in the dark. Deep trays.
After sowing	Keep moist, not wet.
Germination time	About **10 days** (Geraldton nursery).
Pricking out	When seedlings have 2–3 leaves, into deep tubes.

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Item	Details
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Slow – est. 6–9 months from sowing (record your own). Plant out in late autumn or winter.

Common problems

Problem	Likely cause and fix
Little germination	Seed from isolated females, or sown too shallow. Collect near males; cover seed.
Seedlings rot	Mix too wet. Use coarse sand and water less.
Slow growth	Normal for spinifex. Full sun, deep tubes.

Notes

Other notes	Rooted runner sections (Cuttings Guide) and tussock division (Division Guide) are alternatives.
Confidence	**Medium.** WA community-nursery method for this species plus trial results for the closely related *S. sericeus*.
Sources	See References: 4, 5, 36, 37

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Spyridium globulosum

Basket Bush

Family **Rhamnaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Moderate	Hot water (or dry heat)	Apr – Jun	2–3 mm (needs dark)	41%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	OK	Best	Best	Best	OK	OK	–	–	–	–

Best window | Workable – slower or less reliable | Avoid / low success

Plant snapshot

Habit	Erect shrub, usually 0.6–5 m.
Flowering	Jun – Nov (white, in woolly heads).
Habitat	Sand on coastal dunes and limestone.
Local note	Recorded in Joondalup and Wanneroo.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Not used	Not documented.	–
Seed	Main method	Hot water; cool, dark conditions to germinate.	This guide
Division	Not used	–	–

Seed collection & storage

Item	Details
Seed collection	Collect fruit as it ripens in late spring–summer; bag branches to catch shed seed.
Extraction & cleaning	Rub and sieve the small seed from the woolly flower heads.
Seed quality	Water-impervious seed coat (physical dormancy) – seed won't take up water until treated.
Storage life	Not well documented; hard-coated seed is usually long-lived when dry.
Typical germination	About **41%** after heat treatment (Alcoa, only 2 tests). About **205 seedlings per gram**. Research recorded high germination after hot water at cool (7/18°C) temperatures.

Sowing & growing on

Item	Details
Best sowing time	**Apr – Jun** – germinates best at winter-like temperatures (7/18°C); warmth suppresses it. Mar, Jul – Aug workable. Avoid Sep – Feb.
Seed mix	Standard sterile native seed mix.
Pre-treatment	**Hot water** (about 90–95°C) breaks the seed-coat dormancy – keep it short, as over-long treatment can kill seed. Alcoa used a dry-heat pulse instead.
Sowing depth & method	Cover about **2–3 mm** – germination is suppressed by light, so keep the seed covered and trays shaded.
After sowing	Water in; keep moist in a cool, shaded spot.
Germination time	Not well documented – allow several weeks and record results.

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Item	Details
Pricking out	When the second pair of leaves appears.
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 4–6 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
No germination	Seed coat not broken, or trays too warm or too light. Re-treat; cover seed; sow in cooler months.
Seed killed by treatment	Hot water too long. Shorten the soak.
Damping off	Wet, crowded trays. Thin sowing, airflow.

Notes

Other notes	One of the few coastal shrubs here that needs cool temperatures *and* darkness to germinate.
Confidence	**High.** Species-specific dormancy research and nursery data available.
Sources	See References: 3, 4, 22, 23

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Stirlingia latifolia

Blueboy

Family **Proteaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Difficult	**Smoke water (trial)**	**Apr – Jun**	**3–5 mm**	**2%**

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	OK	Best	Best	Best	OK	–	–	–	–	–

Best window · Workable – slower or less reliable · Avoid / low success

Plant snapshot

Habit	Erect, lignotuberous shrub, 0.2–1.5 m. Resprouts after fire.
Flowering	Aug – Oct (yellow to red-brown).
Habitat	White, grey, yellow-brown or black sand, sometimes with lateritic gravel.
Local note	Recorded in the Wanneroo LGA. Widely bush-picked for the dried-flower trade.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Difficult	Best from new shoots after fire; tissue culture works better.	–
Seed	Difficult	About 2% in nursery tests; smoke water (trial).	This guide
Division	Not used	Lignotuberous; not documented as a method.	–

Seed collection & storage

Item	Details
Seed collection	Collect dry, mature heads after flowering (late spring–summer).
Extraction & cleaning	Rub heads over a sieve to free the hairy nuts.
Seed quality	Viability is often low (a seed supplier quotes 20–40%) – cut-test a sample and expect many empty nuts.
Storage life	Not well documented. Store dry and cool.
Typical germination	Very low: about **2%** untreated (Alcoa, 6 tests), about 1.8 seedlings per gram – roughly 55 g of seed per 100 seedlings.

Sowing & growing on

Item	Details
Best sowing time	**Apr – Jun** (cool, about 15–23°C). Mar and Jul workable.
Seed mix	Free-draining, pathogen-free, low-phosphorus mix.
Pre-treatment	**Smoke water** (24 h soak or first watering) – *Stirlingia* is one of the hard-to-germinate genera that responds to smoke. Research found seed buried in soil for longer germinated even without smoke, so older seed may do better. Treat as a trial.
Sowing depth & method	Cover about **3–5 mm**. Sow generously in deep trays.
After sowing	Keep moist, not wet. Keep trays going for months.
Germination time	Slow and erratic.

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Item	Details
Pricking out	At the cotyledon stage, into deep tubes.
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Slow – allow 6+ months (record your own).

Common problems

Problem	Likely cause and fix
Almost no germination	Normal for this species. Try smoke water and older seed; sow large amounts.
Seedlings collapse	Damping off. Sterile mix, good airflow.
Slow growth	Normal. Very low-P feed only.

Notes

Other notes	Conventional cuttings are poor. Research found the best material is new shoots after a fire, and tissue culture does better still – a specialist job.
Confidence	**Medium.** Nursery tests (6) show very low germination; smoke and burial research published. No reliable nursery method yet.
Sources	See References: 3, 4, 60, 61, 62, 63

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Templetonia retusa

Cockies Tongues

Family **Fabaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Easy–Moderate	Hot water (or boil 30 s); smoke	Mar – May	3–5 mm	42%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	OK	Best	Best	Best	OK	OK	OK	Best	OK	–	–

Best window · Workable – slower or less reliable · Avoid / low success

Plant snapshot

Habit	Much-branched shrub, 0.3–4 m.
Flowering	Apr – Nov (red, sometimes white-yellow).
Habitat	White sand, red clay, gravel and limestone; coastal cliffs, hillsides and plateaus.
Local note	Recorded in the Wanneroo LGA. Tough on coastal limestone.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Good	Firm current season's growth is successful.	→ Cuttings Guide
Seed	Main method	Hot water + smoke water; 8–12 days.	This guide
Division	Not used	Woody shrub.	–

Seed collection & storage

Item	Details
Seed collection	Pods ripen and split in late spring–summer. Bag developing pods and collect once they split.
Extraction & cleaning	Rub pods and sieve or winnow.
Seed quality	Hard-seeded. Float test: discard seed that floats.
Storage life	**Long** – seed keeps its viability for many years (ANPSA).
Typical germination	About **42%** after boiling-water treatment (Alcoa nursery average, 14 tests). About **20 seedlings per gram** – roughly 5 g of seed per 100 seedlings.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May** and **Sep**. Feb, Jun – Aug and Oct workable. Avoid Nov – Jan.
Seed mix	Standard sterile native seed mix.
Pre-treatment	**Hot water:** simmer at 95°C for 2 minutes, pour over the seed and soak 24 hours, then **soak 24 hours in smoke water** (Geraldton method). Alcoa used a 30 s boil. Sow the swollen seed.
Sowing depth & method	Cover about **3–5 mm**. Direct sow 2 seeds per tube and thin.
After sowing	Water in; keep moist, not wet.
Germination time	About **8–12 days** (Geraldton nursery).
Pricking out	Early, at the cotyledon to first true leaf stage.

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Item	Details
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 3–5 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Low germination	Seed coat not broken. Re-treat unswollen seed.
Swollen seed rots	Mix too wet. Water less.
Seedlings collapse	Damping off. Sterile mix, good airflow.

Notes

Other notes	Cuttings of firm current season's growth are also successful (see the Cuttings Guide).
Confidence	**High.** Nursery germination data (14 tests) plus a WA community-nursery method.
Sources	See References: 3, 4, 5, 41

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Threlkeldia diffusa

Coast Bonefruit

Family **Amaranthaceae (chenopods)**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Easy–Moderate	24 h smoke-water soak	Apr – Jun	1–2 mm	Not recorded

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	OK	Best	Best	Best	OK	OK	–	–	–	–

Best window · Workable – slower or less reliable · Avoid / low success

Plant snapshot

Habit	Much-branched, prostrate to erect perennial, 0.1–0.4 m, with semi-succulent leaves.
Flowering	Oct – Nov (tiny, green), followed by small red to black fruits.
Habitat	White or grey sand over limestone, coastal areas and saline flats.
Local note	Recorded in the City of Joondalup.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Trial	No published cutting data; trial protocol.	→ Cuttings Guide
Seed	Main method	Smoke-water soak; 10–15 days in cool weather.	This guide
Division	Not used	Not documented.	–

Seed collection & storage

Item	Details
Seed collection	Collect the small fruits in summer once they turn red to black.
Extraction & cleaning	Dry the fruits; sow them whole.
Seed quality	Cut-test a few fruits to check they are filled.
Storage life	Not well documented. Store dry and cool.
Typical germination	Not recorded.

Sowing & growing on

Item	Details
Best sowing time	**Apr – Jun** – germinates in cooler weather. Mar, Jul – Aug workable. Avoid Sep – Feb.
Seed mix	Standard sterile native seed mix, extra free-draining.
Pre-treatment	**Soak in smoke water (1:9) for 24 hours** (Geraldton method).
Sowing depth & method	Cover about **1–2 mm**.
After sowing	Water in; keep just moist – semi-succulent seedlings rot if wet.
Germination time	About **10–15 days** (Geraldton nursery).
Pricking out	When the second pair of leaves appears.

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Item	Details
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 3–4 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
No germination	Too warm. Sow in the cooler months.
Seedlings rot	Too wet. Water less, improve drainage.
Leggy seedlings	Not enough light. Move to full sun sooner.

Notes

Other notes	Softwood cuttings are a trial alternative (see the Cuttings Guide).
Confidence	**Medium.** WA community-nursery method available; no nursery test data.
Sources	See References: 4, 5

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Thysanotus manglesianus

Mangles' Fringed Lily / Climbing Fringed Lily

Family **Asparagaceae (Lomandroideae)**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Moderate–Difficult *	**Smoke water**	**Mar – May**	**Lightly covered**	**Not recorded (relative: 49%)**

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	Best	Best	Best	OK	OK	OK	–	–	–	–

Best window | Workable – slower or less reliable | Avoid / low success

Plant snapshot

Habit	Twining, leafless perennial herb with tuberous roots; stems 0.2–2 m.
Flowering	Aug – Nov (purple, with fringed petals).
Habitat	White or red sand, loam, clay, laterite and granite.
Local note	Recorded in the Wanneroo LGA on FloraBase, and listed as indigenous at Warwick Bushland, Joondalup. Dies back each year and grows new stems in autumn.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Not used	Not grown from cuttings.	–
Seed	Main method	Smoke water; sow fresh seed (within 6 months).	This guide
Division	Not used	Tuberous roots – division not documented.	–

Seed collection & storage

Item	Details
Seed collection	Estimate: Nov – Jan, as the capsules dry after flowering – the capsule splits inside the old flower parts. Plants set little seed (related species), so collect from many plants.
Extraction & cleaning	Dry in paper bags, rub and sieve.
Seed quality	Round seed about 1.5 mm across with a straw-coloured aril; *Thysanotus* seed is usually black (genus).
Storage life	**Short** – sow within 6 months of collection (related *T. patersonii*).
Typical germination	No germination % recorded; Alcoa's tests average about **50 seedlings per gram** (5 tests). Its close relative *T. patersonii* gave about 49% untreated (7 tests). Smoke strongly promotes germination in *Thysanotus* (Kings Park).

Sowing & growing on

Item	Details
Best sowing time	**Mar – May** (estimate). Jun – Aug workable. Avoid Sep – Feb.
Seed mix	Standard sterile native seed mix, free-draining.
Pre-treatment	**Smoke water** – 24 h soak (1:9) or water the trays with it.
Sowing depth & method	Sow thinly and cover lightly.
After sowing	Keep moist through the cool months.

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Item	Details
Germination time	About **2–7 weeks** in the related *T. patersonii* (Kings Park: 15–46 days).
Pricking out	Early, at 1–2 leaves, before the tuberous roots form – handle gently.
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Slow – seedlings may die back in their first summer. Record your own.

Common problems

Problem	Likely cause and fix
Seedlings die back in summer	Normal – it dies back each year. Keep pots barely moist until new stems appear in autumn.
Very few seedlings	Use smoke water and fresh seed, and sow generously.
Seedlings collapse	Damping off. Sterile mix, airflow.

Notes

Other notes	Dividing the tuberous roots is not documented – grow from seed. It twines, so give potted plants a stake.
Confidence	**Low.** Nursery seedlings-per-gram data (5 tests) but no germination %. Method based on its close relative *T. patersonii* and smoke research on the genus.
Sources	See References: 3, 4, 49, 143, 149, 152, 153

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Trachymene pilosa

Native Parsnip / Dwarf Parsnip

Family **Araliaceae (formerly Apiaceae)**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Easy–Moderate	None; smoke optional	Apr – May	Lightly covered	41%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	OK	Best	Best	OK	–	–	–	–	–	–

Best window · Workable – slower or less reliable · Avoid / low success

Plant snapshot

Habit	Softly hairy annual herb, up to 0.3 m.
Flowering	Aug – Oct (small, white to pale pink).
Habitat	Woodland and heath; seedlings come up in numbers after fire.
Local note	Recorded at Warwick Bushland (Joondalup), where seedlings were common after fire. An annual – it flowers the spring after sowing, then dies.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Not used	Annual.	–
Seed	Main method	Annual – no pre-treatment; about 41% (Alcoa).	This guide
Division	Not used	Annual.	–

Seed collection & storage

Item	Details
Seed collection	Estimate: Oct – Dec, as the fruit heads dry after flowering. Each fruit splits into two halves; the outer half is bristly.
Extraction & cleaning	Rub dry heads and sieve; sow the fruit halves whole.
Seed quality	Fruit halves are 2–2.5 mm long. Seed lots tested fully viable in a 2025 study.
Storage life	Not documented. Store dry and cool; sow the following autumn.
Typical germination	About **41%** untreated (Alcoa, 15 tests), about **294 seedlings per gram**. In a 2025 lab study untreated seed gave 16–18%, and neither gibberellic acid nor a smoke chemical improved it.

Sowing & growing on

Item	Details
Best sowing time	**Apr – May**, after the first autumn rains. Mar and Jun workable. Lab germination was similar at 20/10°C and 25/15°C.
Seed mix	Standard sterile native seed mix.
Pre-treatment	No pre-treatment was used in tests (41% at Alcoa). The 2025 study found the seed fully viable but only 16–18% germinated, so much of each lot is dormant. A seed supplier recommends smoke, and *Trachymene* is listed as smoke-responsive – trial smoke water on part of a batch.
Sowing depth & method	Cover lightly. Sowing 2–3 fruit halves straight into each cell avoids pricking out.
After sowing	Keep moist.

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Item	Details
Germination time	First seedlings after about **9–13 days** in the lab and 15–24 days (mean 19) in 3 Kings Park lots.
Pricking out	If tray-sown, prick out small, at the second-leaf stage. Thin direct-sown cells to 1–3 seedlings.
Shade house & hardening	Short – a week or so in the shade house, then full sun.
Time to planting size	Plant out small, about **6–10 weeks** after sowing (estimate), so plants establish in winter and flower in spring.

Common problems

Problem	Likely cause and fix
Patchy germination	Seed lots vary. Record each lot; compare smoked and untreated trays.
Leggy seedlings	Too much shade. Move to full sun early.
Seedlings collapse	Damping off. Sterile mix, airflow.

Notes

Other notes	Annual *Trachymene* was among the most common plants coming up from transferred topsoil in Perth Banksia woodland restoration, though numbers fell by the third year.
Confidence	**Medium.** Nursery data (15 tests), Kings Park emergence records and a 2025 lab study.
Sources	See References: 3, 4, 43, 49, 110, 143, 146, 149, 151

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Tricoryne elatior

Yellow Autumn Lily

Family **Asphodelaceae (Hemerocallidoideae)**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Difficult	**Smoke water (trial)**	**Mar – May**	**Lightly covered**	**1%**

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	**Best**	**Best**	**Best**	**OK**	**OK**	**OK**	–	–	–	–

Best window · Workable – slower or less reliable · Avoid / low success

Plant snapshot

Habit	Rhizomatous perennial herb, 0.1–0.5 m (to 0.6 m).
Flowering	Yellow, all year (FloraBase: Jan – Dec).
Habitat	Sand, loam and lateritic soils, sometimes wet.
Local note	Recorded in Joondalup and Wanneroo. Also called Yellow Rush-lily.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Not used	Not grown from cuttings.	–
Seed	Difficult	Dormant seed; about 1% in nursery tests. Smoke may help.	This guide
Division	Possible	Usually divided in eastern Australia – small rhizome clumps.	–

Seed collection & storage

Item	Details
Seed collection	Fruit ripens 2–3 months after flowering and is **still green when the seed is ripe** – check for black, firm seed, then pick capsules by hand into a paper bag. Seed is shed 3–14 days after it matures.
Extraction & cleaning	Stand capsules upside-down in a paper bag; thresh lightly once dry, and sieve.
Seed quality	Black, egg-shaped seed about 3.5 mm long. Viability is usually high (76% in WA smoke research, RIRDC).
Storage life	Not documented.
Typical germination	Very low: about **1%** untreated (Alcoa, 2 tests), about 2.4 seedlings per gram. Fresh seed is dormant; smoke may help.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May** (estimate). Jun – Aug workable. Avoid Sep – Feb.
Seed mix	Standard sterile native seed mix, free-draining.
Pre-treatment	**Smoke water** (24 h soak, 1:9) as a trial.
Sowing depth & method	Sow thinly and cover lightly.
After sowing	Keep moist, not wet. Keep trays a second season – dormant seed may come up later.
Germination time	Not documented.
Pricking out	At 2–3 leaves, gently.

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Item	Details
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 5–6 months from germination (record your own).

Common problems

Problem	Likely cause and fix
Very few seedlings	Normal – seed is dormant. Use smoke and keep trays; try division instead.
Seedlings rot	Too wet. Water less; free-draining mix.
Seedlings collapse	Damping off. Sterile mix, airflow.

Notes

Other notes	Usually propagated by **division** in eastern Australia – it spreads from short rhizomes. See the Division Guide's list of other species.
Confidence	**Medium.** Nursery data (2 tests) and eastern Australian growing notes agree germination is poor.
Sources	See References: 3, 4, 110, 131, 132

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Trymalium ledifolium

Star Buckthorn

Family **Rhamnaceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Moderate	Hot water	Apr – Jun	2–3 mm (needs dark)	22%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	OK	Best	Best	Best	OK	OK	–	–	–	–

Best window · Workable – slower or less reliable · Avoid / low success

Plant snapshot

Habit	Erect, multi-stemmed shrub, 0.3–2.5 m (var. *ledifolium*).
Flowering	Jul – Nov (white-cream).
Habitat	Calcareous grey sand; limestone ridges, outcrops and dunes.
Local note	The local plant is var. *ledifolium*, recorded in Joondalup and Wanneroo. The other two varieties are single-stemmed and grow inland on laterite or clay loam.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Trial	Not documented.	–
Seed	Main method	Hot water, then sow cool and covered; about 22% (Alcoa).	This guide
Division	Not used	Shrub – not divided.	–

Seed collection & storage

Item	Details
Seed collection	Fruits **Oct – Dec** (Flora of Australia). The fruit is small (1.5–3.5 mm), greenish or black – bag branches or collect as it darkens, before the fruitlets fall.
Extraction & cleaning	Dry, rub and sieve. The fruit breaks into hard, bony, one-seeded fruitlets that don't open – sow these as they are (the seed inside is 1–1.8 mm).
Seed quality	Water-impervious seed coat (physical dormancy) – seed won't take up water until treated (WA research).
Storage life	Not documented; hard-coated seed is usually long-lived when kept dry.
Typical germination	About **22%** after boiling water (Alcoa, 43 tests), about **165 seedlings per gram**. In a WA study of six Rhamnaceae including this species, hot-water-treated seed reached over 70% at cool (7/18°C) temperatures; warmth and light suppressed germination.

Sowing & growing on

Item	Details
Best sowing time	**Apr – Jun** – germinates best at winter-like temperatures (7/18°C); warmth suppresses it. Mar, Jul – Aug workable. Avoid Sep – Feb.
Seed mix	Standard sterile native seed mix.
Pre-treatment	**Hot water** – Alcoa dipped seed in boiling water for 30 seconds. Keep it short; over-long treatment can kill seed.
Sowing depth & method	Cover about **2–3 mm** – light suppresses germination, so keep the seed covered and trays shaded.

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Item	Details
After sowing	Water in; keep moist in a cool, shaded spot.
Germination time	Not well documented – one Kings Park lot showed first seedlings at 38 days. Allow several weeks and record results.
Pricking out	When the second pair of leaves appears.
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 4–6 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
No germination	Seed coat not broken, or trays too warm or too light. Re-treat; cover seed; sow in cooler months.
Seed killed by treatment	Hot water too long. Keep the dip short.
Damping off	Wet, crowded trays. Thin sowing, airflow.

Notes

Other notes	Handle like *Spyridium globulosum* (also Rhamnaceae): hot water, then sow cool and covered. Cuttings are not documented.
Confidence	**Medium.** Nursery data (43 tests, boiling water), Flora of Australia fruit notes and WA research on seed dormancy in Rhamnaceae.
Sources	See References: 3, 4, 22, 143, 157, 160

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Waitzia suaveolens

Fragrant Waitzia

Family **Asteraceae**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Easy–Moderate	Smoke water (optional)	Apr – May	Surface – light cover	21%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	OK	Best	Best	OK	–	–	–	–	–	–

Best window · Workable – slower or less reliable · Avoid / low success

Plant snapshot

Habit	Annual herb to 0.6 m.
Flowering	Sep – Dec or Jan (white, pink, purple or yellow papery heads).
Habitat	Widespread across the south-west, including the Swan Coastal Plain.
Local note	Recorded in the Wanneroo LGA. An annual – it flowers the spring after sowing, then dies.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Not used	Annual.	–
Seed	Main method	Annual everlasting – sow after the first autumn rains.	This guide
Division	Not used	Annual.	–

Seed collection & storage

Item	Details
Seed collection	Collect heads in early summer once they dry and the fluffy seed starts to lift.
Extraction & cleaning	Rub heads gently over a sieve. Seed is very fine.
Seed quality	Fine seed – a little goes a long way.
Storage life	Store dry and cool and sow the following autumn.
Typical germination	About **21%** (untreated, Alcoa nursery average, 2 tests). About **967 seedlings per gram** – roughly 0.1 g of seed per 100 seedlings.

Sowing & growing on

Item	Details
Best sowing time	**Apr – May**, after the first autumn rains. Mar and Jun are workable.
Seed mix	Standard sterile native seed mix.
Pre-treatment	None required. Smoke water in the first watering may help.
Sowing depth & method	Mix the fine seed with dry sand and surface-sow; cover very lightly or not at all. A pinch per tube avoids pricking out.
After sowing	Water in with a fine mist; keep moist.
Germination time	About **2–4 weeks**.
Pricking out	If tray-sown, prick out small, at the second-leaf stage. Thin direct-sown cells to 1–3 seedlings.

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Item	Details
Shade house & hardening	Short – a week or so in the shade house, then full sun.
Time to planting size	Plant out small, about **6–10 weeks** after sowing, so plants flower in spring.

Common problems

Problem	Likely cause and fix
Seed washed into clumps	Bottom water or use a very fine mist.
Leggy seedlings	Too much shade. Full sun early.
Plants flower in the tube	Held too long. Plant out while small.

Notes

Other notes	Often direct-seeded into revegetation sites after the first autumn rains.
Confidence	**Medium.** Nursery germination data (2 tests); sowing follows standard practice for annual everlastings.
Sources	See References: 3, 4, 30

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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Xanthorrhoea preissii

Balga / Grass Tree

Family **Asphodelaceae (Xanthorrhoeoideae)**

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Easy–Moderate	None	Mar – May	3–5 mm	47%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	Best	Best	Best	OK	OK	OK	–	–	–	–

Best window · Workable – slower or less reliable · Avoid / low success

Plant snapshot

Habit	Grass tree to 5 m, trunk over 3 m on old plants. **Very slow** – about 1.5 cm of trunk a year.
Flowering	Jun or Aug – Dec (white-cream spikes).
Habitat	Grey to black sand, laterite and granite; ranges, coastal plain and near watercourses.
Local note	Recorded in the Wanneroo LGA and common on the coastal plain. Susceptible to *Phytophthora* dieback – strict hygiene.

Propagation options

Method	Suitability	Notes	Details in
Cuttings	Not used	Not possible.	–
Seed	Main method	No pre-treatment; about 47% (Alcoa). Very slow growing.	This guide
Division	Not used	Not divided; bush transplants often fail.	–

Seed collection & storage

Item	Details
Seed collection	The black seeds are released from the spike when ripe (summer). Cut spikes as capsules begin to split, or bag them.
Extraction & cleaning	Dry spikes on a tarp or in bags; knock out the black seeds and sieve.
Seed quality	Cut-test a sample – good seed is firm and white inside.
Storage life	Not well documented. Store dry and cool.
Typical germination	About **47%** untreated (Alcoa nursery average, 42 tests). About **45 seedlings per gram** – roughly 2.2 g of seed per 100 seedlings.

Sowing & growing on

Item	Details
Best sowing time	**Mar – May**. Jun – Aug workable. Avoid Sep – Feb.
Seed mix	Free-draining, pathogen-free mix.
Pre-treatment	**None needed.**
Sowing depth & method	Cover about **3–5 mm**. Direct sow 2–3 seeds per deep tube.
After sowing	Keep moist, not wet.
Germination time	Starts in about **3–6 weeks**, but some seed can take much longer – keep trays going.
Pricking out	Avoid if possible – sow direct into deep tubes. Otherwise prick out at the first leaf.

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Item	Details
Shade house & hardening	About **2 weeks** in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 12 months or more – very slow (record your own).

Common problems

Problem	Likely cause and fix
Seedlings yellow and die	Dieback or waterlogging. Strict hygiene, pots off the ground, free-draining mix.
Very slow growth	Normal for grass trees. Don't overfeed.
Damping off	Sow thinly; airflow.

Notes

Other notes	Not divided. Moving established plants from the bush has a high failure rate and is only worth doing with a large root ball where plants would otherwise be destroyed.
Confidence	**High.** Large nursery data set (42 tests); growth and handling notes published.
Sources	See References: 3, 4, 49, 85, 86, 87, 97

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

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120	Crisp M.D. et al. (2017) Phytotaxa 300 – Daviesia revision (D. decurrens pods, seed and fruiting time)	tb.plazi.org
121	State Flora SA – coastal garden plants (Leucopogon parviflorus: buffers coastal wind; birds and lizards eat the fruit)	stateflora.sa.gov.au/news/coastal-garden-plants
122	Seeds of South Australia – Atriplex cinerea (collection Oct – Mar, drying, storage in the fruit)	seedsofsa.au (read via syzygium.xyz mirror)
123	Understorey Network (Tasmania) – Atriplex cinerea (washing salts out of the fruit; 2–3 seeds per pot; seed shed 3–14 days after maturity)	database.understorey-network.org.au/plant/atriplex-cinerea

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#	Source	Address
124	Martyn A.J. et al. (2009) – seed dormancy in Rutaceae: smoke and gibberellic acid (lists Philotheca spicata as smoke-responsive)	publikationen.ub.uni-frankfurt.de/files/29327/martyn_et_al_2009_rutaceae.pdf
125	Understorey Network (Tasmania) – Philotheca verrucosa (genus: seed released explosively; ripens over 1–2 weeks)	database.understorey-network.org.au/plant/philotheca-verrucosa
126	AusGrass2 – Amphipogon turbinatus (description: rhizomes, panicle, three-awned lemma)	ausgrass2.myspecies.info/node/297
127	Brundrett M. – Banksia woodland restoration (RIAWA presentation; Amphipogon turbinatus among the best-performing understorey species)	riawa.com.au/assets/documents/Mark-Brundrett-Banksia-Woodland-Retsoration.pdf
128	Keighery G.J. & Muir W. (2005) Nuytsia 15 – Burchardia congesta reinstated for WA plants; seed maturity and release at Woodvale	library.dbca.wa.gov.au/static/Journals/080057/080057-15.038.pdf
129	Understorey Network (Tasmania) – Burchardia umbellata (3-month after-ripening; germinates below about 18°C; 4–8 weeks)	database.understorey-network.org.au/plant/burchardia-umbellata
130	Greening Australia – About Burchardia umbellata (role of smoke ambiguous; seed handling)	greeningaustralia.org.au/publications/about-burchardia-umbellata
131	Greening Australia – About Tricoryne elatior (fruit green when ripe; dormant seed; usually divided)	greeningaustralia.org.au/publications/about-tricoryne-elatior
132	Understorey Network (Tasmania) – Tricoryne elatior (very difficult from seed; shed 3–14 days after maturity)	database.understorey-network.org.au/plant/tricoryne-elatior
133	Keighery G.J. (2002) Nuytsia 14 – Laxmannia (western species in mature seed in January)	library.dbca.wa.gov.au/static/Journals/080057/080057-14.008.pdf
134	Maher K. (2009) – Gnangara restoration species report (Burchardia and Laxmannia propagated from seed)	library.dbca.wa.gov.au/FullTextFiles/922997.pdf
135	Friends of Kings Park – Laxmannia squarrosa (good container plant; light sandy soils)	friendsofkingspark.com.au/product/laxmannia-squarrosa/
136	West Tamar Landcare – Burchardia umbellata (watering dormant plants rots the tubers)	wtlandcare.org
137	NZ Plant Conservation Network – Atriplex cinerea fact sheet (female flowers occasionally on male plants; most stock female)	nzpcn.org.nz
138	Seeds of South Australia – Amphipogon strictus (25% viability in an SA seed lot)	seedsofsa.au (read via syzygium.xyz mirror)
139	Seeds of South Australia – Melaleuca lanceolata (large hard capsules lower on the stems; drying, sieving; non-dormant; 85–95% viable)	seedsofsa.au (read via syzygium.xyz mirror)
140	Turner S.R. et al. (2009) Annals of Botany 104 – water gap in physically dormant seeds of Dodonaea petiolaris (germination temperatures; D. aptera seed coat after a 30 s hot-water dip)	pmc.ncbi.nlm.nih.gov/articles/PMC2749536
141	Closs J. (with West J.), ANPSA Australian Plants Online (Sep 2002; from Australian Plants, Dec 1993) – Dodonaea: boiling water, male and female plants, cuttings with 2,000 ppm IBA	anpsa.org.au/APOL27/sep02-1.html
142	APACE WA – Dodonaea aptera (green flowers; ripe four-angled fruit in early summer)	apacewa.org.au/?p=612
143	Kullmann W.H. (comp.) (c. 1981) Seed Germination Records of Western Australian Plants. Kings Park Research Notes No. 7 (seed lots germinated; days to first seedling)	library.dbca.wa.gov.au/static/Journals/080234/080234-07.pdf
144	Atlas of Living Australia – Eryngium pinnatifidum subsp. Palustre (Priority Three) and subsp. Umbraphilum (Priority Two)	bie.ala.org.au
145	Seeds of South Australia – Eryngium ovinum (related species: non-dormant, 88%; collection and cleaning with a rubber bung)	seedsofsa.au (read via syzygium.xyz mirror)
146	Saraeian Z., Farrell C., Hirst M. & Williams N.S.G. (2025) Australian Journal of Botany 73, BT25002 – seed mass, germination and climate in 58 Australian annual species (Hyalosperma cotula, Podotheca angustifolia, Trachymene pilosa)	connectsci.au/bt/article/73/8/BT25002

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#	Source	Address
147	Seeds of South Australia – Hyalosperma demissum and H. semisterile (genus: collect Oct – Dec; drying, sieving; non-dormant)	seedsofsa.au (read via syzygium.xyz mirror)
148	Plummer J.A. & Bell D.T. (1995) Australian Journal of Botany 43 – temperature, light and GA3 in the germination of everlasting daisies (best at 10–20°C)	connectsci.au/bt/article/43/1/93
149	Friends of Warwick Bushland (Joondalup) – indigenous plants list 2024 (Eryngium pinnatifidum, Podotheca angustifolia, Thysanotus manglesianus) and trail notes (annual seedlings after fire, incl. Trachymene pilosa)	friendsofwarwickbushland.com
150	Seeds of South Australia – Podotheca angustifolia (collection, cleaning, −18°C storage, germination 96% / 52%)	seedsofsa.au (read via syzygium.xyz mirror)
151	Banksia woodland topsoil-transfer study, Perth (bioRxiv 2020) – annual Austrostipa, Podotheca and Trachymene most common at first	biorxiv.org/content/10.1101/2020.07.09.195131
152	Dixon K. & Roche S. – Smoke stimulates the germination of many Western Australian plants (SGAP Journal 1995; ANPSA Australian Plants Online, Jun 1996): smoke promotes Thysanotus, suppresses some paper daisies	anpsa.org.au/APOL2/jun96-6.html
153	Understorey Network (Tasmania) – Thysanotus patersonii (smoke may help; sow within 6 months; plants set little seed)	database.understorey-network.org.au/plant/thysanotus-patersonii
154	The Seed Collection – Yanchep Rose, Diplolaena angustifolia (85 seeds per gram; 30–70 days at 22–26°C; smoke may help)	theseedcollection.com.au/Yanchep-Rose
155	ANPSA – Diplolaena microcephala profile (seed difficult; cuttings of firm current season's growth)	anpsa.org.au/plant_profiles/diplolaena-microcephala/
156	Wikipedia – Melaleuca lanceolata (resistant to salt spray; windbreak and shelterbelt use)	en.wikipedia.org/wiki/Melaleuca_lanceolata
157	Wikipedia – Trymalium ledifolium (sometimes called Star Buckthorn)	en.wikipedia.org/wiki/Trymalium_ledifolium
158	Flora of Australia online – Diplolaena angustifolia (Wanneroo north to near Eneabba)	profiles.ala.org.au/opus/foa/profile/Diplolaena%20angustifolia
159	Flora of Australia online – Dodonaea aptera (capsule 5–6 mm, lobed not winged; Shark Bay to Cape Freycinet)	profiles.ala.org.au/opus/foa/profile/Dodonaea%20aptera
160	Flora of Australia online – Trymalium ledifolium (fruits Oct – Dec; fruit 1.5–3.5 mm; bony one-seeded fruitlets; seed 1–1.8 mm)	profiles.ala.org.au/opus/foa/profile/Trymalium%20ledifolium

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Revision log

Version	Date	Changes
1.0	Sep 2026	First edition: Allocasuarina fraseriana, A. humilis, Anigozanthos manglesii, Banksia attenuata, B. menziesii, Daviesia nudiflora, Gompholobium tomentosum, Hovea pungens.
1.1	Sep 2026	Added 15 species: Austrostipa flavescens, Banksia sessilis, Bossiaea eriocarpa, Calothamnus quadrifidus, Corymbia calophylla, Dianella revoluta, Eucalyptus gomphocephala, E. marginata, Kennedia prostrata, Melaleuca huegelii, M. systena, Olearia axillaris, Petrophile linearis, P. macrostachya, Spyridium globulosum.
1.2	Sep 2026	Added 18 species: Allocasuarina lehmanniana, Atriplex isatidea, Carpobrotus virescens, Daviesia divaricata, D. triflora, Grevillea vestita, Leucophyta brownii, Macrozamia fraseri, Myoporum insulare, Podotheca gnaphalioides, Ptilotus drummondii, P. manglesii, P. polystachyus, Rhagodia baccata, Spinifex hirsutus, S. longifolius, Threlkeldia diffusa, Waitzia suaveolens. Added a ‘Propagation options’ table (cuttings, seed, division) to every species page.
1.3	Sep 2026	Added 13 species: Acacia lasiocarpa, A. pulchella, A. truncata, Eucalyptus decipiens, Gastrolobium capitatum, G. nervosum, Hardenbergia comptoniana, Hovea trisperma, Jacksonia furcellata, J. sericea, J. sternbergiana, Nuytsia floribunda, Templetonia retusa.
1.4	Sep 2026	Added 7 species: Banksia dallanneyi, B. grandis, B. prionotes, Hakea lissocarpha, H. prostrata, H. trifurcata, Stirlingia latifolia.
1.5	Sep 2026	Added 11 species: Acanthocarpus preissii, Ammothryon grandiflorum (Schoenus grandiflorus), Clematis linearifolia, Conostylis aculeata, Exocarpos sparteus, Ficinia nodosa, Lepidosperma gladiatum, Mesomelaena pseudostygia, Opercularia vaginata, Pigea calycina (Hybanthus calycinus), Santalum acuminatum.
1.6	Sep 2026	Added 12 species: Acacia cochlearis, Anigozanthos humilis, Austrostipa elegantissima, Conostylis candicans, C. setigera, Eremaea pauciflora, Frankenia pauciflora, Guichenotia ledifolia, Petrophile axillaris, P. brevifolia, Poa poiformis, Xanthorrhoea preissii.
1.7	Oct 2026	Added 7 species: Daviesia decurrens, Haemodorum laxum, Leucopogon parviflorus, Melaleuca cardiophylla, Orthrosanthus laxus, Patersonia occidentalis, Ricinocarpos undulatus.
1.8	Oct 2026	Added 6 species: Amphipogon turbinatus, Atriplex cinerea, Burchardia congesta, Laxmannia squarrosa, Philotheca spicata, Tricoryne elatior.
1.9	Oct 2026	Added 9 species: Diplolaena angustifolia, Dodonaea aptera, Eryngium pinnatifidum, Hyalosperma cotula, Melaleuca lanceolata, Podotheca angustifolia, Thysanotus manglesianus, Trachymene pilosa, Trymalium ledifolium.

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